

Fadhili Combined Heat and Power Plant – EPC Contract

Exhibit B, Part 2 Sponsor's Functional Requirement

TABLE OF CONTENTS

1.	12	
2.	13	
2.1.	13	
2.2.	13	
2.3.	13	
2.3.1.	Mode of operation	12
2.3.2.	Availability and reliability	13
2.4.	15	
3.	16	
3.1.	17	
4.	18	
4.1.	18	
4.2.	18	
4.2.1.	Equipment identification system	17
5.	21	
5.1.	21	
5.2.	21	
5.3.	21	
5.4.	21	
5.5.	21	
5.6.	21	
5.7.	22	
5.8.	22	

5.8.1.	Environmental and Social Impact Assessment	19
6.	23	
6.1.	23	
6.1.1.	General requirements	20
6.1.2.	Noise limits	21
6.1.3.	Frequency variations	21
6.1.4.	Air intake systems and filtration	21
6.1.5.	Fire/Gas detection and protection systems	22
6.1.6.	Exhaust ducting	22
6.1.7.	Maintainability	23
6.2.	27	
6.2.1.	General requirements	23
6.2.2.	Performance Requirements	24
6.2.3.	Non-Return Valves	24
6.2.4.	Protective Devices	25
6.2.5.	Bearings & Lubricating Oil System	25
6.2.6.	Barring Gear	26
6.2.7.	Steam Turbine Control & Monitoring Systems	26
6.2.8.	Steam Turbine Start up System	26
6.2.9.	Condensate extraction pumps	26
6.2.10.	Condensate tank	27
6.2.11.	Drain pot and drain pumps	27
6.2.12.	Vacuum Raising and Maintaining Plant	27
6.2.13.	Steam Connections and Drains	27
6.2.14.	Noise Limits	28
6.2.15.	Turbine Pipework and Drains	28
6.2.16.	Maintainability	28
6.3.	33	
6.3.1.	General requirements	28
6.3.2.	HRSG drums and headers	30
6.3.3.	Superheater, evaporator and economiser banks	30
6.3.4.	Steam temperature and output control	31
6.3.5.	Valves and accessories	31
6.3.6.	Feed Water control stations	31
6.3.7.	Supplementary firing system	31
6.3.8.	Blowdown equipment	32
6.3.9.	HRSG drum local water level gauges and remote indicators	32
6.3.10.	Stacks	32
6.3.11.	Insulation	33
6.3.12.	Cleaning and preservation	33

6.3.13. Steam connections and drains	33
6.4.	39
6.4.1. General	33
6.4.2. Steam turbine by-pass	34
6.4.3. Reserve Boiler(s)	34
6.5.	41
6.6.	45
6.6.1. Feedwater pumps	38
6.6.2. Condensate system	39
6.6.3. Condensate polishing system	39
6.7.	46
6.7.1. Closed loop cooling water systems	39
6.8.	47
6.8.1. Fuel Gas systems	40
6.8.2. Additional requirements	42
6.8.3. Gas Buffer Station	42
6.9.	50
6.10.	50
6.11.	50
6.12.	52
6.13.	52
6.14.	53
6.14.1. General requirements	45
6.14.2. Separate Fire Compartments	46
6.14.3. Fixed Wet Pipe Sprinkler Systems	46
6.14.4. Fixed Deluge Systems	46
6.14.5. Fixed CO2 Extinguishing System	47
6.14.6. Gas Detection System for Natural Gas	47
6.14.7. Fire Detection Systems	47
6.14.8. Smoke Ventilation	48
6.14.9. External Hydrants and Hose Reels	48
6.14.10. Portable Fire Extinguishers	48
6.15.	57
6.15.1. Emergency/black start diesel generator	48
6.15.2. Compressed air system	49

6.15.3. Nitrogen system / preservation of the Plant	50
6.15.4. Lifting / Hoisting Equipment	50
6.16. 60	
6.16.1. Piping and tubing	51
6.16.2. Pumps	53
6.16.3. Valves and Actuators	54
6.16.4. Heat Exchangers	56
6.16.5. Thermal Insulation	56
6.16.6. Vibration	57
6.16.7. Protection against Corrosion	57
6.17. 68	
6.17.1. Welders	58
6.17.2. Welder's Tests	58
6.17.3. Welding Procedures	58
6.17.4. Welding and Heat Treatment	58
6.17.5. Site Weld repairs	60
6.17.6. Welding Electrodes and Consumables	60
6.17.7. Inspection of Welding	60
6.18. 72	
7. 74	
7.1. 74	
7.2. 74	
7.2.1. Design requirements	62
7.2.2. Generators and generator excitation systems	65
7.2.3. Generator circuit breaker	67
7.2.4. Power transformers	69
7.2.5. MV system neutral earthing resistors	72
7.2.6. MV and LV AC switchboards	72
7.2.7. Busbar and busduct systems	73
7.2.8. Electrical system control, indications and alarms	74
7.2.9. Low voltage AC and DC sub-distribution boards	75
7.2.10. Electric motors	75
7.2.11. DC supply systems and un-interruptible power supplies	77
7.2.12. Cables and installation	77
7.2.13. Communications / telecommunications	81
7.2.14. Lighting requirements	82
7.2.15. Earthing system design, bonding and lightning protection	83
7.3. 99	

8.	100	
8.1.	100	
8.1.1.	Software security management	84
8.1.2.	Lifetime system maintenance support	85
8.1.3.	Engineer's programming terminal	86
8.2.	103	
8.3.	104	
8.3.1.	General requirements	87
8.3.2.	Availability - Reliability - Maintenance	90
8.3.3.	Typical functions of the DCMS	90
8.3.4.	Execution times of the main DCMS	90
8.3.5.	Checking of measurements to DCMS	91
8.3.6.	DCMS control functions	91
8.3.7.	DCMS supervision functions	91
8.3.8.	DCMS event chronological recording	92
8.3.9.	Measurement supervision	92
8.3.10.	Alarm function	92
8.3.11.	External links – design requirements	94
8.3.12.	Plant Information (PI) System	94
8.3.13.	Reserve capacity	94
8.4.	113	
8.4.1.	General requirements	95
8.4.2.	Safety Systems	95
8.4.3.	User interface	96
8.4.4.	Signal interfaces	97
8.4.5.	Grounding	97
8.4.6.	Safety Integrity requirements	97
8.4.7.	Fault tolerances of the SRS	98
8.4.8.	Field devices	98
8.5.	117	
8.6.	117	
8.7.	118	
8.7.1.	Field mounted instruments	100
8.7.2.	Final control elements	109
8.8.	133	
8.8.1.	I&C cabling	112
8.8.2.	Cubicles and racks	112

9.	136	
9.1.	136	
9.2.	136	
9.3.	136	
9.3.1.	Geotechnical / Foundations	115
9.4.	138	
9.4.1.	General	115
9.4.2.	Administration and social building	116
9.4.3.	Workshop and warehouse buildings/areas	116
9.4.4.	Gate house	117
9.4.5.	Central Control and Switchgear building	118
9.4.6.	Laboratory	118
9.4.7.	Other buildings	119
9.4.8.	Design criteria / structural design	119
9.5.	143	
9.5.1.	Structural framework for buildings	120
9.5.2.	Flooring	120
9.5.3.	External Wall Cladding	121
9.5.4.	Roof Cladding	121
9.5.5.	Internal Walls and Partitions	121
9.5.6.	Doors	121
9.5.7.	Windows	121
9.5.8.	Stairways	121
9.5.9.	Internal Finishes	122
9.5.10.	Furnishing and Fittings	123
9.5.11.	Fire protection	147
9.5.12.	Sanitary and cold water plumbing	123
9.6.	148	
9.6.1.	Transformer compound design	124
9.6.2.	Turbine foundations	125
9.6.3.	HRSG foundations	125
9.6.4.	Roads	126
9.6.5.	Paved areas	126
9.6.6.	Permanent fences	127
9.6.7.	Cable trenches	127
9.6.8.	Site installation	127
9.6.9.	General levelling and grading of Site	128
9.6.10.	Fuel Oil Storage	128

9.7.	153	
9.8.	153	
9.8.1.	General	128
9.8.2.	HVAC automation	129
9.9.	155	
10.	156	
10.1.	156	
10.2.	156	
10.3.	156	
10.4.	157	
10.4.1.	Overall Maintainability	132
10.4.2.	Maintenance Access Criteria	133
10.4.3.	Operating and Troubleshooting Criteria	134
10.5.	160	
11.	161	
12.	161	
13.	161	
14.	162	
14.1.	162	
14.1.1.	Contractors Central Administration Area	136
14.1.2.	Contractor's Accommodation	136
14.1.3.	Owner's Accommodation and Facilities	136
14.1.4.	Site Records Office	137
14.1.5.	Telephones & IT	137
14.1.6.	First Aid	137
14.1.7.	Site security	138
14.1.8.	Contractor's Storage Areas	138
14.1.9.	Sign Boards	138
14.1.10.	Site Cleanliness and Waste Disposal	138
14.1.11.	Sanitation and Drainage	139
14.1.12.	Sanitary Facilities	139
14.1.13.	Scaffolding	140
14.1.14.	Use of Motor Vehicles and Parking	140

14.2.	167	
14.2.1.	HAZOP study	141
14.2.2.	Safety Management	142
14.2.3.	Safety Plans and Method Statements	142
14.2.4.	Safety Rules	142
14.2.5.	Safety Personnel	143
14.2.6.	Safety Training	143
14.2.7.	Safety Meetings	143
14.2.8.	Typical Items for Inclusion in a Site Safety Plan	143
14.2.9.	Permit to Work system	145
14.2.10.	Obstruction to access	145
14.2.11.	Clear road access	146
14.2.12.	Site buildings	146
14.2.13.	Site labour	146
14.2.14.	Sanitary and health control	146
14.2.15.	Dumping	146
14.3.	174	
14.4.	175	
14.5.	175	
14.6.	176	
14.6.1.	Monitoring of Environmental Impacts	148
14.6.2.	Approvals Procedure	148
15.	177	
16.	177	
17.	178	
17.1.	178	
17.2.	179	
17.3.	179	
17.3.1.	Notification Points for Inspections and Tests (NPIT) (or “Witness Point”)	151
17.3.2.	Hold Points (HP)	152
17.4.	180	
17.5.	181	
17.6.	181	

18.182	
18.1.	182
18.1.1.	Documentation Review Procedure 154
18.1.2.	Drawings 155
18.2.	185
18.3.	189
18.4.	189
18.5.	190
18.6.	190
18.6.1.	General Requirements 160
18.6.2.	Operation Manuals 161
18.6.3.	Maintenance Manuals 161
18.7.	192
18.8.	192
18.8.1.	Cogeneration plant concept 163
19.195	
19.1.	195
19.2.	195
19.2.1.	Project schedule 165
19.3.	195
19.3.1.	Levels required are: Planning Network Programmes 165
19.3.2.	Progress Measurement 166
19.3.3.	Progress Reporting 166
19.3.4.	Progress Meetings 168
19.3.5.	Project Management and Co-ordination Procedures 168
19.3.6.	Communications 168
19.3.7.	Diaries for Daily Work Measurements for construction site 168
20.199	
20.1.	199
21.201	
21.1.	201

22. 204

1. GENERAL INTRODUCTION AND SUBJECT

This Exhibit B, Part 2 (Sponsor's Functional Requirements) defines the general and specific requirements for the Works in addition to the requirements in given in Exhibit B, Part 1, which is a strict copy of the Saudi Aramco and SEC technical specifications of the Master Offtake Agreement, of Fadhili Combined Heat and Power Plant (FCHPP). This Exhibit B, Part 2 must also be read in conjunction with the terms and conditions of the Agreement.

The intent of this specification is to define the scope of Works and activities to be undertaken under an Engineering Procurement and Construction (EPC) term sheet to procure for the Owner a modern, functional, well-designed power station meeting the performance requirements and being capable of continuous, efficient and reliable operation of the Plant with minimum maintenance.

The overall plant and individual systems and equipment supplied by the Contractor shall be of proven, robust and reliable design with appropriate levels of redundancy and factors of safety built-in and incorporate protective systems and devices.

It is not the intent of this specification to define all details of design and construction of the complete Works. However, it shall be recognised that the whole of the Works shall conform in all respects to high standards of engineering, design and workmanship, be in accordance with Good Industry Practices and shall be capable of performing in continuous commercial operation with very high availability whilst meeting the Contractor's guarantees and complying with all environmental and statutory regulations. No defect, error or omission in this document in relation to any design, specification or method set out herein shall relieve Contractor from his responsibility to provide a fully operational power generation Plant capable of delivering the defined capacities over the specified range of operating conditions.

In the event of any conflict with the requirements of Exhibit B, Part 2 SFR (Sponsor's Functional Requirements) and Exhibit B, Part 1 (Technical Specifications), Exhibit B, Part 1 will take precedence.

2. BASIC DESIGN / FUNCTIONAL REQUIREMENTS

2.1. Plant configuration

Refer to Exhibit B, Part 1.

2.2. General

The Contractor shall provide a Plant designed to meet all design requirements and conditions given in this Exhibit B and elsewhere in the Agreement.

The entire design and manufacture shall conform to the best current engineering practice to provide high availability and reliability of power, steam and water at minimum cost. Each of the several parts of the Plant shall be of the manufacturer's standard design, provided that this design is in general accordance with this Exhibit B and the Agreement and shall use components proven to be satisfactory by previous experience.

The design dimensions and materials of all parts shall be such that they will not suffer damage as a result of stresses under the most severe service conditions. The materials used in the construction of the Plants shall be of the highest quality and selected particularly to meet the duties required of them. The plants shall be designed and constructed to minimise corrosion. Workmanship and general finish shall be of the highest class throughout.

All equipment shall be designed to minimize the risk of fire and damage that may be caused in the event of fire.

The equipment shall also be designed to prevent ingress of all vermin, accidental contact with electrical live parts and minimize the ingress of dust and dirt.

2.3. Plant operating conditions

Generally the Plant design shall follow and meet the operational and functional requirements specified in Part B1 of Exhibit B and also the following requirements shall be incorporated in Plant design.

2.3.1. Mode of operation

Refer to Exhibit B, Part 1

2.3.2. Availability and reliability

The Contractor shall demonstrate during the design with necessary details and references of the recorded operational reliability of the key equipment and systems to be provided.

Throughout the design phase the Contractor shall take into account in his design the independency of each Group (if applicable) from the other groups and avoid any design leading to “single point” failure causing the shutdown of the whole Plant or loss of total power, steam and water generation.

The Plant shall be designed to achieve a high degree of reliability and availability and such that the impact of a failure of any single piece of auxiliary equipment on the output of the Plant will be minimal.

To that effect:

- all equipment shall be based on proven technology and shall not be a prototype;
- all fleet wide modifications in design and construction recommended by the Gas Turbine (GT) and Steam Turbine (ST) manufacturers after award of the Contract up to Plant final takeover shall be immediately informed to Owner and Contractor shall implement the same free of cost and without any time impact to the Owner. The details of modifications, schedule and mode of implementation shall be mutually discussed and agreed upon with the Owner
- all the latest upgrades / improvements / safety modifications announced by the manufacturer (TIL / ECN etc.,) before and during the contract period up to Plant final take over shall also be applicable for this contract and implemented by the Contractor free of cost and without any time impact to the Owner at shop or at site with the consent of The equipment installed at the Plant shall be designed with sufficient margins to cope with the most stringent load configurations within the range of ambient conditions
- considering any unfavourable subsoil conditions and any seismic requirements, special attention shall be paid to foundations design and settlements evaluation;
- the critical components (those components that are necessary to support the power and steam train of the Plant in order to reliably maintain its rated power production and steam production to comply with the required annual availability) shall be designed with adequate redundancy, in the design of systems, and shall be supported through the on-site spare parts program and
- two-of-three logic shall be used for critical trip signals

2.4. Plant Design Lifetime

The Plant shall endure throughout its service life the load changes defined in such a manner that no stresses exceeding the permissible limits will be caused in the components. The Plant shall be matched to the application so that with operation and maintenance in accordance with Good Industry Practice, an operational life of 25 years can be achieved from COD. A minimum Design life criteria of at least 25 years (from COD) shall be applied to materials operating in the creep region (pressure parts, steam turbine, gas turbine, HRSG, Reserve boiler, main steam piping etc.,...)

The Contractor shall state and specify the calculated number of Warm/Hot/Cold starts throughout the plant life incorporated in his design reflecting the assumed operation regime of the Plant.

Number of Plant starts in combined cycle mode per year for the Plant design shall be, as a minimum, as follows:

- maximum 60 starts per year for the water-steam cycle equipment (incl. HRSGs, steam turbine, and condenser) out of which, on average per Gas Turbine:
 - 10 cold starts per year
 - 18 warm starts per year
 - 28 hot starts per year
- number of starts per year for the GTs: according to the Long Term Services Agreement (LTSA)

The Contractor shall use such design criteria as to enable a Plant operation of 25 years (from COD), assuming an operation and maintenance plan in accordance with the operating regime as proposed by the Contractor in this Agreement, the O&M Manuals and/or Good Industry Practice where this Agreement and/or the relevant O&M Manuals are silent. Without prejudice to main body of this Agreement, it is understood that certain components or parts thereof (including but not limited to parts of the turbines) are subject to wear and tear and/or usually have a shorter life span, thereby requiring servicing and/or replacement on a more frequent basis.

3. SCOPE OF WORKS

Contractor shall be fully responsible for engineering, design, detailed engineering, expediting, inspection, permitting, insurance, procurement, manufacturing, fabrication, factory testing, painting, packing, shipping, transport to site, site storage, erection, construction and erection, protection and preservation, testing and pre-commissioning, commissioning, and performance testing, required documentation, site investigations, site development and all related civil works of the complete Plant and all related support facilities and other services as specified or necessary for the generation and supply of electricity, steam and water and safe and reliable operation.

The Contractor shall include in his scope of work described in his proposal all the facilities, equipment, works and services necessary for complete, safe and prudent operation and maintenance of the Plant, even if certain essential works are not expressly stated in this Exhibit B, Part 2 and Part 1. The Contractor shall be responsible for supplying all equipment/systems, as necessary to make the Plant completely functional. Additionally, the Contractor shall provide unpriced purchase order copies including scope of services (both for the service and supply part) and technical specifications for all major equipment suppliers such as GT, HRSG, ST, Reserve Boiler(s), Transformers, Fuel Gas System, DCMS etc., to the Owner during detailed engineering stage.

In addition to the Scope of Work specified in this Exhibit B, Part 2 and Part B1, the minimum scope for work shall also include, but not limited to the following sections:

- delivery of technical files and engineering documentation as required by the Environmental Consultant who will be in charge of the EIA (Environmental Impact Assessment)
- site investigations and surveys
- coordination through the Owner of plans and activities with Saudi Aramco and SEC during design, construction and commissioning of the Project
- importation and export of goods required to complete the Works, inclusive of customs clearance in any jurisdiction
- management and control of the works, including control and supervision of the sub-contractors
- obtaining all Consents and Permits for activities at Site, related to labor, construction, handling, storage, commissioning, testing, as per Exhibit L

The Contractor shall be deemed to have included in the Scope of Services any additional services necessary to meet the Plant design, construction, performance, operation, maintenance, and environmental criteria, but which are not specifically identified in the present document, or in the EPC Term sheet.

3.1. Battery limits / Terminal points

The terminal points and delivery limits are as specified in Exhibit B, Part 1.

4. CODES, STANDARDS AND REGULATIONS

4.1. General

In addition to the Codes and Standards specified in Exhibit B, Part 1, the following National Standards must be applied:

- Saudi Arabian Grid Code

The Contractor is responsible for the coherence of the codes and standards system chosen as reference, especially when codes and standards are from various origins.

The Contractor's activities in relation to the Plant shall be carried out in accordance with Good Industry Practice. Such practices include, but are not limited to, ensuring that:

- adequate materials, resources construction utilities and supplies are available to meet the needs at the relevant time
- sufficient engineering, design, construction and safety personnel are available and are adequately experienced and trained to design, construct, start-up and test the Plant, as well as each of their component parts, properly, efficiently and within the applicable laws, manufacturer's guidelines and specifications
- appropriate monitoring and testing is done during the design, manufacturing, erection and commissioning to ensure equipment is functioning as designed and to provide assurance that it shall function properly
- appropriate protective devices and design features are provided to ensure that safe, reliable, long-term operation of the Plant can be achieved
- any equipment is operated in a manner consistent with the applicable laws, manufacturer's guidelines and specifications and is safe for workers, the general public and the environment

4.2. System of Units

SI units shall be used for this project and all materials, fittings, components, items of Plant and equipment used in the Plant.

SI units shall be used in all correspondence, documentation, calculations, drawings, measurements etc. If reference has to be made to non-standard units, the SI units shall be quoted followed by the non-standard units in brackets.

However, British units (with International System of Units (SI units) shown parenthetically) shall be used for all documentation sent to Saudi Aramco

4.2.1. Equipment identification system

Equipment numbering shall be in accordance with a clear, concise and systematic numbering system that is applied consistently across all the plant areas and one that can be easily incorporated into the Owner's supplied maintenance management system (MMS). e.g. KKS equipment numbering system.. Contractor shall use KKS in order to make the identification compatible with the MMS. No alternate identification system is permitted.

All major equipment shall also be identified by name plates indicating manufacturer's name, type of equipment, serial number, construction year and main design data.

A consistent system of labelling shall be proposed by the Contractor for the approval of the Owner's Representative. The proposed material, size and exact inscription as well as proposals for the arrangement shall be submitted to the Owner's Representative for approval.

The labels shall be written in English. Warning and safety notices shall be in Arabic and English.

Safe Working Load indications on lifting equipment shall be in Arabic and English and in S.I. units.

A piping identification system subject to Owner's Representative's approval shall be applied for the Plant piping using colour strips around the pipes, as well as arrows showing flow direction.

A labelling system for the cables, control panels and instrumentation, shall be provided and shall be submitted to the Owner's Representative's for its approval .

The labelling system for cubicles and equipment inside cubicles shall be in accordance with manufacturer's standard. The interconnection drawings are related to this standard

As a minimum requirement, labels shall be provided for:

- colour identification code for conductors of all electrical cables shall comply with the required IEC standards however; for GTG package only, the cables may be provided pre-wired with color codes per NEMA / ICEA.
- all instruments, control switches, push buttons, indication lights, breakers, etc
- all construction units and all separate construction units (cubicle, panel desk, box, etc)
- all equipment inside cubicles, panels, boxes, etc(including terminal blocks)

- all cables installed (power, instrument, control, telephone- and special cables)
- all wiring (cables and inside panel wiring) shall be labelled

Contractor shall ensure that all numbering and nomenclature of high voltage apparatus will be in accordance with the Plant identification system of SEC / Saudi Aramco.

Contractor shall properly coordinate between the GT Original Equipment Manufacturer (OEM) codes and general equipment identification system.

5. PROJECT AND SITE DATA

5.1. Site location and area / layout

The location in Saudi Arabia of the Fadhili Gas Complex of Saudi Aramco is shown in the map attached as Annex 1 to Part IV (Drawings and Diagrams).

The Complex is located in the vicinity of Saudi Aramco's gas fields, which are shown in Annex 2 to Part IV (Drawings and Diagrams) of the RFP.

The tentative Layout of the Cogeneration Plant can be seen in Annex 5a to Part IV (Drawings and Diagrams) of the RFP.

5.2. Access

Refer to Exhibit B, Part 1.

5.3. Site Conditions and Technical Limits

Refer to Exhibit B, Part 1 and Part IV Formsheets H.

5.4. Topography and soil conditions

Refer to Exhibit B, Part 1.

5.5. Seismic data

Refer to Exhibit B, Part 1.

5.6. Utilities for construction

During the construction phase, all utilities for construction shall be managed by the Contractor themselves.

5.7. Initial stock of Chemicals and lube oil

Contractor shall provide initial stock of all Process Chemicals including Lube oil (full tanks) up to COD.

5.8. Environmental protection regulations

In addition to the requirements of Exhibit B, Part 1, the Plant shall also comply with environmental standards and safeguards required by local laws as well as those embodied in the “Equator Principles”, and if applicable, those required by the countries of any Export Credit Agencies (“ECAs”) involved in the project,.

Shall also be applicable:

- The World Bank guidelines for air emissions
- The Laws of the Kingdom of Saudi Arabia

5.8.1. Environmental and Social Impact Assessment

Owner will prepare an Environmental and Social Impact Assessment (EIA) as per the requirements of Exhibit B, Part 1. The Contractor shall provide all data as necessary for the EIA study. In case the EIA report suggests any material changes to the plant; Contractor shall discuss the same with the Owner and do the needful.

6. MECHANICAL TECHNICAL REQUIREMENTS

6.1. Gas Turbine and Generator (GT) Units

6.1.1. General requirements

The GT shall be capable of being automatically started, and the generator synchronised to the electrical system and loaded to full load by an automatic start-up sequence from the Plant main central control room (CCR). The gas turbine shall be capable of being automatically unloaded from the full load, disconnected from the grid and shut down safely with an automatic sequence.

The GT shall be newly manufactured and shall be complete with ancillary plant, and shall comply with the following requirements:

The GT shall be located outdoors in weatherproof acoustic enclosures and shall be of the Gas fired, heavy duty, industrial, single shaft type directly coupled to a 60Hz generator.

The GT shall be capable of start-up, shutdown and continuous operation on both Sales Gas and Low BTU Gas throughout their complete range. Refer to Exhibit B, Part 1 for the fuel specifications.

The GT shall be provided with ultra-dry low NOx combustion systems for burning Sales Gas and Low BTU Gas. The GT shall comply with the emissions requirements specified elsewhere in Exhibit B, Part 1.

The GT control system shall incorporate an interface with the DCMS to facilitate integrated control of HRSGs throughout all operational manoeuvres.

GT shall also be capable of coming to rest without bearing or other damage on loss of cooling water or main AC power to its oil systems.

Permanent equipment for both on-load and off-load compressor cleaning shall be provided for gas turbine, allowing for the connection of common mobile washing equipment. One set of mobile washing equipment shall be supplied at each site. The Contractor shall detail on-load and off-load procedures for wet cleaning of compressor.

The GT shall be equipped with either Start up motor or Static Start system. The gas turbine shall be provided with a full set of automatic control and protection equipment incorporating governor, temperature control and supervisory functions. Vibration levels shall conform with the ISO standard 10816, Zone B. ISO 7919 zone A/B can be used as a guideline. GT OEM package will meet A/B boundary +25% as stated in DIN ISO 7919-4 Acceptance criteria (Section 4).

The GT shall be equipped with turbine inlet air cooling by means of an evaporative cooler

Lifting gear, such as lifting beams, slings, jacks etc. required for removing complete compressor-turbine rotor, generator rotor, combustion and not gas parts, the upper portion of casings and other heavy equipment shall be provided. Turning gear system shall also be capable of manual turning.

Means shall be provided for powered turning of the rotors for a sufficient period following normal or emergency shutdown to ensure satisfactory cooling of components. The turning gear shall be interlocked such that loss of lubricating oil shall immediately stop the gear. It shall not be possible to start the turning gear without adequate lubrication oil supplies.

Jacking facilities shall be provided for all turbine and generator bearings if necessary.

The GT generator end windings shall use wet ties free of any vibration issues. The OEM shall perform bump test and phase loop at the factory. The generator shall be fitted with new design SRS, wet tie (full wrap/frap) support. OEM must ensure that the generator has undergone a bump test or full end winding frequency analysis report at the factory after assembly.

As a backup to the powered system, means shall be provided for turning of rotors by hand during maintenance and for emergency use.

The GT shall be suitable for island mode operation and black start.

Necessary OEM service support during erection, commissioning, and operational test period shall be provided.

Contractor should consider adequate time buffer and resources (including Gas Turbine OEM personnel as required) for timely commissioning and testing (including fine tuning of the ULN combustors for the Sales gas and Low – BTU gas) of the Gas turbines.

6.1.2. Noise limits

The gas turbine generator will be provided with individual suitable weather proof acoustic hoods/enclosures to meet noise limits specified in Exhibit B, Part 1.

6.1.3. Frequency variations

Attention is drawn to the requirement in Section 7 (Frequency variations in transmission system performance characteristics). The Contractor should clarify the GT capability under these conditions and in accordance to grid code.

6.1.4. Air intake systems and filtration

The gas turbine shall be equipped with a high efficiency of F9 class as per EN779-2012, high-level air inlet filter unit and intake silencing.

The gas turbine intake air filtration system shall consist of minimum two stage filters and of the pulse-clean type. The type of filter element selected shall be capable of operating without maintenance for a minimum two-year period before a media change is required as per manufacturer's standards under normal ambient and operating conditions and under all conditions as specified in Exhibit B, Part 1. It shall be possible to operate the pulse-clean system without shutting down or reducing power output on the associated unit. In case of GT shutdown, the pulse cleaning shall be from the Plant compressed air system. Necessary design and interconnection shall be made by the Contractor.

The selection and design of the air filtration system shall be such as to ensure that sand, water droplets, dust and other harmful contaminants in the ambient air be filtered to a level which will not be detrimental to the life of the gas turbine unit under the most adverse ambient conditions at the plant site when the plant is operating continuously at maximum output with the necessity for blade cleaning reduced to a minimum.

The number of access points and penetrations into the air inlet system for maintenance and inspection shall be minimised. Any door or hatch shall be capable of being securely air-tight locked, and interlocks shall be provided to prevent any attempted start with any door or hatch not properly closed.

Particular attention shall be paid to the structural materials and paint used for the construction of the air filtration plant in view of the extensive temperature deviations and climatic conditions. Rain hoods shall be provided on any horizontal air filter inlets.

Silencing shall be provided as required in the inlet system in order to meet the specified noise limits.

6.1.5. Fire/Gas detection and protection systems

The entire gas turbine generator installation, including all auxiliary enclosures, shall be protected from fire with a complete automatic fire protection system. The system shall be complete in all respects and shall include cylinders, cylinder racks, piping, valves, nozzles, detectors, fire alarm actuating system, and all other equipment necessary for the safe and efficient operation of the units.

The fire detectors shall be wired as class A loops.

The extinguishing agent shall be carbon dioxide (CO₂) or foam as per requirements of NFPA 750.

The fire protection and detection system provided with the gas turbine generator shall interface, and be connected to, the Plant's main fire protection and detection system.

6.1.6. Exhaust ducting

The gas turbines shall be provided with the exhaust gas system direct to the waste heat recovery boiler. The whole of the exhaust system shall be designed to withstand a back pressure of not less than 50 mbar.

A series of test points shall be arranged in the ducts at positions such that the distribution and temperature of the exhaust gas can be determined and the exhaust gas sampled in accordance with the agreed test code.

Gas tight access doors shall be provided to give access to ducts for inspection and maintenance. They shall be not less than 600mm x 600mm size. Interlocks shall be provided to prevent any attempted start with the doors open.

Silencing shall be provided as required in the ducting or stack in order to meet the specified noise limits.

6.1.7. Maintainability

The maintainability features incorporated in the design and construction of the gas turbine generator units shall include the following items:

The formulae and a chart showing the variation in recommended running hours between mandatory inspection/overhauls in relation to the number of normal and emergency trips and starts, fired hours on part load, base load and peak load continuous operation of fuel gas.

Equivalent Hours formulae (if applicable) are to be supplied to relate the emergency trips, at various loads, and starts to the total hours between maintenance inspections. For this purpose a start shall mean a run-up to at least the self-sustaining condition.

Identification of maintenance procedures that can be performed within the enclosure, if provided, and which procedures are required to be performed with the enclosure removed.

Details of the features of the gas turbine generator unit, such as modular construction, that can reduce down time and allow ease of maintenance, thus improving plant availability.

The weight of individual major components shall be stated. If overhead lifting equipment is required, the capacity and type of equipment required and clearance around the enclosure for unimpeded operation and movement shall be stated. The necessary special lifting equipment shall be provided within the maintenance tools and equipment supplied.

Details of the extent of dismantling of the equipment for mandatory inspection/overhaul, and lay-down area required.

The Contractor shall fulfil all the requirements of the OEM as specified in Marginal Conditions and Storage requirements of Attachment 5 and 6, Enclosure - 3 to this Exhibit B, Part 2

6.2. Steam Turbine (ST) Units

6.2.1. General requirements

The ST and its ancillary plant and equipment shall provide continuous, safe, reliable and economical operation without undue heating, vibration, or noise. It shall be located indoors and designed for ease of operation and maintenance and shall have rapid start-up and shutdown capability. Appropriate concepts shall be included in the design so as to minimise blade erosion due to exhaust steam wetness.

An exhaust hood spray cooling shall be supplied, if necessary, to limit the exhaust temperature rise under conditions of low exhaust flow operation.

The ST shall incorporate an interface with the DCMS to facilitate integrated control of the Plant throughout all operational manoeuvres.

The ST shall also be capable of coming to rest without any damage on loss of cooling and/or main AC power to its oil systems.

The control system shall include features to provide automatic intervention to limit maximum thermal stress levels to established threshold values.

The steam turbine unit and all auxiliary plant shall be provided with all equipment necessary to monitor all significant operational data.

The HP steam header shall feed two identical condensing steam turbine-generator units. Each steam turbine shall be fitted with a steam extraction to feed the FGP.

Overloading requirements specified in IEC60045 shall be taken into account. The turbine main valves shall be of the fail-safe type.

On-load testing of emergency stop valves and of the simulated over-speed trip gear operation shall be possible.

All fleet wide modifications in design and construction recommended by the ST manufacturer after award of the Contract up to Plant final takeover shall be immediately informed to the Owner and the Contractor shall implement the same free of cost and without any time impact to the Owner. The details of modifications, schedule and mode of implementation shall be mutually discussed and agreed upon with the Owner.

All the latest upgrades/improvements/safety modifications announced by the manufacturer (TIL/ECN etc.) before and during the contract period up to Plant final take over shall also be applicable for this contract and implemented by the Contractor free of cost and without any time impact to the Owner at shop or at site with the consent of owner.

6.2.2. Performance Requirements

The Steam Turbine is expected to operate at base load and at part load and other operational loads as specified in Exhibit B, Part 1.

Contractor shall supply the following performance information during the design review phase if not already submitted in its proposal:

- performance degradation on output and heat rate
- cold, warm, and hot start up times
- start-up / shutdown curves;
- minimum stable generation
- performance correction curves of heat rate/efficiency vs load/exhaust steam pressure
- supplier recommended maintenance schedule (frequency, type and duration of outages)

6.2.3. Non-Return Valves

Proven fast-acting forced-closing non-return valves shall be provided at steam extractions in close proximity to the Steam Turbine.

6.2.4. Protective Devices

The following turbine protection devices shall be included in the supply as a minimum:

- local trip push button
- remote emergency push button
- low bearing oil pressure trip
- bearing drain oil temperature or high bearing metal temperature trip
- over-speed protection system (duplex)
- high back pressure
- high exhaust temperature (alarm only)
- fire detection
- axial shift protection
- turbine shaft vibration
- bearing metal temperatures (alarm only)

Arrangements shall be provided for on-load testing of these devices.

6.2.5. Bearings & Lubricating Oil System

The bearings of the turbine and generator shall be readily accessible, adjustable and replaceable without risk of damaging adjacent glands, and that the lower halves can be removed and replaced without lifting the shaft more than a small amount or disturbing major components.

Arrangements shall be made for visible indications of the oil flow from each bearing pedestal and mounting of dial-type thermometers.

A pressure lubrication oil system shall be furnished for the Steam Turbine Generator to supply oil at a suitable pressure as applicable to the Steam Turbine, shaft driven auxiliary equipment, gearing, generator, continuously lubricated couplings, governing and hydraulic systems. The system shall include all vents, drains, valves, pipes, instrumentation and controls as required.

The lubricating system shall cater for loss of the main lubricating oil pump whilst on load, momentary loss of AC power whilst on load (such as occurs during a change of supply from unit board to station board) and safe shutdown without any AC power at all. The provision of a DC type pump as emergency shutdown in the event of an AC failure is required.

Following erection, the lubricating oil system shall be flushed until the level of cleanliness demonstrated to Owner's Engineer is acceptable. Acceptance will not be unreasonably withheld.

Oil leakages shall be contained and prevented from entering the surface water drainage system

The arrangement of the lubricating oil system shall comply with IEC45, and in particular multiple oil coolers and filters shall be provided of sufficient capacity so that one (1) oil cooler or filter can be taken out of service while the turbine remains in operation. Emergency DC lubricating oil pump and AC jacking oil pumps shall be provided for both the Steam Turbine and its Generator.

6.2.6. Barring Gear

Barring gear shall be provided to rotate the turbine generator rotors continuously during periods of shutdown to ensure even cooling. The barring gear shall be arranged for both automatic and manual engagement, which shall be selectable. The barring gear shall automatically dis-engage on turbine starting under steam.

Manual barring by means of a lever fitted to the turning gear motor shall be provided. A limit switch should lock out the electric turning operation if the lever is attached.

6.2.7. Steam Turbine Control & Monitoring Systems

The control system shall include features to provide automatic intervention in order to limit maximum thermal stress levels.

The Steam Turbine unit and all auxiliary plant shall be provided with all equipment necessary to monitor and record performance and all significant operational data.

For the turbine drains faulty drain operation shall be indicated on the DCMS.

6.2.8. Steam Turbine Start up System

The steam turbine steam start up systems shall include steam bypass stations to facilitate steam and metal temperature matching during turbine start-up. The steam turbine bypass system of each unit shall be designed to accommodate 100% of the maximum continuous rated flow of its respective HRSG or as specified in Exhibit B. Part 1 whichever is more stringent at valves wide opened, thus keeping the steam generator running at full load without blowing the safety valves in case of a turbine or generator trip at full load

6.2.9. Condensate extraction pumps

In addition to the requirements of Exhibit B, Part 1, the pumps shall be provided with a minimum flow protection system. The pumps shall operate under a controlled condensate level which shall ensure cavitation free operation. Care shall be taken in the selection of materials for the pumps to obviate erosion and corrosion of any part.

All parts of the pump in contact with feed water shall be of stainless steel construction. Parts subject to wear shall be fitted with renewable liners and sleeves.

The pumps shall be provided with mechanical shaft seals and seal water coolers, if required by the equipment manufacturer.

The electric drive motor shall be suitable for at least 3 starts per hour without degradation.

The condensate extraction pumps shall be accessible via overhead crane or suitable lifting beams, without the requirement to move major pipework.

6.2.10. Condensate tank

The condensate shall be drained by gravity from the heat exchangers into a condensate collection tank (hot well).

The tank shall be provided with all necessary instruments, fittings and connections including a drain to waste connection. All instruments and valves shall be arranged to be easily accessible for operation, adjustment and maintenance.

6.2.11. Drain pot and drain pumps

The exhaust steam duct is drained through a drain pot arranged in the horizontal section of the exhaust steam duct.

Piping and accessories, valves, filters, instrument, joints, etc., shall be included in the ACC Supplier scope of supply.

2 x 100% drain pumps, complete with motors shall be provided.

6.2.12. Vacuum Raising and Maintaining Plant

The air extraction plant shall be designed to maintain vacuum in the steam turbine condenser. The start-up or hogging evacuation raising equipment shall be designed in accordance with HEI standards and shall be capable of quickly extracting air and incondensable gases from the turbines, condensers to raise vacuum from atmospheric pressure to a vacuum at the minimum delay that is possible and at a rate to commensurate with unit synchronization. Means shall be provided for discharging the air and gases to the atmosphere and retaining water condensed from the air stream.

Standby capacity shall be proposed to ensure no loss of vacuum as a result of plant failure.

The design duty air removal capability (with dry air) shall be consistent with or better than that theoretically determined using an internationally recognised standard (HEI or equivalent).

The glands and seats of all valves and any other areas subject to sub atmospheric pressure shall be suitably sealed to avoid the ingress of air. The connection at each valve shall incorporate an isolating valve and union coupling to facilitate removal for maintenance. Water used for this duty shall be demineralised water.

6.2.13. Steam Connections and Drains

The steam and feed water piping systems shall incorporate recognised features for rapid detection and prevention of water induction into the turbine from main, extraction and exhaust steam lines, spray attemperators, gland sealing systems and turbine drain lines (to ASME Standard TDP-1-1998 requirements, or equivalent alternative code approved by Owner).

6.2.14. Noise Limits

If the noise level requirements applicable to each Steam Turbine Generator cannot be achieved with the standard Steam Turbine alone, then suitable acoustic hoods/enclosures or treatment shall be provided and installed with the Steam Turbine Generator unit if required to meet noise limits.

6.2.15. Turbine Pipework and Drains

The pipework and drainage system shall allow for safe and efficient start-up and operation of the Steam Turbine Generator, controlled remotely and locally. For the turbine drains, faulty drain operation shall be indicated on the DCMS.

A clean drains collecting vessel complete with transfer pump shall be provided to handle drains from the Steam Turbines.

6.2.16. Maintainability

Information regarding the maintainability features incorporated in the design and construction of the Steam Turbine Generator unit shall be provided during the design review phase. The following items shall be included:

- the weight of individual major components, and the lifting facilities required for removal. If overhead lifting equipment is required, the capacity and type of equipment required and clearance around the enclosure for unimpeded operation and movement shall be stated. Any necessary special lifting equipment shall be provided within the maintenance tools and equipment supplied
- details of the extent of dismantling of the equipment for mandatory inspection/overhaul, and lay-down area required
- the lay-down area in the turbine hall showing the location of the turbine components (e.g. rotor, casing, etc.) which are to be placed during maintenance
- maintenance Materials, manpower and skill set requirements to be stated

6.3. Heat Recovery Steam Generators (HRSG)

6.3.1. General requirements

Heat recovery steam generator (HRSG) shall be complete with ducting, chimneys, mountings, integral valves and pipes and all other items associated with the unit to allow reliable operation.

The HRSGs shall be of proven design and in accordance with the requirements of relevant ASME standards where applicable or equivalent international standard. The HRSG shall be ASME stamped. The HRSGs shall incorporate supplementary firing for meeting the steam requirements as set forth in Exhibit B, Part 1. HRSG shall be entirely suitable for the exhaust gas temperatures and mass flow rates of the gas turbine resulting after supplementary firing (at full load and all required loads and ambient conditions).

The HRSGs and all associated facilities (e.g. Gas supply system, steam supply system etc.) shall therefore be capable of accommodating the following operating cases:

- continuous operation of HRSG without supplementary firing
- continuous operation of HRSG with minimum supplementary firing enabling fast increase of steam generation

- within design limits operation of supplementary firing according to steam demand by steam generation control or by steam pressure control; continuous operation of HRSG with maximum supplementary firing

The HRSG shall be designed for cold end corrosion issues and so that exhaust gas temperature is above the acid dew point when the gas turbine is fired on Gas.

Inspection doors and platforms shall be provided to allow ready access to the HRSG gas path, tubing and other pressure parts.

Gas duct linings and insulation shall use materials with proven capability at the maximum gas temperatures anticipated, together with design margin to allow for uneven temperature distribution downstream of supplementary firing burners. The duct geometry downstream of the supplementary firing burners shall be designed so as to ensure that no direct flame impingement on heat exchange surfaces or other metallic surfaces can occur.

The pinch point for the HRSGs shall be optimised at the design point and shall be less than 36°F (20°C).

The HRSGs shall be designed for outdoor installation and all equipment and mountings shall be entirely weather-proof and suitable for the aggressive environment. The main pressure parts shall be arranged in the form of a self-contained construction and shall be of the self-draining, horizontal gas pass type.

The water side of the HRSG shall be chemical/acid cleaned after completion of erection in accordance with accepted chemical cleaning procedures.

Regarding HRSG pressure retaining parts material selection shall be in line with requirements and stipulations of ASME I (pressure vessels) and ASME B31.1 (power piping).

All fleet wide modifications in design and construction recommended by the HRSG manufacturer after award of the Contract up to Plant final takeover shall be immediately informed to the Owner and the Contractor shall implement the same free of cost and without any time impact to the Owner. The details of modifications, schedule and mode of implementation shall be mutually discussed and agreed upon with the Owner.

All the latest upgrades/improvements/safety modifications announced by the manufacturer (TIL/ECN etc.) before and during the contract period up to Plant final take over shall also be applicable for this contract and implemented by the Contractor free of cost and without any time impact to the Owner at shop or at site with the consent of owner.

Necessary OEM service support during erection, commissioning, and operational test period shall be provided

6.3.2. HRSG drums and headers

HRSG evaporative section shall be provided with steam drum(s), each of adequate size to ensure stable operation under all normal load conditions and other single item trip conditions. Drum internals shall include primary and secondary steam and water separators to limit the mechanical carry over. The moisture carryover from the drum to the first stage of the superheater will be less than 0.1%. The moisture carryover after the first stage of the superheater should not be detectable.

Steam drum shall be tested hydrostatically at shop and at site. Butt welds for alloy steel and carbon steel will be 100% Radiographically tested.

Contractor shall ensure that the drum has as a minimum two (2) minute hold up time between normal working level (NWL) and low-low level condition (LLC = low-low cut off).

6.3.3. Superheater, evaporator and economiser banks

Each element and tube bank shall contain tubes on a horizontal and vertical pitch, adequately supported and of shallow depth to enable easy cleaning. Any fins used shall be solidly attached to the tube throughout the fin length.

All superheater tubes shall be solid drawn seamless and not electric resistance welded. The tubes shall be of minimum 3.2mm thickness. However a minimum of 2.7mm seamless tubes are acceptable, subject to the:

- contractor providing the necessary calculations and documentary evidence for Owner's approval during detailed engineering
- contractor confirming and providing evidence of commercial availability of these tubes in the market; and
- confirmation by the Contractor to HRSG standards and codes as asked for in this document

As an exception to the above paragraph, it is acceptable only for the LP super heater tubes to be ERW (electric resistance welded) with a minimum tube thickness of 2.4mm subject to:

- contractor providing the necessary calculations and documentary evidence for Owner's approval during detailed engineering
- contractor confirming and providing evidence of commercial availability of these tubes in the market; and

- confirmation by the Contractor to HRSG standards and codes as asked for in this document

All the HRSG heating surfaces shall be designed taking into account the requirement for off-load water washing and a number of local drain points and external hose connections shall be provided.

The design shall incorporate measures to prevent sagging and vibrations occurring in the tube banks.

6.3.4. Steam temperature and output control

Steam temperature control equipment shall be provided to ensure that the HRSG outlet steam temperature is the required steam temperature at the Interface to the existing systems of the FGP processing Plant within the acceptable temperature range at all generation and supply possibilities/conditions.

6.3.5. Valves and accessories

Boiler and superheater safety valves for all steam circuits shall be provided and shall be specified in accordance with design code recommendations. Each safety valve shall be fitted with its own vent-pipe, silencer and suitable drain and shall be designed and constructed to prevent back-pressure from interfering with the proper operation of the valve. The safety valves shall be designed and installed according to ASME Section I

An electrically operated relief valve shall be provided on the HP steam main before boiler stop valve for the purpose of venting steam during start-up and pressure excursions. This valve may discharge into the silencer of one of the safety relief valves as long as this does not impair the operation of the safety relief valve. The electrically operated safety relief valve is in addition to spring operated safety relief valve before boiler stop valve.

One electric motor or hydraulically operated stop valve and one non-return valve, together with one motor operated drain valve shall be installed in all main steam lines. Main steam stop valve shall be provided with a smaller electric motor operated by-pass valve or other integral means to enable downstream warming and pressure equalisation. High pressure steam from the boilers shall be routed to steam headers to the FGP interface point.

Double isolation valves operating as master/martyr shall be provided on all boiler emptying, venting and drain lines. Main vent and drain valves shall be motorised for remote operation. In the case where motor operated drain/vent valves are provided, there will be one motorised valve and one manual valve in series.

All pressurised main drains in the boiler area shall be connected to the HRSG blowdown vessel/tank.

Adequate access doors and access platforms shall be provided in the upper and lower sections of the HRSG for internal inspection.

6.3.6. Feed Water control stations

The control of feed water flow to HRSG shall be achieved by feed regulating valve(s) capable of controlling between 0% and 100% flow.

6.3.7. Supplementary firing system

The supplementary firing system shall utilise grid burners.

The supplementary firing system shall be designed for a steam generation capacity as requested in Exhibit B, Part 1 over and above the base ranges at RSC, OP A and OP B points and various scenarios as specified therein.

Operation of supplementary firing should be possible at any case of GT partial loads and base loads.

The burners shall be designed to provide maximum efficiency and uniform heat distribution to the HRSG. The burner system shall be of a proven design and include equipment necessary for its safe, efficient and reliable operation.

All igniters and flame monitors shall be capable of being maintained or replaced whilst the unit is on line.

6.3.8. Blowdown equipment

All HRSG circuits shall be fitted with blowdown facilities which shall be designed to operate on both intermittent and continuous basis. The blowdown lines shall be designed for a minimum of 3% mass flow (at MCR).

Before being discharged to the pre-treatment plant (or) wastewater piping, the HRSG blowdown water shall be cooled so that the wastewater temperature at the interface point shall not exceed 140°F (60°C), for which necessary HRSG blow-down coolers shall be provided.

6.3.9. HRSG drum local water level gauges and remote indicators

HRSG steam drum shall each be furnished with a minimum of two fully (on line) maintainable local water level gauges at each end of the drum and as per ASME Section I. These shall be of the bicolour type.

The steam drum shall also be fitted with an electronic water level gauging system of the hydrastep type, or alternative proven system including LCD type that is configurable to read in both percentage and engineering units. The system provided shall be complete with provision of hard-wired level indication in the central control room.

Dedicated drum level measurements shall be provided for drum level control which will also use steam flow and feed flow meters and temperature and pressure compensation appropriate to maintaining control of level within the defined range under all operating conditions and transients in normal operation.

6.3.10. Stacks

Each HRSG shall have a single flue stack of > 60m height, subject to confirmation by Saudi Aramco/Owner based on the air emissions assessment. The diameter shall be such that it is self-sustaining, i.e. its lift due to temperature differential shall be equal to or greater than the sum of all losses.

The stacks shall be designed and constructed in accordance with DIN, EN, BS or ASME STS-1 or an equivalent approved standard if acceptable as per Exhibit B, Part 1. The stacks shall be provided with suitable earthing bosses and earthed.

The stacks shall be clad and insulated where required for personnel protection. Optionally expanded metal instead of insulation can also be provided where required for personnel protection in the Stacks. The style and finish shall be as approved by the Owner. Alternatively; the main stack can be insulated up to the stack damper if similar references in Saudi Arabia are provided by the Contractor and is as per the relevant standards.

The efflux velocity shall not be less than 15m/sec at normal operating loads (60-100% GT load with or without supplementary firing) and shall be designed throughout the life of the Plant.

Aircraft warning lights as required shall be provided on the Stacks to meet local regulations.

Provision shall be made for the permanent monitoring of stack outlet gas emissions in accordance with environmental requirements specified in Exhibit B, Part 1. In addition, the stack shall be provided with test points at positions such that the distribution and temperature of the exhaust gas can be determined in accordance with the agreed test code.

6.3.11. Insulation

The use of refractory materials, if required by the design, shall be limited to an absolute minimum in the area of the gas turbine exhaust ducting and supplementary firing duct.

The HRSG structure shall be designed to give adequate support to the HRSG body under the most adverse operating conditions taking into consideration expansion and wind loading. The duct and casing part of the HRSG shall be insulated internally. Insulation materials shall be non-carcinogenic such as mineral wool, ceramic fibre and stainless steel liner plate and carbon steel casing. The insulation shall be adequate to minimise insulation losses and permit safe working conditions.

6.3.12. Cleaning and preservation

A complete boiler off-load cleaning, filling and preservation system shall be provided.

6.3.13. Steam connections and drains

The steam and Feedwater piping systems shall incorporate recognised features for rapid detection and prevention of water carry over.

6.4. Main steam systems

6.4.1. General

Refer to Exhibit B, Part 1.

The live steam system design shall permit the HRSG and the steam turbines to be operated without limitations at specified load. It shall also make possible the bypass operation of each steam turbine feeding directly the FGP. It shall also ensure the HRSGs stability during start and stop and when a turbine trip or the load rejection occurs or when the Plant is running on ST bypass mode.

With the common header configuration feeding both steam turbines, the proper heating measures for cold start of one ST rest of the plant being in full operation shall be provided.

Each GT shall be capable of continuous operation in combined cycle mode also in case of a part of the other GTs are not in operation. The stopping and start-up of one GT/HRSG unit while any of the other GT/HRSG unit(s) remain(s) in combined cycle mode shall be possible and is a normal operation mode. Specific steam by-pass station is to be provided to enable smooth start-up/stop of a GT/HRSG unit when the other unit(s) is (are) already in operation.

6.4.2. Steam turbine by-pass

The capacity and design of the ST bypass system shall enable continuous GT operation at base load also in the event ST has been tripped and/or when the ST is not in operation. The design of the bypass system, bypass valves and their control capability shall control a smooth LP steam pressure transient in the event of ST trips (and all scenarios 1-11 as mentioned in Exhibit B, Part 1) in order to continue to feed the FGP.

The bypass system shall include fast action isolating valves, pressure reducing valves with de-superheating sections and modulating valves with atomizers for injection of spray water.

The by-pass system shall be operated automatically during start-up/shutdown, in the event of a trip or sudden ST load reduction or in case of lower electrical demand. They shall be dedicated special bypass valves specifically designed for steam bypass applications with hydraulic or pneumatic actuation. The steam turbine bypass controls shall be integrated with the overall Plant DCMS.

The Contractor shall indicate in his proposal his successful experience in using bypass systems of similar design and shall indicate the design aspects that prevent damage from vibration, thermal shock, and pressure waves resulting from shock waves at the different pressure reduction stages and pressure reducing valve instabilities. Special attention shall be given to effective vibration damping design measures.

6.4.3. Reserve Boiler(s)

A Reserve Boiler(s) inclusive of, burner(s), fuel and combustion air delivery systems, burner management systems, electrical control and feedwater systems shall be specifically engineered as a packaged system for the facility

The Reserve Boiler shall be of proven design and in accordance with the requirements of relevant ASME standards where applicable such as ASME I (pressure vessels) and ASME B31.1 (power piping) and shall be ASME certified. The Reserve Boiler(s) shall provide reliable and consistent performance to the operating parameters as defined in Exhibit B, Part 1.

The reserve boiler shall have dual fuel capability to fire both Low BTU and Sales gas, all inherit safety devices, an individual CEMS and meet the emission and stack limits as per Exhibit B, Part 1. Forced draft fans shall provide necessary combustion air. Feedwater shall be supplied from the deaerator via a dedicated feedwater pump. The reserve boiler shall be kept in warm standby as per Exhibit B, Part 1 when in stand-by with a steam bleed from the main steam system.

Inspection doors and platforms shall be provided to allow ready access to the Reserve Boiler's gas path, tubing and other pressure parts.

Gas duct linings and insulation shall use materials with proven capability at the maximum gas temperatures anticipated.

The Reserve Boiler(s) shall be designed for outdoor installation and all equipment and mountings shall be entirely weather-proof and suitable for the aggressive environment.

The water side of the Reserve Boiler(s) shall be chemical/acid cleaned after completion of erection in accordance with accepted chemical cleaning procedures.

Materials operating in the creep region shall have a minimum design life criteria of at least 25 years (from COD).

All modifications in design and construction recommended by the Reserve Boiler(s)' manufacturer after award of the Contract up to Plant final takeover shall be immediately informed to Saudi Aramco and the Contractor shall implement the same free of cost and without any time impact to the Owner. The details of modifications, schedule and mode of implementation shall be mutually discussed and agreed upon with the Owner.

All the latest upgrades/improvements/safety modifications announced by the manufacturer (TIL/ECN etc.,) before and during the contract period up to Plant final take over shall also be applicable for this contract and implemented by the Contractor free of cost and without any time impact to the Owner at shop or at site with the consent of owner.

Necessary OEM service support during erection, commissioning, and operational test period shall be provided.

6.5. Air Cooled Condenser

Contractor shall provide correction curves for gross power output of the Steam turbine versus ambient conditions of dry bulb temperature and relative humidity. In addition, the ACC shall be capable of handling steam turbine bypass operation during start-up, shut-down and/or emergencies (shall include a reserve of at least 10% to cover possible steam turbine overloads). The orientation and layout of the ACC shall be such that the adequate air circulation is possible around the ACC.

Each ACC fan shall be individually isolatable. Access ways, galleries, platforms and ladders shall be provided for safe and efficient operation and maintenance including maintenance of fans, gearboxes, bearings etc. The ACC shall comprise a number of cells mounted on a hot dip galvanised steel support structure. Internal partition walls shall be located between each fan section to reduce mal-distribution of air flow. Lockable access doors shall be provided in the partition walls (normally kept closed and locked).

The ACC shall be of proven design with supplier references demonstrating the same. The ACC shall be designed for full vacuum to an approved heat exchanger design code as applicable such as Heat Exchanger Institute (HEI) Standards for Steam Surface Condensers, ASME Boiler and Pressure Vessel Code etc.

The ACC shall be of the forced draught type, located on a steel structure at ground level. It shall be capable of condensing steam from the steam turbine and turbine drain flash vessel (if provided) at all ambient conditions and at all loads.

The ACC shall be arranged to minimise the recirculation of cooling air.

During steam turbine trip transient/start-up phase, the whole steam flow from the by-pass system will be discharged to the ACC.

During the by-pass conditions, the ACC shall be able to maintain the pressure such that no limitation occurs for the steam turbine capabilities.

In addition to the requirements of Exhibit B, Part 1; the ACC shall be designed to be capable of condensing the entire steam flow dumped by the by pass system following a steam turbine trip, for indefinite time.

Finned tubed bundles shall be used within the ACC. The tube – fin material combination shall be of Welded carbon steel tube + carbon steel fins (both hot dipped galvanized). In case Contractor proposes Aluminium coated (Aluminized steel) tubing with Aluminium fins then the same is acceptable only for single row type condensers.

Wind walls shall be installed around the perimeter of the ACC structure, minimally extending from fan deck level to the top of the heat exchanger bundles. Each ACC shall be equipped with systems such as high pressure spray to enable cleaning of the outer surfaces of the tubes.

Fans shall be of single-speed type. The Contractor shall develop the operating concept for ACC during the detailed design phase.

The ACC shall be arranged together with condensate extraction pumps, condensate control valve including a bypass control valve with manual valve, in addition to steam (if applicable) and water side isolating valves. The ACC shall be arranged to operate automatically.

The design and workmanship shall be such that the pumps are readily assembled or dismantled and that they shall operate satisfactorily under all operating conditions without undue wear, vibration and cavitation.

The pumps and motors shall be located at ground floor level outdoors and close to the ACC condensate tank. The configuration must be based on the optimal suction requirements.

Basket strainer with fine mesh/screen shall be supplied on the pump suction to remove suspended solids in the condensate, particularly during start-up (commissioning period). After commissioning this shall be replaced by the Contractor with coarse mesh / screen.

In case load restrictions or damages to the ACC are observed before the end of the warranty period while the plant is operated within the specified design range, the Contractor shall improve the design with adequate means (e.g. wind screens, different fan design, etc) to avoid such restrictions or damages, at no cost to the Owner. The proposed corrective actions and the timing of installation shall be subject to Owner's approval.

An ACC cleaning system should be included.

The contractor should provide correction curves of ACC performance in function of wind velocity, direction and ambient temperature.

6.5.1.1. ACC STRUCTURAL SUPPORTS

An integrated structure carrying the condenser, fans and motors, plenum chamber, access galleries, staircases, platforms and maintenance runway beams shall be supplied and secured to ground foundation concrete supports. Steelwork, platforms, ladders and stairways shall be hot dip galvanized.

The height of the structure shall take account of the air velocity when operating with all fans in service. The ACC shall be designed with a fan deck at least 16m high to minimize air recirculation.

The structural steelwork shall be capable of supporting the ACC at all operating, maintenance and testing loads and shall comply with the wind speed design loadings given for the Site. Where stress analysis calculations are performed, the location of anchors, guides or restraints shall be based on the results of the final stress analysis. Such anchors, guides and restraints shall be designed for the maximum forces and moments encountered.

The civil engineering and structural works shall be designed in accordance with accepted codes and standards of skill, care and diligence.

The seismic design, both for structural sizing and operating behavior, shall comply with the minimum seismic requirements provided in Exhibit B, Part 1.

The design, fabrication, testing, installation and fixing method of all steelwork including structural members, checker plate flooring, hand-railing, ladders, open grid flooring and stair treads shall be in accordance with recognized codes and standards for steelwork.

The necessary signalizations shall be foreseen.

Stairs shall be preferred to ladders wherever possible. Where ladders are used they shall be installed only in areas where only occasional access is required or where space limitations so dictate.

6.5.1.2. MECHANICAL AND ELECTRICAL REQUIREMENTS

Rotating parts and relevant transmission components shall be designed to limit the noise according to the environmental regulations and as specified in Exhibit B-Part 1, whichever is more stringent.

Each cell shall have a motor driven forced draught fan driven by an electric motor, connected to the reduction gear by a shaft. Fan electric motors shall have half shrouded, stay put, emergency stop buttons located within 5m of the motor location. These buttons shall require a key to reset. 3NC contacts are required (230VAC or 220VDC insulated)

The electric motors shall be totally enclosed and fan cooled, fitted with lubricated antifriction bearings and moisture resistant insulation, designed for continuous outdoor service, with mechanical protection IP55 as a minimum, insulation Class F.

The motor and the reduction gear shall be designed in order to withstand the maximum electrical power with adequate margin.

Platform and ladders must be made to allow periodical check, lubrication of motor bearings and of gearbox bearings.

The motors will be selected taking into account the driven load characteristic; the motors will have at least a 10% operating margin on power output.

6.6. Condensate and feedwater systems

6.6.1. Feedwater pumps

The feedwater pumps shall comply with the requirements of Exhibit B, Part 1.

The boiler feedwater pumps shall be capable of supplying the maximum continuous rating water flow to each drum, at a pressure 3 percent higher than the setting of HRSG drum safety valve.

Suction and discharge isolation valves and a non-return discharge valve shall be installed for each pump. Each pump shall be protected during periods of low flow with independent, minimum flow recirculation paths returning to the feedwater tank. The energy of the recirculation fluid shall be reduced through a pressure drop across the minimum flow valve and through an orifice/backpressure device located at the inlet to the feedwater tank. The minimum flow valve shall be designed for severe service, with provisions to minimize cavitation. The orifice/backpressure device shall prevent flashing in the recirculation piping downstream of the minimum flow valve.

Each pump shall be fitted with a suction strainer complete with manual blowdown. Strainers shall be equipped with commissioning and operating inserts. The strainer shall have differential pressure indication, with automatic pump changeover on high strainer differential pressure alarm.

Vibration monitoring and protection shall be provided.

A transient analysis shall be performed for the pump suction to ensure that the feedwater tank volume, elevation, suction line sizing, and layout maintain net positive suction head (NPSH) at all times and loads.

The pumps shall withstand any possible thermal shock without damage or leakage. The operating speed range of the pumps shall avoid all natural structural resonances. All parts of the pump in contact with feed water shall be corrosion resistant steel. Cooling of the feed water pumps and auxiliaries shall be provided by the closed cycle cooling water system.

6.6.2. Condensate system

Refer to Exhibit B –Part 1.

The condensate system includes return from the ACCs and also from the FGP. Condensate return from the ACCs shall mix with FGP return condensate which has been treated in the CPP. The final condensate shall be directed to the deaerator/feedwater tanks (with consideration of balancing the water in each water/steam cycle if necessary).

FGP return condensate is at high temperature ($<100^{\circ}\text{C}$) and must be cooled prior to entering the CPP. A Fin Fan Cooler shall be considered with N+2 modules where any one module can be isolated for repair without impacting the full condensate flow. The fin fan cooler shall be designed to meet CPP inlet temperature requirements of 60°C at design conditions.

Because the FGP condensate return temperature is $<100^{\circ}\text{C}$, it is not possible to consider the condensate storage tank as part of the normal operating cycle as the water will flash if it goes to atmospheric pressure. Condensate storage tank should therefore be connected either before or after the CPP where the condensate temperature is low. Additionally; condensate storage should be connected away from the normal path of the condensate.

The condensate coming from the ACC and from the condensate polishing system shall be directed to the deaerator/feedwater tanks (with consideration of balancing the water in each water/steam cycle if necessary).

6.6.3. Condensate polishing system

Refer to Exhibit B –Part 1.

6.7. Cooling systems

6.7.1. Closed loop cooling water systems

Refer to Exhibit B, Part 1.

The closed circuit cooling water system shall be provided to remove the heat dissipated by Plant components. Cooling water will not be provided. This system shall be used to meet cooling requirements associated with the gas turbine generators, as well as sampling panels, lube oil, HRSG blowdown (if necessary) and other miscellaneous loads.

The closed circuit cooling water system will include:

- cooling of the closed circuit system shall be through air to water fin fan coolers (total number (n) of fans to be determined during detail design however; the design shall be inclusive of two additional redundant fans (i.e. n+2) including all margins.)
- two (2) x 100% or three (3) x 50% capacity cooling water pumps. These pumps will circulate cooling water from the fin fan heat exchangers to various locations and back to the heat exchangers.
- one (1) closed cooling water head/storage tank to provide makeup water and be capable of performing both functions of a head tank and storage/surge tank.)
- a chemical dosing system, including chemical addition tank to inhibit corrosion in the closed cooling water system

The cooling system shall be designed in accordance with relevant international codes for Fin Fan Coolers for operation of the plant at maximum continuous rating for the entire ambient temperature range specified in Exhibit B, Part 1.

Demin water pump shall be used as makeup water for filling the closed circuit cooling water system and for covering losses. If Feedwater is used, the necessary treatment facilities shall be provided by the Contractor. Contractor shall consider automatic filling method in case of requirement during black start.

The system shall be capable of providing the required system pressure and volume control.

6.8. Fuel systems

6.8.1. Fuel Gas systems

The Plant shall be designed to fire both Sales gas and Low BTU gas ("Gas"). Both of these gases will be supplied by Saudi Aramco at the Delivery Point as per the Fuel gas quality specifications and as per the conditions specified in Exhibit B, Part 1.

The scope of work for each of the Gas supply systems includes, but is not limited to, the following:

- on the inlet pipe, one isolating valve (manual) + one emergency stop valve which can be closed from the CCR as per MFS (Exhibit B, Part 1 of)
- if required by the gas turbines, one compressors section (3 x 33 % plus 1 x 33 % in hot stand-by Two gas coolers (2 x 50% ; + by-pass) used to avoid too warm gas during the start-up or low load operation (if necessary – to be examined)
- filtering section (3 x 50% (or) 2 x 100% filters)
- metering sections compliant with the (Exhibit B - Part 1)
- fuel gas heating sections (3 x 50% (or) 2 x 100% heat exchangers or water bath heaters – if necessary)
- if required; dew point heater (1 x 100%) to be used only for start-up
- necessary piping including piping between the Delivery Point and the GT/HRSG/Reserve Boiler
- gas pressure and temperature control facilities for all gas feed points as required
- pressure reducing sections with 3 x 50% or 2 x 100% lines; each line consists of one manual isolating valve, one gas operated slam shut valve, one gas operated active pressure reducing valve, one gas operated monitor pressure reducing valve, one check valve, one manual isolating valve. If necessary a smaller line shall be installed for start-up or low load.

- gas conditioning and filtering facilities
- separate Gas metering facilities for Gas supplied to HRSGs for supplementary firing and Reserve Boilers
- gas back-up metering facility

The Contractor shall supply all equipment, valves & fittings, controls, piping, drain collection and disposal, gas vent stack (cold vent stack or flare system), vents etc. as necessary to make the fuel gas supply system complete and functional.

The total Gas flow will be measured by the Gas Metering System to be provided by the Contractor.

Separate Gas meters shall be supplied by the Contractor to measure the Gas consumed for supplementary firing on HRSG and for Reserve Boilers.

The Gas conditioning and filtering skid shall include carbon steel filters/separators installed downstream of Plant Gas Delivery Point. The filters/separators shall be fabricated per ASME VIII to remove solid particles of micron size 3μ with an efficiency of 100% filtrate rate and 0.5μ at 99.5% or better and liquid particles of micron size 3μ with an efficiency of 100% filtrate rate and 0.3μ at 99.5% or better, to the requirements of the gas turbine equipment supplier. Liquid drain from the filter separators shall be routed to the fuel gas condensate drain tank from where it will be truck-loaded out for disposal. Level-indicating transmitters and switches shall provide level indication to the control room operator. The isolation valves for filter separators shall be double seat ball valves.

Gas pressure control stations shall be provided for each of the Plant power and steam Unit. The Gas pressure to the Plant power and steam units shall be controlled within the appropriate operating range via pneumatically operated control valves. The pressure control station shall be provided with necessary control valves. A pneumatically operated high-pressure shutoff valve will also be provided to close upon high downstream Gas pressure and will minimize the amount of Gas vented to atmosphere when the safety relief valve lifts. Triple redundant pressure transmitters shall be provided downstream of the regulating station to provide a highly reliable Gas pressure control system that will minimize unnecessary turbine trips. The Gas System shall be provided with ESD valves upstream to the fuel gas system.

Extended capacity strainer shall be provided to remove impurities from the Gas. A pressure switch shall monitor the differential across the strainers will signal an alarm to the central control room via DCMS for the pressure drop across the strainer indicates cleaning is required.

If required, Gas heaters and heating water boosting pumps shall be provided in order to control Gas temperature to maintain variation of Modified Wobbe Index (MWI) within +/- 5% to enable the gas turbines to operate at full load for both the Gas qualities within the specified ranges in Exhibit B, Part 1.

Scrubbers shall be provided to remove liquid inside fuel gas. In case the gas pressure is lower than heating liquid, scrubbers will also be installed downstream to the fuel gas heaters to prevent any moisture carry over from any tube leak in fuel gas heater. Liquid drain from the scrubber shall be routed to a fuel gas scrubber condensate drain tanks from where it will be truck-loaded out for disposal.

For each gas turbine, 2 coalescent final filters (2 x 100%) shall be foreseen (if required by the gas turbine manufacturer).

Piping after last scrubber to gas turbine shall be of stainless steel, if recommended by the manufacturer.

6.8.2. Additional requirements

The Contractor shall provide the facilities required to test the quality of the fuel gas including:

- quality: calorific heating value (higher and lower)
- on-line gas composition analysis
- gas buffer station

6.8.3. Gas Buffer Station

The gas buffer station shall satisfy the requirements as defined in Exhibit B, Part 1.

Generally, if there is a sudden interruption of Low BTU gas supply or sudden gas pressure reduction, the gas buffer station shall enable a smooth fuel changeover from Low BTU to Sales gas.

Incoming gas shall be analysed to determine the Wobbe Index upstream of the buffer station to identify the characteristics of the gas. The buffer station shall be sufficiently sized to allow the gas turbine control system sufficient time to react to the fuel quality changes and therefore enable the smooth fuel changeover from Low BTU to Sales gas.

The buffer station shall be of a proven design and suitable for the harsh climatic conditions, the working temperatures and pressures of the gas. The vessel class shall be suitable for the application.

6.9. Demineralization plant / distribution

Refer to Volume 1, section 4.4.10 of Exhibit B, Part 1 with consideration of the Ion Exchangers alternative.

The quality of the product water from the demineralisation plant shall be according to VGB R 450 L.

The minimum time between regenerations shall be 24 hours at full demineralization plant output while the regeneration time does not exceed 4 hours. The regeneration equipment is common for all trains (acid and caustic storage tanks, pumps, neutralization pit with a capacity equal to 1.5 times the regeneration effluent volume).

The demineralisation plant shall have at least a 24 hours storage capacity (two tanks).

Each tank vent is equipped with a CO₂ trap with an air filter of a replaceable type (if necessary - to be finalized during detailed design).

Chemicals storage capacity related to demineralisation shall be 14 days of maximal consumption or 1.5 times the unloading tanker capacity. Tanks shall be equipped with bund wall for a volume equal to 110% of the tank.

6.10. Potable and sanitary water / Service water

Refer to Exhibit B-1.

The potable water system is sized for 100 people.

6.11. Chemical dosing system for water/steam cycle

The Plant shall be equipped with chemical dosing systems to control the boiler and Feedwater chemistry. The chemical dosing system to control the boiler and Feedwater chemistry will be operated based on chemistry parameters of HRSG water. Since the steam will be used in Saudi Aramco processing facilities, the chemical dosing agents chosen have to be agreed with Saudi Aramco. However, the following shall be considered as preliminary:

- ammonia NH₃
- Oxygen scavenging chemical, only during malfunction
- trisodiumphosphate Na₃ PO₄

The Contractor may propose also other chemicals if he can demonstrate long term economical, successful and safe utilization in other plants. A fully automatic chemical dosing system shall be provided for ammonia and oxygen scavenging. For trisodiumphosphate, manual dosing is acceptable as long as on-line phosphate analyzer is implemented .

The operational parameters of quality of the steam leaving each HRSG shall meet the parameters specified in IEC/TS 61370. The make-up water quality shall be produced accordingly.

All chemical dosing systems shall be supplied and provided as complete systems.

Each injection shall allow the conditioning of the boiler within 30 minutes and maintain sufficient excess of the chemicals in the steam drum.

Each chemical dosing system shall comprise all equipment for the safe, reliable and efficient operation of the dosing systems, including bunds, segregated chemical drains, appropriate ventilation and emergency eye wash and deluge shower units.

Each chemical dosing system shall consist of two (2) x 100% chemical dosing pumps and one (1) chemical dosing tank.

The chemical dosing tank shall combine solution mixing and preparation and shall be made of stainless steel or synthetic material. It shall have a working capacity of 4 days of continuous use for each additive. The tank shall be provided with electric agitator as necessary.

The dosing pumps material shall be AISI 316L (1.4404) for the casing. Two (2) pumps per chemical shall be provided.

Injection pipes and valves are made of stainless steel AISI 316L (1.4404).

Bulk storage for each chemical shall be provided with a storage capacity of at least 7 days with the Plant at full load. For chemicals in liquid or gaseous state bulk storage tanks sized to unload a full standard road tanker shall be provided Plus 4 day's storage capacity. Required unloading arrangements and interconnections for auto transfer between the storage tanks and dosing tanks shall also be provided.

- scale Control - Phosphate
- pH Control - Ammonia

6.12. Sampling system for water/steam cycle

The Plant shall be equipped with a system for the continuous extraction and monitoring of selected samples from the HRSG steam and water circuit. A sample cooler shall be provided for each circuit. The selection of sample locations and analysers shall ensure adequate observation and control of steam and water chemistry limits. The measurements from the analysers shall be recorded on the power station DCMS. A manual sampling facility shall be provided for each sample. As a minimum, the following analysers shall be provided:

- Feed water:
 - DO (Dissolved Oxygen),
 - pH & Specific Conductivity,
 - cation Conductivity meter,
 - silica Analyser
- Main steam
 - pH & specific cation conductivity
 - silica Analyser
 - degassed cation conductivity meter and sodium
- Saturated steam (drum)
 - specific cation conductivity
 - pH Analyser
 - degassed Cation Conductivity meter
- Drum water
 - pH & Specific Conductivity
 - phosphate analyser (HRSG drum sample (Blowdown water))

The sampling system shall control the water-steam-cycle by providing samples from different points of the process. Samples will be taken both continuously and intermittently. The samples to be taken continuously shall be received, cooled, analysed and values displayed and transmitted to the DCMS. Intermittent samples will be analysed in the laboratory.

The samples with coolers shall be collected to one location on a central sampling rack close to the laboratory.

6.13. Waste water treatment

Refer to Exhibit B, Part 1.

6.14. Fire fighting system

6.14.1. General requirements

For technical requirements refer to Exhibit B, Part 1.

The fire protection and fire-fighting system for the Plant shall be designed in compliance with Saudi Aramco Standard SAES-B-017 and NFPA requirements/guidelines. In case of discrepancy the Saudi Aramco requirements prevail on any other document.

The fire protection and fire fighting system shall include, among others

- Water storage as required by NFPA but not less than two hours
- one (1) Fire fighting pumping station comprising of at least one (1) main fire water pump (electrical motor), one (1) main fire water pump (Diesel engine driven), jockey pumps, accumulators (as applicable) piping and valves
- underground main loop encircling the Plant and hydrants and hose rail stations inside installations
- sprinkler / deluge systems according to local regulations
- dry end pre-action type sprinkler on turbine and generator bearings
- leak confinement and foam systems on oil and diesel storage equipment
- sprinkler systems on main and auxiliary transformers
- the necessary mobile/portable extinguishers
- separate fire compartments in the buildings
- fire endurance of building bearing structures and stair cases
- fixed wet pipe sprinkler systems
- fixed deluge systems
- fixed CO2-systems
- gas detection systems
- fire alarm systems
- smoke ventilation
- hydrants and hose reels
- wet rising mains
- portable fire extinguishers

Clean agent suppression systems required for critical areas shall be designed in accordance with NFPA 2001 and NFPA 850. Central bank system or Independent systems will be confirmed during detail design and are subject to the approval of the Owner.

6.14.2. Separate Fire Compartments

The Plant shall be divided into fire compartments. Special attention shall be paid to fire barriers and penetration seals in separating walls and floors all over the Plant.

From every exit areas there will be two exit routes to the exit stairways or directly outdoor or to another fire compartment.

Exit routes will be provided with emergency and exit lighting and panic bars. The exits and the access to them shall be marked.

6.14.3. Fixed Wet Pipe Sprinkler Systems

When the temperature of fire exceeds the opening temperature of the sprinklers they shall open and spray water to extinguish the fire.

System valve position indicator shall be installed for alarm annunciation in the control room to indicate system operation.

Cable spreading rooms and cable tunnels shall be protected as per NFPA850 clause 7.8.2.1 with automatic sprinklers systems.

6.14.4. Fixed Deluge Systems

A fixed deluge system shall be installed for the following areas:

- lubricating oil system (if not protected by CO₂ system)
- steam turbine bearings protection shall be proposed, automatic but with operator local manual remote activation
- main transformers
- auxiliary transformers

A deluge system employs open spray nozzles connected via a piping system to water supply mains through a deluge valve. The system shall be equipped with a manual by-pass valve located on a safe distance.

Detection shall be achieved by fusible plug or EFD (Electrical Fire Detector) in accordance with SAES. Other means may be approved by the Chief Fire Prevention Engineer.

Pressure switches shall be installed for alarm indications in the control room.

6.14.5. Fixed CO₂ Extinguishing System

The GT compartment shall be provided with its own dedicated fire detection and CO₂-systems.

CO₂-systems shall operate automatically when detectors indicate fire.

In the event of alarm activating, an audible and visual indication shall be provided in the control room and in the concerned area.

6.14.6. Gas Detection System for Natural Gas

The hazardous areas (all the spaces where gas may be accumulated with particular attention to the GT enclosure as well as like areas of gas pipe routes and the pressure reduction station) shall be monitored by a gas detection system.

The gas detection system shall cause various alarms and controlling measures to be initiated at detected levels of LEL (Lower Explosive Limit – to be demonstrated by the Contractor and agreed with the Owner during the detailed design stage).

A low LEL alarm shall cause the ventilation rate of respective area to be increased to raise level to provide maximum dilution of the detected leak. A coincident high LEL alarm from two or more detectors shall cause the CCGT Plant to be shut down and gas supplies to be isolated.

The following signals shall be transmitted to the control room:

- equipment faulty
- equipment deactivated
- the first level alarm
- the second level alarm

The system shall have an uninterrupted electric power supply.

6.14.7. Fire Detection Systems

Fire detection system shall be installed to provide a high level of protection to both personnel and equipment. The system shall comprise automatic detectors (type according to the risk), manual call points, alarm bells, remote alarm annunciation and detection systems shall be installed for the following areas:

- admin facilities
- control and electrical building and all equipment rooms
- power block buildings
- water treatment plant offices, workshop and warehouse fire alarm/control panel.

Other facilities and outdoor areas shall be provided with manual call points.

In the event of a fire alarm being activated, an audible and visual indication shall be provided. System control panel shall be provided in the control room.

6.14.8. Smoke Ventilation

Depending on the case, roof and wall openings, windows and hatches as well as specially designed smoke vents and fans may be used.

Specially designed smoke vents shall be operated automatically by the elevated temperatures generated during a fire. They shall also be opened manually and from a remote control panel.

The total area of the smoke vents shall be approximately 1% from the bottom area of each smoke-ventilated space.

6.14.9. External Hydrants and Hose Reels

Hose reels for the ground floor levels shall be located in accordance to mandatory norms and standards.

Hose reels shall be installed at floor and service platform levels (outside each protected stairwell) throughout the height of all building in accordance to mandatory norms and standards. These hose reels shall be served by the site fire water main in order to provide the necessary pressure at high level to ensure acceptable operation.

External hydrants shall be positioned around the Plant in accordance with local regulations.

6.14.10. Portable Fire Extinguishers

Fire extinguishers of an appropriate type shall be located throughout the areas as required by NFPA and local fire regulation.

The type of extinguishers will be determined according to the risk. The following types can be used:

- dry powder
- CO₂
- water/foam

At least one portable foam unit shall be provided for each GT area.

6.15. Auxiliary mechanical systems

6.15.1. Emergency/black start diesel generator

Combined emergency/black start diesel generator(s) may be provided for safe shutdown and start-up of the plant without grid power. The units shall consist of a diesel engine directly coupled to generator, mounted on common steel base with a fuel tank and a self-contained enclosure. In case of failure of normal power supply, the emergency/black start diesel generator shall fast start automatically to supply power to the essential loads of the plant which are to be run to take the plant to a safe shutdown condition. Later, the emergency/black start diesel generator shall cater to the start-up of the GT up to grid synchronization and minimum load.

All operational logic and interlocks for the generators shall be provided in the local control panel and DCMS. The generators will be controlled by a PLC (local control panel). Additionally, each generator shall also be able to start / stop and be feedback monitored from the DCMS. The power from the generators is fed to the switchgear through an automatic transfer switch.

The diesel generators shall be of minimum IP Protection class IP 23DW and terminal box enclosure should be class IP 54. The diesel generator shall be installed indoors.

Emergency diesel generators shall be designed for class F insulation and shall be operated with class B temperature rise.

6.15.2. Compressed air system

Refer to Exhibit B, Part 1

Instrument air has an absolute priority and any perturbation / problem in the service air part of the system may not have any influence on the instrument air part of the system.

Complete Plant service air system for maintenance with sufficient distribution loops (e.g. at different levels of the Plant equipment).

Compressed air systems supplying technically oil free air to DIN ISO 8573-1 shall be provided, and sized to provide a reliable supply of instrument and service air as applicable in the correct quantity and conditions for the various users in each area. The quality of air shall be in accordance with ISA-S7.0.01-1996

The equipment shall be designed with appropriate redundancy and shall consist of 3 x 100% heavy duty air compressors (100% means normal maximum air consumption for plant), 2 x 100% air chillers, 2 x 100% air dryers, air filters, air receiver (2 x 100%), control system and pipework distribution system. Each control air receiver shall have sufficient capacity to enable the Plant to be shut down in a safe and controllable manner and enable standby compressor plant to come into operational service without reduction in air supply. An interconnection shall be provided between the instrument and service air pipelines upstream of the air dryers on each system. The air distribution bus main shall be of stainless steel.

The compressed air systems shall include (i) General Service Air and (ii) Control and Instrumentation Air, configured to ensure a sustained supply of compressed air to the plant.

The control and instrumentation air quality shall meet the following requirements:

- compressor intakes to contain particles no greater than 3 microns
- the product air shall:
 - •contain no desiccant fines
 - •contain no solid particles greater than 1 micron
 - •contain no more than 1 ppm weight/weight of oil at 20C and line pressure
 - •have a final dew point at line pressure of the lower of -10C or 10K below the lowest ambient temperature

6.15.3. Nitrogen system / preservation of the Plant

The Plant shall be designed in such a manner that preservation of all parts is enabled without any danger of corrosion. The Contractor shall provide a system for the preservation of the Plant for a short term standby and provisions for a long term standby.

Rack mounted high pressure portable cylinders sets and distribution lines to the boilers for conservation during extended out-of-service periods for maintenance. The manifold for the connection of the racks of bottles will be installed at ground level. The number and capacity of the cylinders will be determined during the engineering phase.

6.15.4. Lifting / Hoisting Equipment

6.15.4.1. MAIN OVERHEAD CRANE FOR TURBINES AND GENERATORS

The Contractor shall provide electric overhead travelling crane for the maintenance of the steam turbine generator units.

The crane shall be designed and constructed in accordance with the requirements of applicable mandatory local norms and standards or equivalent international standard and shall be complete in all respects for safe and efficient operation.

If required a traverse, to facilitate the separate lifting of generator stator and rotor assemblies and component parts, shall be supplied. The lifting beam shall be attached to the main hook of the crane.

The control of each crane motion will be arranged to incorporate a steady creep speed.

The crane shall be complete in every respect including access platforms, ladders, supports, protective covers, motors, control gear and wiring with main isolator and longitudinal isolated gantry conductors, collectors/supports, and longitudinal crane rails complete with holding down bolts and any other auxiliary apparatus incidental thereto for the safe and efficient operation of the crane.

A warning alarm shall be mounted on the electrically operated crane bridge in an easily accessible position. The alarm shall be initiated by the crane operator from the radio remote control to give warning of impending long travel motion.

The crane shall be fitted with a nameplate showing the manufacturer's name, the date of manufacture, a serial or identification number and the rated capacity of the crane.

6.15.4.2. OTHER LIFTING / HOISTING EQUIPMENT

Complete lifting equipment required for operation, maintenance, removal, and reinstallation of the equipment shall be designed, specified, procured, developed, fabricated, furnished, delivered, and installed.

Scope shall include turbine hall overhead travelling crane, and boiler feed pump overhead monorail(s) and the monorails in other areas required for the maintenance of the equipment, including but not limited to bridge, trolley, rail sweeps, motors, motor starters, gears, brakes, wheels, stops, bumpers, handrails, cables, wiring, panels, lighting, conductors, power feed connectors, pendant controls, radio controls, hoists, hooks, ropes, limit switches, etc., as needed to make a complete system.

The lifting access bay in the boiler structure servicing the boiler structure floors shall be provided with an electric operated controlled hoist with motor operated trolley for boiler top.

For indoor auxiliaries and machinery where the weight of a single part to be handled exceeds 10 tons, a minimum of a monorail with motorized trolley and motorized hoist shall be provided.

For indoor machines or indoor equipment, where the weight of a single part to be handled is 5 tons to 10 tons, a minimum of a monorail with geared trolley and manual hoist shall be provided and where the weight of a single part to be handled is less than 5 tons, only an overhead steel beam (monorail) or eye-plate shall be provided for supporting a temporary geared trolley and manual hoist or lifting equipment.

The bridge and gantry cranes and trolleys shall be sized to lift the heaviest piece in the equipment to facilitate their erection and maintenance. The turbine building crane shall have both main and auxiliary hooks.

The bridge and gantry crane wheel loads shall be as specified by the crane manufacturer. Impact loads from cranes shall be in accordance with AISC Standard.

Monorail hoist loads shall include lifted load, hoist weight, trolley weight and impact. Impact for monorail beams shall be 25% of the lifted load in the vertical direction. Transverse and longitudinal loads from trolley to the trolley beam shall be in accordance with trolley manufacturer's recommendation. The cranes and trolley shall have stops to prevent over travel.

6.16. General Mechanical Requirements

6.16.1. Piping and tubing

The quality of the pipes shall be guaranteed with appropriate NDT-methods. In addition to the material used, also the location, correct dimensioning and quality of supports as well as flexibility of the pipework shall be taken into account when designing the piping and piping layout. Expansion bends or loops shall be provided where required.

The design of all pipes, bends and fittings shall comply with norms and standards as per Exhibit B -Part 1

Piping for high-pressure services shall be cylindrical and uniform in section.

All atmospheric blow-down lines, drains, vents, sampling lines etc. belonging to the water-steam cycle shall be equipped with double shut-off valves if pressure exceeds 40 bar(g).

When locating the valves and the measuring points (temperature sensors, pressure taps etc.) the movements of the pipes with respect to the steel construction shall be taken into consideration.

Generally, clean drains shall be returned back to the process but considering at all times the operational requirements for chemical purity of the water-steam cycle. Such drains include e.g. steam lines, turbine drains, by-pass lines etc. Alternatively, for steam line drainage between FCHPP and up to FGP interface point only, the use of steam traps with soakaway pits is acceptable

Flanges shall be the same materials as the joining pipe or fitting. Where unavoidable for dissimilar joining flanges within the GTG Package only (not applicable for interface between Contractor and GTG Package where flanges of similar material shall be used) dissimilar flanges may be used as per applicable codes.

All steam mains shall be erected with a continuous slope in the direction of the flow so that the condensed steam will flow towards the drainage points.

The pipes shall be equipped with necessary venting, draining and sampling points including valves. Venting valves shall be collected to one group and they shall be provided with drainage to the drainage tank. Subject to the approval of Saudi Aramco, warm up vent with silencer (if required) is acceptable only for steam headers between FCHPP and FGP interface point.

All HP steam stop valves shall be equipped with by-pass valves. In case Contractor is of the opinion that a bypass valve is not required for any particular valve then, the list of such valves along with sufficient technical details shall be provided to the Owner during detailed design stage for the Owner's approval.

All fire fighting system piping underground shall be in GRP or any other adequate material suitable for that function. Special caution shall be taken during installation and erection to avoid any internal tension and leaks.

The effect of water hammering shall be taken into account when designing the location of drain lines.

All drain pipes discharging to a drain header shall be connected at an angle to the header so as to discharge in the direction of flow in the header. All drainpipes shall have an approved continuous slope towards the discharge ends.

Labels indicating the source of drain for each respective pipe entering a drain vessel shall be attached to the pipe.

The piping support structure design shall be such that no harmful vibrations occur during operation of the Plant. The movements of the pipework during standstill (cold) and during the normal operation (hot) shall be taken into account. Supports shall not be welded to the bottom flange of beams.

Pipes and fittings shall be so supported that the pipes may expand and contract freely.

All high pressure drain pipes on the inlet of drain valves, together with the drain pipes, shall be designed for the same pressure and temperature as the valve, pipe or vessel which they drain and shall be subject to the same specification and tests relevant thereto.

The Contractor shall provide a piping specification for each piping system for Saudi Aramco's approval.

Except where particularly specified in other clauses of the Specification, the materials of construction of pipework shall be selected in accordance with the following duties.

This following table applies for the Power and Water sections of the plant:

Service	Material
Steam and condensate	Alloy and Carbon steel
Chemical injection	316L stainless steel or A106GR.B.S./A53GR.B.E (if suitable) or MDPE may be considered for some LP piping applications subject to approval of SA.
Potable water (Drinking water) for underground: For above ground:	GRP or rubber lined carbon steel 316L stainless steel
Fire fighting	Underground : HDPE Aboveground : A53GR.B.E Galvanised (All in accordance with NFPA)
Waste Water common discharge lines	GRP or A53GR.B.E
Instrument air	304L stainless steel
Service air	304L stainless steel

Where GRP pipe is used above ground it is to comply with the requirements of ANSI B31.3. Where GRP pipe is used above ground design shall meet full vacuum conditions except for gravity flow and non-pressurised piping such as discharge outfall drain piping.

The flow velocities shall be as per Exhibit B, Part 1.

Pipe materials are subject to the Owner's approval.

6.16.2. Pumps

The Contractor shall match his pump characteristics to the pipe system network to achieve the high pump efficiency and reliability.

In normal operation, it shall be possible to start and stop the pumps under any operating conditions without any special precautions such as venting or warming.

The service life of the mechanical shaft seals shall be at least 20,000 operating hours and 40,000 operating hours for the bearings.

To avoid vibration problems all large pumps, including feedwater pumps, shall be located on separate foundations.

The impeller of the single stage pumps shall be readily withdrawable from the pump casing without the need to transfer the pump and with minimum dismantling and disconnection work from the pipework.

The pump design shall ensure that alignment is maintained between the various assemblies by recesses, spigots and dowels and shall be such that all components liable to wear can be replaced.

Components of the multistage pumps shall be permanently marked with the manufacturer's number and where dowels are not used, permanently marked for correct assembly. The pump casing and the impeller shall have detachable wear rings.

Where grease points are necessary they shall be fitted with removable screwed plugs which shall be accessible without removing guards.

Bearings with bearings which require a continuous supply of lubricant shall incorporate a means of monitoring such a supply, by temperature rise as appropriate for the type of bearing employed; separate monitors being fitted for each bearing feed or housing.

If sealing water is needed, flow shall be shown by sight glasses.

Special care in the selection of materials shall be taken in order to avoid binding and electrolytic action between the shaft sleeve and the mechanical seal components, particularly where long periods of idleness are inherent in the duty cycle as in the case of standby and storm pumping.

The highest point on the pump casing shall be fitted with a manual air release cock having a removable handle or an automatic air release valve with a lock-able isolation valve as specified (or according to the Manufacturer standard).

When required, pumps shall have necessary hoisting and dismantling facilities, like hoisting beams, for maintenance purposes.

The test performed to verify the trouble free operation, of pumps shall at least include verification of pumps performances and validation of pumps curves including required NPSH curve, vibration measurements, etc). Saudi Aramco and/or their representative shall be informed reasonably in advance and allowed to witness the tests.

6.16.3. Valves and Actuators

The design, structure and material of the valves shall be suitable for their particular operational purpose.

Valves and devices shall be located in an ergonomically correct way in respect to the maintenance platforms considering that maintenance shall be possible without removing the valves (except butterfly valve) from the pipe.

The valve and seat material shall comply with the operational and corrosion conditions for which they are intended.

The upper structure of the steam valves shall be designed in the manner that, at different operational temperatures of the spindle and cover, there is sufficient spindle force left to maintain the seat pressure required. In motor-controlled valves the seat force shall not vary according to the temperature so that the operating mechanism could not open cold valve.

The valves and the accessories shall be mounted so that pipe reactions do not hamper perfect valve operation and tightness.

The valves shall be well accessible for maintenance and operation. The larger valves (e.g.. boiler safety valves) must allow for the use of a crane or a hoist during maintenance. When necessary, the valve shall be equipped with an extended spindle. The valve hand-wheel shall not be higher than 1700mm above the operating level.

The following minimum data shall be permanently marked on the valve body:

- DN and PN
- material
- manufacturer's mark
- identification marks of the certificates
- flow direction

Generally (subject to Owner's Design Review and wherever they are required to be operated during start-up, shut-down and operation) stop valves larger than DN400 shall be equipped with actuators in order to enable operations from the main control room via DCMS.

Each valve must display with a clear text giving the identification number of the valve and markings of the "CLOSED-OPEN" positions with the direction arrows, and local position indicators.

The direction of rotation of valve hand-wheels (when looking at the face of the hand-wheel) shall be clockwise to close valves.

The position of all control valves shall be indicated on the DCS operating station.

The electrical valve-actuator combination has to be designed in such a way that the actuator does not damage the valve even if wrongly controlled with stall torque at 10 % overvoltage. There shall be one adjustable torque switch for both directions to prevent damage in the event of jamming.

Valve sizes shall be chosen according to pipe sizes, with the exception of control valves.

In the case of strong cavitation the control valves shall be of the multistage construction type.

The bonnets of the high-pressure (> 40barg) gate valves shall be of self-sealing construction type.

Stop valves, which operate at under pressure, must have either seal water supply, or some other system that minimises air leakage through the stem packing.

Intake and outlet nozzles of globe valves shall be located on the same centre line of the valve casing.

All valves have to be designed for full shut-off pressure difference.

The electric actuators shall be provided with anti-condensation heaters when mounted outdoors.

The electric actuator shall be self-locking. The motor and control gear shall be placed in such a position relative to the valve that there is no possibility of leakage of steam or water from the valve joints or glands blowing on to the motor or control equipment. A hand operating facility shall be provided and be so arranged that the mechanism will disengage automatically when the motor operates.

The actuator shall have an adjustable mechanical position indicator so that it may readily be seen whether the valves are open or shut and in the case of those valves fitted with extended spindles, indicators shall be fitted both to the extended spindle and to the valve spindle.

The Contractor shall propose to Owner criteria to select either pneumatic or electrical actuator depending on the needs of the process in question (process media: steam, water, air, gas, oil), location of the valve (outdoors/ indoors), space available, maintenance needs, purpose of function (regular closing or control, fast closing or control, safety position requirement).

All blow-out and safety valves shall be equipped with efficient silencers.

6.16.4. Heat Exchangers

If applicable, plate heat exchangers shall be designed with tie-rods the length of which would permit an increase of 25% of the exchanger surface.

6.16.5. Thermal Insulation

Thermal insulations shall be designed, selected and installed in compliance to mandatory local norms and standards, or if not applicable, equivalent international standard.

All lagging adjacent to hot surfaces shall be capable of withstanding the maximum surface temperature obtainable in service without deterioration or breakdown of the material

The valve boxes and the covers of manholes shall be easily dismantled and reinstalled. The same is also valid for the boiler parts, which shall be regularly checked. The demountable insulation boxes shall be strengthened. It shall be possible to change temperature thermo-elements without dismantling the insulation.

The penetrations of hot pipes (e.g. blow-out piping of safety valves) shall also be insulated. The risk of ignition of construction materials of the boiler/turbine house shall be taken into account, and the insulation material used shall be non-combustible.

All exposed portions of the Plant which are subject to a temperature exceeding 60°C and are likely to come in direct contact with personnel shall be provided with approved thermal insulation so that the temperature on the outside of the insulation does not exceed 60°C when the ambient temperature is 35°C.

The insulation of the boiler shall be made up of two layers of the insulating wool mat, when applicable, and the seams of the separate layers shall be overlapped. The cladding shall be aluminium sheeting. The insulation of the turbine shall be made up of either at least two layers insulation wool mat or spray-on insulation material, and it shall be covered so that it prevents moisture and oil penetration into the insulation material.

Lagging shall be properly secured in position and it shall not loosen or suffer damage when subjected to vibration. Turbine cylinders shall be provided with fixing points for the insulation.

The insulation works shall be designed so that insulation will not get wet during storage or installation.

Piping and vessels shall be insulated with prefabricated mineral wool elements. The insulation of all piping, vessels, valves and fittings shall be finally covered (e.g. aluminium for outdoor application or galvanised steel sheeting for indoor application) and completely weatherproofed on any outdoor installations.

In all cases, the use of Asbestos is forbidden.

6.16.6. Vibration

All rotating equipment shall comply with the requirements for vibration magnitude stipulated in ISO 10816 Parts 1 – 7 as applicable. For combustion gas turbines, air compressors, and vertical pumps Zone B shall apply. Emergency Diesel Generator shall comply with ISO 8528-9, part 9. For all other equipment ISO 10816 parts 1-7, Zone A shall apply.

For only GT, ISO 7919 zone A/B is acceptable as a guideline.

6.16.7. Protection against Corrosion

6.16.7.1. PAINTING / SURFACE TREATMENT

As per Codes and Standards (refer to section 4), Exhibit B, Part 1 and Good Industry Practice.

The Contractor shall be aware of the corrosion problems to be faced on Site, especially with the equipment installed outdoors, including incidence of dust and sand storms. The coating specification provided by Contractor shall consider the following corrosion categories (according to ISO 12944):

- at least C4 with high durability for all other outdoor steel elements including the civil structures and the Plant equipment (e.g. HRSG, pumps, etc)
- at least C3 with high durability for all the indoor steel structures and plant equipment

6.16.7.2. GALVANIZATION

In addition to the requirements of Exhibit B, Part 1, galvanising work shall conform to and shall be performed by the hot dip process, unless otherwise specified. Two coats of cold galvanising paint shall touch up damaged parts of the galvanising surface.

The following standards or equivalent shall be applied; JIS H8641 Zinc Hot Dip Galvanizing, BS 729, Swedish Standard SIS 055900.

6.16.7.3. CATHODIC PROTECTION

In addition to the requirements of Exhibit B, Part 1, cathodic protection (sacrificial anode type or equivalent as approved by the Owner) shall be applied for all buried metallic pipes and all metallic storage tanks for the design life.

6.17. Welding

6.17.1. Welders

All welders and welding operators shall be qualified for the work for which they are engaged. Pressure Part Welders shall hold current welder's qualification certificates in accordance with the latest edition of the ASME Boiler and Pressure Vessel Code Section IX, "Qualification Standard for Welding Procedures, Welders and Welding Operations" or approved equivalent.

In the case of pressure parts and pressure part attachment welds, each approved welder shall have a unique identification number which shall be indicated on all welds made by him and on all permanent test records applicable to those welds including radiographs.

All welders shall be under the supervision of a competent welding engineer who shall demonstrate, to the Owner's satisfaction, a complete system for the control and treatment of consumables, deployment of welders, use of approved procedures, heat treatment, non-destructive testing, including spectrographic material checks on weld and parent metals, repairs and general quality control.

6.17.2. Welder's Tests

Sample welds shall be carried out on specimens of similar shape, thickness and chemical analysis to the material to be welded, and in a position which shall represent as far as possible the worst conditions under which actual welding shall take place, as laid down by the approved Standard/Code.

The method of preparation of the test pieces and the standards to be used in examination of them shall be according to the agreed Standard or Codes, or as may be agreed with the Owner. All expenses of every kind in connection with welder's tests shall be borne by the Contractor.

6.17.3. Welding Procedures

Welding procedures including those proposed in repair situations shall be qualified as required by the governing design code or, where no appropriate standard is thereby indicated, by Section IX of the ASME Code or approved equivalent and all qualified procedures shall be submitted to the Owner prior to the commencement of any welding.

6.17.4. Welding and Heat Treatment

All welding procedures, welders and operators of machine welding equipment shall be qualified. All shop and field welding procedures, welders and operators of machine welding equipment for welding of pressure parts shall be qualified in accordance with the latest edition of the ASME Boiler and Pressure Vessel Code Section IX, "Qualification Standard for Welding Procedures, Welders and Welding Operations" or equivalent.

All welds in structural steel shall be in accordance with AWS D1.1 and other applicable codes.

All electrodes and/or welding rods and the fabrication procedure shall be in accordance with the applicable codes and standards. Before any welding is commenced, the Contractor shall submit a list of all electrodes and consumables to be used to the Owner.

6.17.4.1. PREHEATING

Where necessary, the work shall be heated prior to welding in an approved manner, and the temperature maintained throughout the operation. The metal temperature shall be checked by means of suitable thermocouples or temperature indicating crayons.

Prior to welding, the work shall be preheated, where necessary, in accordance with applicable codes and standards, and the temperature shall be maintained throughout the operation.

6.17.4.2. POST WELD HEAT TREATMENT

Regardless of the requirements of the specific code or specifications applicable to the design and/or construction of the plant and associated pipework, it is a mandatory requirement of this Specification that all welds shall be subject to appropriate post weld treatment with the following exceptions:

- austenitic stainless steels
- plain carbon steels containing less than 0.25 per cent carbon at thicknesses up to and including 30 mm
- low alloy steels containing $\frac{1}{2}$ per cent molybdenum and $1\frac{1}{4}$ per cent chromium, $\frac{1}{2}$ per cent molybdenum up to and including thicknesses of 12.5 mm provided that:
 - a hardness traverse across the weld zone gives no single result greater than 240 hv10
 - no fillet weld leg is greater than 8 mm
 - a minimum preheat of 200°C is used for welding

Plain carbon steels shall contain no more than 0.25% carbon.

All post weld heat treatments of materials and components for use at temperatures in excess of 400°C shall be so designed as to guarantee maximum creep properties in the parent materials and weldments.

The soaking times at the selected heat treatment temperatures shall be:

- low alloy steels not containing vanadium, 5 minutes per mm of thickness with a minimum time of 120 minutes
- low alloy steels containing vanadium and high (11 to 15 per cent) chromium steels containing vanadium, 180 minutes regardless of thickness

In the event that any post weld heat treatment is carried out at a temperature in excess of the original tempering temperature of the parent materials, a re-assessment of the design strength limit stresses for the materials will be required. This may, in the absence of acceptable mechanical test data for the specific condition, require re-determination of the relevant mechanical properties of the materials.

The requirements for post weld heat treatment do not affect or replace in any way the requirements for maintenance of welding preheats or post welding heating for the purposes of diffusing hydrogen from the weldment.

Post weld heat treatment, where applicable, shall be conducted in accordance with applicable codes and standards. Records of post weld heat treatment shall be made available for inspection when requested.

After completion of welding, fabricated parts shall be stress relieved in accordance with the applicable codes and standards.

In the case of IPB welds, AWS or other relevant international Standards shall apply.

6.17.5. Site Weld repairs

Where repairs on major items of plant are found to be necessary by welding, dimensioned sketches of the excavated areas shall be prepared and the proposed repair procedure submitted to the Owner before any repair is commenced. Under no circumstances shall the repair be commenced until the agreement of the Owner has been obtained. The areas shall be subjected to a suitable non-destructive test and on completion of the weld will again be non-destructively tested. In the event that the repair requires all or any part of the item to be stress-relieved, the non-destructive tests shall be repeated on any part deemed to have been affected by the stress relief.

In the case of weld cracking, the Owner may require a full investigation to be made of the cause of the cracking prior to any approval of repair procedures. The cost of all such investigations and reports is to be wholly borne by the Contractor.

Weld repair of cast iron parts is not permitted under any circumstances.

6.17.6. Welding Electrodes and Consumables

Before any welding is commenced, the Contractor shall submit a list of all electrodes and consumables to be used to the Owner.

6.17.7. Inspection of Welding

The extent of weld inspection and the final weld quality shall comply with the applicable codes and standards. Records of all welder qualifications, weld procedures, and weld inspection and repair reports shall be maintained. The Contractor shall maintain, both at works and on site, adequately indexed records of all welding, weld inspection and repairs, which shall be available to the Owner, together with facilities for viewing of X-ray photographs.

The views taken by X-ray shall be to a procedure and resolution agreed with the Owner and shall incorporate the appropriate image quality indicator.

Adequate safe storage facilities and appropriate safety regulations shall be provided for gamma-ray units, if used.

6.18. Galleries, Platforms and Staircases

Galleries, staircases, platforms, other related steel constructions shall be designed according to the mandatory local norms and standards, or if not applicable, equivalent international standard (ISO) related to directions of industrial safety and in the way that all places that are necessary for maintenance (e.g. valves, meters) are easily accessible. Vertical ladders are not allowed as a fairway to places which must be entered during normal operation, but they shall be allowed for escape ladders, access to stack and for access to not frequently used equipment (e.g. remote sensors).

Steelwork in close proximity with hot air or gas or subject to radiation from any part of the Plant shall be adequately protected and insulated. On completion of the work, all necessary platforms, ladders and stairways for the Plant and ancillary equipment shall be provided to give safe access to all equipment, valves, and instrumentation.

Platforms shall be constructed using galvanised open steel flooring and shall cater for the relevant loading for maintenance. The flooring shall be bolted to the framework using approved bolted clips. The platform framework and the corresponding nuts and bolts shall be galvanised. The minimum width of the galleries for emergency exits and the minimum free height of these galleries shall be according to relevant standards. Exit routes shall be clearly marked with colours according to applicable norms and standards.

Platforms shall be provided around equipment as required for maintenance work, testing, inspection and safe operation. Platforms shall be sized to facilitate safe ergonomic operation of manual valves and equipment between waist and shoulder level of a standard man. Relief valves shall be grouped together and an access platform provided. Operating valves which cannot be serviced from a platform shall be operated by other approved means.

All platforms to be visited on a daily basis shall be provided with stairways, except where a permanent access ladder is the only feasible means of access. All interior and exterior platforms, gratings, checker plate, stairways and handrails shall be galvanised.

Platforms shall be constructed using galvanised open steel flooring or a suitable grade of GRP grating in areas of high salinity, to the relevant ISO, British Standards or ASME and shall cater for the relevant loadings for maintenance. Generally, no frequently accessed platform shall be less than 1m wide and all platforms shall be extended up to the equipment, valves and instrumentation that they serve.

All edges of floors, platforms and walkways shall be provided with curbs or kicking strips.

7. ELECTRICAL TECHNICAL REQUIREMENTS

7.1. Introduction

The scope covered by this section of the specification relates to all equipment not specifically covered in Exhibit B, Part 1. In some cases certain duplication may be apparent but has been included in order to provide completeness.

The Contractor shall provide the complete electrical system for the Plant, including power to the various auxiliary buildings, interface works and the mechanical equipment in his scope of supply. The Contractor shall be responsible for the design, installation and commissioning of all cabling systems in the Facility (including Plant interconnections), together with all cabling between items of equipment in his supply. The Contractor shall make his own assessment of the electrical requirements. This is particularly relevant where low voltage loads are concerned.

The numbering and nomenclature of high voltage apparatus will be in accordance with the Plant Identification System of SEC/SA.

7.2. Scope of electrical equipment

7.2.1. Design requirements

Auxiliary power systems shall be arranged to minimize operational constraint and be capable of maintaining or rapidly restoring the full continuous power output of the generators and steam raising plant in the event of a single component or circuit failure, except for a generator unit.

The electrical distribution system shall follow a unitized philosophy but with necessary interconnections to allow for rapid restoration. LV Switchgear shall be fed from the MV/LV transformer and LV emergency switchgear and LV interconnections at the LV emergency switchgear.

Interconnectors between unit boards may be closed until the particular unit has developed sufficient voltage to be self-sustaining.

All electrical equipment shall be designed so as to withstand abnormal system voltages as defined in this Specification.

Rated voltage of electrical equipment

HV Switchyard	380 kV
Generator	18kV or as per OEM's specification
Isolated Phase Duct	18kV or as per OEM's specification
MV switchgears	13.8kV
MV auxiliaries (motors > 200 kW)	13.2kV
LV boards	480V
LV auxiliaries	460V
- Motors 0.25 kW through 200 kW	
- All motor operated valves	
UPS system 3-phase (UPS)	480V (charging supply) /120V (output)
DC system	125 VDC
Emergency Lighting	125 VAC (via 125VDC/125VAC inverters)
Protection and Switchgear Controls	125 VDC
Small Power	230V 1-phase
Motors < 0.25 kW	230V 1-phase
Cubicle Lighting	230V 1-phase
Cubicle Anti-condensation Heaters	230V 1-phase
Motor Anti-condensation Heaters	230V 1-phase
Field Instrumentation	230V 1-phase
General purpose	230V 1-phase

All AC equipment, except where otherwise specified shall be capable of continuous operation at a voltage level in the range 90% to 110% of the nominal voltage, and short-time operation at 80% of the nominal voltage during motor starting without deterioration. However auxiliary systems shall be designed such that a motor terminal voltage on start-up shall not fall below 85% of nominal voltage. Under exceptional circumstances motor terminal voltage may reach 80% provided there is sufficient margin of torque to accelerate the respective drive. DC equipment shall be capable of normal operation in the range of 110 percent to 85 percent of nominal voltage without deterioration.

DC equipment shall be capable of normal operation in the range of 110% to 85% of nominal voltage without deterioration.

During Plant Unit start up and shut down the gas turbine generator MV system shall normally be energized either from the unit auxiliary transformer or start up transformer.

The electrical equipment shall be designed to ensure continuity of operation of the main plant under all working conditions at the Plant Site. The plant shall also be capable of meeting all transient, abnormal and load shedding scenarios. In addition it shall be arranged to facilitate inspection, maintenance and repairs.

All electrical equipment shall be type and routine tested in factories. Type test will not be repeated if type test certificates of the same equipment type are available.

All reasonable precautions shall be taken in the design of equipment and of works to ensure the safety of personnel concerned with the operation and maintenance of works. Safety, isolation, locking and interlocking facilities shall comply with the relevant standards and requirements.

Electrical equipment located in hazardous areas shall be provided with special enclosures suitable for the hazard classification of the areas according to the guidelines provided in statutory regulations and codes.

The final temperature attained as a result of the passage of short circuit current shall not cause permanent damage or deterioration sufficient to reduce the normal operating characteristics below those specified.

Anti-condensation heaters shall be provided for motors above 20kW rating, motorized valve actuators, switchgear and control panels, except where located in air-conditioned rooms. Except for motors which have heaters operated by the motor controller, the other heaters shall be permanently connected without thermostats and energized via a local isolator supplied from a local distribution board.

Electric motors shall be designed for class F temperature rises, with acceptable temperature rises limited to class B over the entire range of ambient air temperatures.

Large motors of 1MW and above shall use high efficiency designs as far as possible.

Station Service and Unit Auxiliary transformers and switchgear shall be designed to allow normal operation with one transformer out of service. A change-over scheme shall allow transfer, without loss of output, between this running condition and normal system operation.

All MV and LV switchgears, DC switchgear equipment, batteries and charger, UPS inverters, distribution equipment and safe AC equipment shall be suitably housed inside fully air-conditioned containments.

Sunshades shall be provided for all cubicles, terminal boxes and enclosures mounted outdoors. OEM (Siemens) standard design only for GTG package for providing no sunshade is acceptable, if the equipment / enclosure is suitable designed for outdoor installation.

A digital fault monitoring and disturbance recording system shall also be provided, which shall cover the entire power station electrical system items, including

- incomers and couplers of all main low voltage (LV) switchgears
- all MV switchgear feeders except motor feeders (motors shall be monitored via the DCMS)
- all HV switchyard bays

All electrical protection relays shall be of the electronic digital type with testing facilities to enable testing of all functions during normal operation without imposing operational restrictions (i.e. leaving other protection functions active).

7.2.2. Generators and generator excitation systems

Generators shall be two pole, 60Hz designed, built and tested in accordance with IEC-60034. The generators shall have a high degree of reliability under steady state condition and shall be capable of confirming recovery under emergency conditions and during transient frequency fluctuations.

Generator shall have a short circuit ratio of not less than 0.5 and shall be capable of withstanding 1.3 times the rated voltage for 30 seconds during load rejection or similar events. The generator sub-transient reactance (saturated) shall be a minimum of 12% (IEC basis).

On line condition monitoring for stator winding insulation and search coil to detect shorting of rotor winding turns shall be provided.

Anti-condensation heaters shall be provided for air circuits, generator windings, excitation systems and control cubicles. Heaters shall be suitably sized to prevent condensation. Heaters shall be switched on automatically when the generators are taken out of service.

The generator neutral point shall be earthed through a distribution transformer and loading resistor to limit the earth fault current to a maximum of 10 Amps.

The following protection shall be provided and shall be duplicated to the extent so that at least two independent devices shall detect a fault.

- generator over excitation (ANSI code 24)
- phase under-voltage (ANSI code 27)

- generator over frequency (ANSI code 81O)
- generator under frequency (ANSI code 81U)
- stator differential (ANSI code 87G)
- overall differential (ANSI code 87TG), CT will be on neutral side
- neutral ground overvoltage (Stator Ground) (ANSI code 59N)
- current unbalance/Negative phase sequence (ANSI code 46)
- loss of excitation (ANSI code 40)
- reverse power/anti-motoring (ANSI code 32)
- monitoring and alarm of generator stator
- voltage transformer supervision (miniature circuit breakers will be used instead of fuses)
- rotor earth fault (ANSI code 64G)
- out of step (ANSI code 78)
- phase time overcurrent (ANSI code 51)
- phase overvoltage (ANSI code 59)
- stator ground protection (3rd Harmonic) (ANSI code 27TN)
- phase distance (ANSI code 21)
- voltage restrained time overcurrent (ANSI code 51V)
- inadvertent energisation (ANSI code 50/27)
- stator earth fault stage 1 and 2 (ANSI 64-1 and 64-2)
- breaker failure protection (ANSI 50BF)

The Contractor shall include additional protection if considered necessary.

The protective relays shall be electronic digital type with self-monitoring functions for testing during normal operating conditions without imposing operational restrictions.

The generators shall withstand the following fault and abnormal conditions without any damage to the stator, rotor, windings and other parts of the generator.

Synchronizing shall be possible with the generator circuit breaker and 380kV circuit breaker.

Synchronising with the combined conditions based on maximum individual synchronising check tolerances of:

- ± 10 degrees phase displacement
- ± 0.7 percent slip
- ± 15 percent voltage

Fire detection and protection shall be provided. Generator shall be provided with required instruments to monitor temperature, oil pressure, vibration etc. These shall be wired out to junction boxes for field connection.

Gas turbine generators and related cooling systems shall be designed to maintain the temperature rise within insulation class B limits throughout the entire operational range and throughout the ranges of ambient temperatures.

The generator stator and rotor windings shall have class F insulation system.

Contractor shall optimise the Generators and as per SFR requirement shall provide suitable air cooled, closed air water cooled or hydrogen cooled.

The generator excitation system shall be complete with a fast response dual automatic voltage regulator, standby manual channel with auto follower and automatic transfer facilities. Monitoring and fault detecting devices with over and under excitation limiters, over-fluxing protection, power system stabiliser, reactive power control, field suppression equipment, dedicated instrument transformers and other required devices shall be provided to ensure a complete and reliable excitation system.

It shall be possible to continuously vary the generator terminal voltage under no load conditions between 50% and 110%. Facilities shall be included to control down to 0% of normal excitation for the purpose of commissioning.

The excitation power shall be derived either from a permanent magnet generator or from a power potential transformer connected to the terminals of the generator or from a SEE transformer which is fed from the 13.8kV MV Switchgear. The excitation system shall be brushless or static as appropriate. It is very important that the generators and the excitation system are suitable for grid connection.

Generator shall have AGC capability with all necessary hardware. Interface details between SEC and the Plant shall be finalized during detailed engineering stage.

Features of the excitation system shall include but not limited to:

- rotor current limiters
- rotor angle limiters
- under excitation alarm
- over excitation alarm
- volt-Hertz limiter and protection
- dual automatic channel
- manual channel
- reactive power control
- power system stabilizer
- field failure detection

- rotating diode monitoring where applicable

Generator compartments should be dust free. Contractor shall ensure that positive pressure in the generator terminal enclosure (GTE) is achieved if the generator is Hydrogen filled.

The design of the system shall be such that it is possible to carry out on-load replacement, and/or adjustment of the brushes. All necessary tools and equipment shall be provided to facilitate this work.

7.2.3. Generator circuit breaker

In particular, the following codes and standards shall be applicable:

- IEEE (ANSI) c37.013
- IEC standards (especially IEC 60056, 60694, 60298, 60129, 60044-1&2, 60099-4 and 60358)

The Gas Turbine shall incorporate a three-pole isolated phase breaker for main alternator on/off switching. These breakers shall be used for synchronising the generator to the grid and shall be designed for the necessary operating conditions and interrupting rate. The generator breaker switchgear panels shall be fitted with the following elements for each pole:

- SF6 alternator breaker in series with a motorized line breaker switch, with the required interlocking devices
- motorized earthing switches, one on the generator side and the other on the transformer side, with the required interlocking devices
- voltage transformers on the transformer side
- current transformer on the transformer side (if required)
- over-voltage discharger (auto-valves) on the transformer side
- protection capacitor on the transformer side
- devices and auxiliary circuits for signals and operations, both electrical and mechanical

Each generator breaker shall include a single-pole (isolated phase) sealed casing. The breaker casing shall be prepared for welding to the isolated phase busbar conductor casings. They shall incorporate viewports to allow visual checking of moving components (isolation switches), without having to open the casing.

The connection of generator breaker system active parts to those of the isolated phase busbars shall be made by means of flexible connection parts.

The breaker shall conduct the maximum generator output current and shall be able to withstand and break the short-circuit currents considering the reestablishment voltages under the most critical operating conditions.

Generator breaker failure protection within the tripping time set in the grid code including balance of equipment tripping shall be provided.

The breaker shall not be able to close if sufficient power is not available for opening. Two independent windings shall be available for opening. One winding is sufficient for closing. It shall be possible to open and close the breaker both locally and remotely.

Voltage transformers shall be installed on the transformer side (for synchronization and protection), with the power and precision required. These transformers shall be wired to the assembly control cabinet. The necessary resistance for protection against transformer resonance shall be fitted inside with open delta secondary. Each of the voltage outputs shall be protected with breaker switches.

To protect against over-voltages and increase the breaker voltage arc extinction effect, capacitor shall be installed on transformer side.

Both sides of the line switch-breaker assembly shall be fitted with blade-type earth switches to permit maintenance work. To this end, the necessary interlocks shall be fitted to both sides (transformer and generator). These connections shall not have current switch ON/OFF capability.

If required by the gas turbine starting system, the assembly will also be fitted with a motorised breaker with remote operation possibility.

The Owner accepts omission of GCBs for 2(two) STGs subject to the Contractor satisfying the requirements as specified in Exhibit B, Part 1 of this Agreement.

7.2.3.1. CONTROL CABINET

The generator breaker system shall include a local operation and control cabinet to house the associated elements.

- control and monitoring components
- “Local-Remote” switching devices
- breaker and isolation switch operation
- wiring interconnection terminals (for local and remote signals, trips and alarms etc)

7.2.3.2. SF6 CONTROL DEVICES

The breakers shall include a densiometer for each independent SF6 gas compartment that permits gas pressure verification with acceptable maximum and minimum limit indications.

These devices shall incorporate independent voltage-free contacts for performing the following functions:

- gas low pressure/density alarm
- gas very low pressure/density interlock
- the operating pressure and density limits shall be established by the manufacturer

7.2.4. Power transformers

7.2.4.1. GENERAL

The design, construction, performance and testing of oil filled transformers shall be in accordance with IEC 60076. Transformer terminals and their provision will comply with IEC 60859.

Cooling shall be ONAN/ONAF/ODAF type for Generator Step-up Transformer (GSUT) and ONAN/ONAF type for Unit Auxiliary Transformer (UAT). The radiators may be mounted separate from the tank or mounted on the transformer tank.

Step-up transformer shall be equipped with on load tap changers. The other transformers shall be equipped with off load tap changers. The tap range, impedance and losses shall be selected to enable full output under normal conditions which allow for the highest and lowest system voltage operation and shall not be restricted over the specified ambient range.

All oil cooled transformers shall have the following as minimum:

- oil conservator
- oil level indication/alarm
- oil sampling device
- silica gel breather
- winding temperature/alarm/tripping
- pressure relief/alarm/tripping
- Buchholz gas and surge protection/tripping/alarm
- oil temperature indicator with alarm/trip contacts

Each transformer shall be fitted with a gas and surge actuated Buchholz relay of approved make and pattern, having alarm and tripping contacts which close simultaneously in response to both liquid surge and low oil level.

Approved facilities shall be provided for inspection, testing and maintenance access to the gas and oil-actuated relays and conservators associated with the transformers.

Transformers shall be located in suitable open concrete enclosures and including oil/water catchment sumps which discharge to the oily water separator drainage system.

All links contained within the transformer tank for earthing and other purposes shall be readily accessible and the links and their fixings shall be of the captive type.

Dry type transformers shall be located in transformer compartments/housings with a protection class of minimum IP23DW.

7.2.4.2. GENERATOR STEP-UP TRANSFORMERS

Generator transformers shall be provided and shown on the single line diagrams. Generator transformers shall be connected to generator using isolated phase busducts. The generator transformer and the isolated phase busduct system shall be designed to transmit the maximum output from the corresponding unit without any limitation and under all ambient conditions.

The windings will be star in H.V, connected to earth directly, and delta in M.V. with a vectorial group of YNd1.

The generator transformer impedance shall be Minimum 14% (IEC basis).

Generator step up transformers shall be equipped with tap changers to be on load type. On load tap changer (OLTC) shall be provided as required by the Saudi Arabian Grid Code setting range shall be a minimum $\pm 10 \times 1.1\%$ of rated voltage.

The OLTC shall be in conformity with IEC 60214 (or equivalent ANSI standard). OLTC shall have been type tested, and related type test certificates shall be supplied with the offer. Only designs which have been type tested in accordance with the relevant IEC standards will be accepted.

All equipment related to the OLTC shall be supplied by the original OLTC manufacturer. This is also applicable for tie-in resistors, if provided, as well as all related accessories. License products etc. are not acceptable for any equipment related to OLTC.

The OLTC shall withstand all kinds of through-fault currents without damage. All OLTC connected to uniformly insulated windings shall have at least same LI and AC voltage withstand levels as the related windings. The OLTC tank should be designed that the OLTC can be removed/replaced without removal of transformer windings/core and top cover of the main tank.

The motor drive, plus all auxiliary equipment for operation of the tap changer, shall be incorporated in a rigid control cabinet, protection class IP56 and shall be mounted onto the transformer tank in a convenient floor height. The driving gear shall be of the belt-type or equivalent dry-type gear. Oil filled driving gears are not acceptable. The electrical supply for the control of the cooling units and for motor drive of OLTC shall be provided by means of two independent feeders (load-break switches) installed in the control cabinet.

To cover the event of an electrical supply failure, automatic switchover from the first feeder to the second feeder is required.

Automatic transfer equipment shall include a time delay relay to prevent immediate transfer from normal to emergency source.

Step-Up Generator Transformer shall be provided with the following minimum electrical protections:

- restricted E/F protection (ANSI code 51TN)
- differential protection (ANSO code 87T)
- overcurrent protection (ANSI code 50/51)
- ground O/C protection (ANSI code 51G)
- overfluxing protection (external) ANSI 24

All transformers to be equipped with of the appropriate class of protection / metering CTs, as required.

7.2.4.3. UNIT AUXILIARY TRANSFORMERS (UAT)

UAT shall be connected to the generator and step-up transformer through the isolated phase busduct system. UAT shall be sized to load the entire electrical loads for it's own unit normal operating allowing the biggest motor start-up and inside permissible voltage range, and during unit start up.

The windings will be delta in HV and star in MV, connected to earth by resistance, with a vectorial group of Dyn11. The HV winding will be provided of a manual tap changer with five positions (off-circuit tap changers, the tappings and range to be +/- 2 x 2.5% of the rated voltage.)

The Unit auxiliary transformers shall be designed for all the ambient conditions at site.

The UAT rated power and impedance should be justified with the electrical calculations.

UAT shall be provided with the following minimum electrical protections:

- differential protection (ANSO code 87T)
- overcurrent protection (ANSI code 50/51) at 18kV side
- ground O/C protection (ANSI code 51G)
- to protect IPB during generator start-up (breaker open), ground detector relay should be included (ANSI code 64) connected to open delta VT on GCB

All transformers to be equipped with the appropriate class of protection / metering CTs, as required.

7.2.4.4. MV/LV TRANSFORMERS

The Contractor shall suitably size ML/LV transformers taking into account power requirements of the Plant.

Main MV/LV transformers should be sized to feed all the main LV switchboard (normal operation with bus tie closed and the other incomer open). MV/LV transformers shall be sized to load the entire electrical loads for it's own unit during normal operating allowing the start-up of the biggest motor and inside permissible voltage range.

Connection between MV/LV transformers and LV switchboards should be through LV copper power busbars.

The numbers and ratings of the various auxiliary transformers required around the plant are to be determined by the Contractor based on his load estimates for each service or building.

The LV transformers have to be designed as oil immersed, three-phase power transformer for outdoor installation. The dry type can be accepted only for indoor installations; however as an exception only for GTG package, outdoor dry type cast coil rated transformers supplied by GTG OEM can be accepted subject to satisfying Exhibit B, Part 1 requirements.

The transformer neutral point on the low voltage winding shall be solidly grounded.

These transformers will step down from the MV to LV and be fitted with off-circuit tap changing facilities of range $\pm 2 \times 2.5\%$ steps

7.2.5. MV system neutral earthing resistors

Where neutral earthing resistors are installed, they shall be of the metal grid type rated for 10 seconds, and bonded to the station earthing system. Each resistor shall be enclosed in a naturally ventilated protected weatherproof cubicle located outdoors alongside their respective equipment.

7.2.6. MV and LV AC switchboards

All main switchboards, MCC, relay panels, control equipment etc. shall be provided with redundant power supplies for control and alarm purposes.

A two position (Local/Remote i.e. Switchgear/CCR) selector switch shall be provided at the switchgear for all switchgear incomer, bus-section, feeder and interconnector circuits respectively for switchgear emergency control and isolated testing/CCR remote control.

Check-synchronising facilities shall be included for the main incomers of the MV systems (at least two different sources). MV transfer between UAT incomer and start-up incomer shall be high speed transfer with short parallel.

The LV switchgear system main incomers, interconnectors and bus-section circuits shall exclude check-synchronising facilities but shall be electrically interlocked to prevent the paralleling of two incoming supplies to a switchboard.

Circuit breakers supplied shall be identical to those used for the type tests for which certificates are submitted.

MV switchgear and all main LV switchgear shall be of withdrawable type. Fault levels and thermal withstand time for MV and LV switchboards shall be in accordance with the technical requirements of the appropriate standards.

7.2.6.1. MV SWITCHGEAR

MV switchgear and control gear shall comply with IEC 60298 and shall be metal-clad.

MV circuit breakers shall comply with IEC 60056 and shall be of withdrawable three pole circuit breakers, using vacuum interrupters type.

The MV switchgears shall be designed for a nominal voltage of 13.8 kV, to be internal arc certified IAC=AFRL according IEC 62271-200 and to withstand a short-circuit current of minimum 10% higher than required by calculations for a duration of three (3) seconds.

The mechanical protection class for all MV switchgears shall be minimum IP41.

Where operation of a switching device can give rise to surge voltages which would be detrimental to the insulation of the equipment controlled, surge suppressers shall be installed in the switchgear to control these voltages to a safe level.

7.2.6.2. LV SWITCHGEAR

LV circuit breakers shall comply with IEC 60947-2. Withdrawable air circuit breakers or moulded case circuit breakers shall be used, unless otherwise approved.

LV switchgear and control gear shall comply with the relevant IEC and/or NEMA standards. It shall be metal-clad and generally of multi-cubicle of Form 4 construction and Form 3b for MCC cubicles.

Panels for different feeder circuits shall be appropriately equipped with control, metering, alarm and protection devices based on Good Industry Practices and current technology.

The construction shall be such that the various components of the switchgear are segregated electrically from each other.

Moulded case circuit-breakers and miniature circuit-breakers may be used if they are properly selected to withstand the maximum specified short-circuit current level and evidence of acceptable performance is provided by the Contractor.

LV contactors used as motor starters shall be of the air-break type with arc shields and to class AC3/AC4 according to IEC standards. It is acceptable for the GTG OEM only to offer 480V AC MCC built according to NEMA standards provided IEC standard requirements are satisfied under NEMA standards.

An earthing bar with a minimum cross-section of 240mm² shall run the full length of the switchboards. The Contractor will connect to this bar all metallic parts not forming part of the live circuits and all CT points to be earthed. Chassis of circuit-breakers shall be directly connected to the earthing bar.

Every low voltage switchboards shall have spare columns with equipped reserve (two breakers) and empty space (for future necessities a 20% free space of reserve must be included)

7.2.7. Busbar and busduct systems

Phase isolated busducts with aluminium enclosures shall be used for the connections between the turbine generators and generator transformer and for unit auxiliary transformer connection. These busducts shall be of pressurised with instrument air (clean and dry) to exclude dust and moisture. For connections between MV switchgear and MV/LV transformers cables shall be provided.

The outer enclosure shall be of the self-supporting type and kept at a slight overpressure to avoid ingress of humidity. The enclosure shall be earthed and fitted with drain plugs at the lowest points.

The generated heat shall be dissipated by natural radiation and convection. No forced air-cooling shall be provided.

The allowable temperatures under full-load operation at site conditions shall be as per the relevant IEC/ANSI Standards

Generator and MV (as applicable) busducts shall be pressurised with instrument air (cleaned and dried air), anti-condensation heaters are not necessary.

Outdoor section of busducts shall be provided with sun shades. Complete steel support structures shall be provided. Busducts shall be terminated using flexible busbars and rubber/neoprene bellows.

7.2.8. Electrical system control, indications and alarms

The Contractor will be responsible for the design, manufacture, supply, installation, testing and commissioning of control, monitoring and protection systems associated with his electrical installation. Control and alarm facilities for the power plant electrical MV, LV and DC systems shall be provided.

The scope of supply shall comprise all necessary control, protection, monitoring and supervisory equipment, cabling and wiring, regardless of whether it is specifically mentioned or not, for safe and efficient operation of the Plant. The control and monitoring of the electrical system and other associated common systems shall include at least the following:

- control and monitoring of MV system via ECMS or Plant DCMS and Digital fault Monitoring System
- control and monitoring of auxiliary power distribution (LV, DC, UPS) via Plant DCMS and Digital fault Monitoring System
- energy meters via Plant DCMS
- alarm facia/windows to be provided on Electrical Panels and Fire Protection Panels
- auxiliary and marshalling cubicles via plant DCMS

MV and LV AC electrical systems (excluding LV sub-distribution systems) shall be monitored and controlled from the CCR including for all transformers, switchgear incomers, bus-section, feeders and interconnector circuits via the plant DCMS. Manual and auto synchronising facilities shall be included for these circuits. The voltage of each switchboard and single phase current of all main circuits (incomers, bus-sections, feeders and interconnectors) shall be displayed locally at each switchgear, and for motors of 55kW and above repeated in the central control room via the plant DCMS.

Motor circuits shall normally be controlled and monitored from the central control room via the plant DCMS, except for station minor plant needing only local control. For motors of 50kW and above the single phase current of each motor circuit shall be displayed locally at each switchboard.

Alternative locations of control of switchgear and motors shall be at the switchgear (under Permit to Work for operation or testing) or plant (for operation or maintenance).

LV sub-distribution systems and the DC systems shall be locally controlled only, but alarms shall be transmitted to the central control room via the plant DCMS.

The switchgear shall include all related circuit prominent coloured LED type status and alarm lamps. The associated protection relays shall also include LED status/alarm indicators or flags.

Communication between the plant DCMS and ESF including the switchyard signal shall be done via a SEC/SA approved method such as a serial link using dual redundant MODBUS protocol for monitoring major switchyard and electrical signals from plant DCMS.

Sequence of Events (SOE), GT and ST operation signals shall be through hardwire interface to the DCMS.

7.2.9. Low voltage AC and DC sub-distribution boards

All distribution boards shall comply with IEC 60439, IEC 60937 and other related IEC standards and, for the enclosure, comply with IEC 60529. They shall be of Form 2 construction as a minimum.

All distribution boards situated outside shall be of the weatherproof enclosure type IPW54. They shall be capable of withstanding the associated fault current until the related protection operates.

The AC sub-distribution boards shall be either single phase or three phase and neutral types, and the DC sub-distribution boards shall be of the two-pole type. All distribution boards shall be rated for the full load current of the incoming supply and equipped with an incomer moulded case circuit breaker (MCCB) isolator and with removable miniature circuit breakers (MCBs) to provide overcurrent protection to each sub circuit.

7.2.10. Electric motors

All motors shall be designed and manufactured in accordance with IEC 60034 and other related internationally recognised standards. Motors in hazardous areas shall be suitably certified.

NEMA is accepted, but Contractor shall ensure:

- the starting current of MV motors does not exceed 5.5 times the rated current
- the starting current for LV motors does not exceed 6 times the rated current
- and prove during detailed engineering that there is no disturbance to the electrical system due to LV motors start ups

Motors of the same type must be fully interchangeable and shall comply as far as applicable with IEC standard motor dimensions. AC motors shall be of the squirrel cage induction type provided with either deep slots or double squirrel cages.

All motors shall normally be of the totally enclosed fan-cooled type or water-cooled type, of protection class IPW54 according to IEC recommendations 60034 and 60529. Where motors are installed outdoors a weatherproof design (minimum IPW55) shall be chosen. Motorised valves shall be to protection class IP65.

Motors shall have Class F insulation but the temperature rise shall not exceed Class B.

DC motors and starters shall be designed for high reliability with direct-on-line starting using a permanently connected series resistance in the armature. Service voltage for GTG OEM DC motors can be 125 VDC.

Emergency stop buttons of the stay-put key-reset red button mushroom head type shall be provided adjacent to all MV motors and all lubricating oil pumps. These buttons shall directly trip the motors, prevent a start until reset and alarm their operation to the central control room. The buttons shall all use a common reset key. This requirement may be relaxed for certain essential loads in the GT package as stopping of lube oil to the generator and turbine will severely damage those units.

The service voltages of the motors will be:

- MV, 3-phase AC, 60Hz, for motors above about 200kW
- 460V, 3-phase AC, 60Hz, for motors below and including about 200kW
Calculations of the voltage drop at start-up shall be made for each motor to additionally verify the capability to cross line starting

The rating of each motor shall be adequate to meet the requirements of its associated load, without infringing upon the service factor as per NEMA and/or ANSI standards

Motors shall be capable of operating continuously under rated output conditions at any frequency between 58 and 62 Hz and/or with any voltage deviation between 90% and 110% of the nominal voltage. A transient overvoltage of 130% of the nominal voltage shall also be sustainable. Further, the motors shall be capable of delivering the maximum torque as required by the driven machine when running at 80% nominal voltage for a period of 5 seconds without damaging, overheating or slipping.

The pullout torque for continuously loaded motors shall be at least 160% of the rated motor torque and for intermittently loaded motors 200% of the rated motor torque.

Fixed speed AC motors shall be designed for direct on-line starting.

The number of starts in one hour for all motors shall be in compliance with relevant standards and shall be determined during the detailed design stage. At the end of the determined number of starts the temperature of the insulated windings shall not exceed the maximum permitted temperature for class B insulation, and the cage windings shall not suffer damaging temperature rise.

Each MV motor shall be capable of enduring two successive starts with the motor initially at operating temperature and the temperature of the winding at the end of start shall not exceed the Class B limit. Each LV motor shall be capable of withstanding three successive starts under the same conditions or once every twenty minutes without damaging temperature rise.

For the start-up of electrical drives of chillers should not perturb the electrical system, therefore. Contractor shall stagger and optimise the chiller start up process and its configuration adequately accordingly.

7.2.11. DC supply systems and un-interruptible power supplies

The Contractor shall suitably size DC and UPS systems taking into account power requirements of the Plant. The Contractor will supply UPS and DC systems to meet all the site DC requirements consisting of at least two (2) x 100% battery chargers, one (1) x 100% Nickel cadmium batteries (use of lead acid batteries (VRLA type) instead of Ni-Cd batteries is subject to approval during design stage), DC distribution panels, two (2) x 100% AC Inverters, and AC Distribution panel for UPS consisting of two coupled bus sections. The 120V Safe AC system shall consist of 2 x 100% invertors each equipped with a static by pass switch and a related safe AC distribution consisting of two coupled bus sections. UPS power for the gas turbines shall be provided from the Plant UPS. AC un-interruptible power supplies are to be provided for equipment requiring continuity of supply or equipment that need supply during emergency operation, when the normal power supply is lost. The rated capacity of the batteries shall be adequate for discharge duration of 1 hour for the emergency consumers. The UPS system shall be also be provided with a bypass static switch, autonomy will be 1 hour.

DC nominal voltage level should be 125VDC and UPS nominal voltage level should be 120V. The chargers shall be designed for operation from the 480V, 60Hz supply. Each charger shall be rated to ensure simultaneously battery charging and the supply of the full DC load. Under normal operating conditions, charging of a fully discharged battery will take a maximum of 10 hours. Each battery charger shall be capable of fully charging the battery on initial installation. Battery chargers shall have facilities for on-line maintenance.

7.2.12. Cables and installation

The Contractor shall undertake all relevant design and engineering of the cable systems for equipment in his supply, and shall prepare the cable connection schematic diagrams, terminal diagrams, cable lists, cable schedule and the cable route plans. All cables shall comply with the relevant IEC standards. It is acceptable for the GTG OEM only to offer cables according to NEMA standards provided IEC standard requirements are satisfied under NEMA standards.

All power cables shall be of cross-linked polyethylene (XLPE) and shall be capable of continuous operation at the highest system voltage specified with a maximum conductor temperature of 90°C, under rated conditions and a maximum temperature under fault conditions of not more than 250°C. Use of PVC cables for LV duty shall be subject to approval.

All power and control cables capacity sizing shall be in accordance with the temperatures given in Exhibit B, Part 1. Cables installed on cable trays that may be exposed to sunlight during operation shall have ultra violet safe insulation.

Control cables shall be multicore, PVC insulated types withstanding without deterioration the conditions prevailing at each installation location. Cables for analogue signals shall have a common screen of metal tape, cores shall be twisted by pairs, for process computers, each pair shall be separately screened. Control and instrumentation cables shall be armoured for outdoor applications and all cables shall be shielded, with a minimum conductor section of 0.5mm², alternatively un-armoured cables for control and instrumentation is acceptable for outdoor application if laid on conduit, trench, duct bank, tray, ladder (i.e., with suitable protection). Cables above ground shall be laid in trays or on ladder racks or conduits and shall be segregated from power cables in accordance with International standards. Cables shall be labelled and conductors shall be identified at terminations by ferrule number or suitable proprietary method.

Every cable end of MV/LV power cables shall be sealed and protected in an approved manner consistent with the voltage levels, cable type, and place of installation.

"Plug-in" cables terminations (if supplied) shall be in accordance with the characteristics of the MV switchgears.

MV and LV cables are generally to be laid in cable trays (inside buildings) and in cable trenches and/or cable trays (outside buildings). Any cables direct buried shall be armoured.

Cables running inside buildings or concrete trenches shall be laid on trays. The trays shall be of adequate strength and size to carry in one layer, the weight, number and configuration required for each tray plus additional cables, not less than 25% by weight and dimensions, in excess of those installed.

Segregation of the various services and voltage levels shall be achieved by use of separate trays and marshalling boxes (when required) separate for each voltage grade of cable used. .

For the data highway communication link, spatially separated routing with the cables running in galvanised steel conduits of approved type shall be supplied and installed.

All power, control and instrumentation cables shall be Flame Retardant Low Smoke type and shall comply with IEC-332 Part 3.

Wherever practical, all electrical/instrumentation circuit shall be routed above grade via a cable tray. For the outdoor areas, underground duct banks and/or cable trenches shall be used where exposed raceway is not practical.

Additionally, Contractor shall consider the required MV and LV cables for the power plants interconnection, all the necessary works will be in Contractor scope (required cable trenches, cable supporting systems etc.) to carry out the interconnections.

7.2.12.1. CABLE TRAYS

Cable trays shall be used to support cables.

Heavy duty, hot dipped galvanized and epoxy coated, ladder type with channel side cable trays shall be used for outdoor cable installations.

Within Indoor areas, tray covers shall be used where physical protection of cable is needed. For outdoor installation, tray covers will be used on the highest level to protect cables from direct sunlight and on other levels to protect from water etc. as required.

Cable tray shall be segregated, top to down, as follows:

- medium voltage power circuits
- low voltage AC power circuits not over than 600VAC
- 230VAC/125VDC control circuits
- analogue low level signal or other analogue signal and 24VDC signal from instrumentation/other devices
- fibre optical cable (as long as it is proven by Contractor that no interference will occur providing signal cable and fibre optical cable in same trays are acceptable)

Low level control cable may be run in the low voltage power cable tray or 230 VAC/125VDC control cable tray when separated by a continuous barrier to prevent mixing.

In a vertical configuration, low voltage power cable trays shall be located below medium voltage power cable trays; control cable trays shall be located below low voltage power cable trays; instrumentation cable trays will be located below control cable trays.

In general, average maintained minimum vertical spacing for cable tray shall be 300mm, measured from the bottom of the upper tray to the top of the lower tray.

At least a 200mm clearance shall be maintained between the top of a tray and beams, piping, etc., to facilitate installation of cables in tray.

The cross member or lung of cable tray shall be spaced maximum of 300mm on centers.

Complete support system for the cable trays including all hangers, braces, brackets, inserts, anchors, fabricated structures, hardware and incidental materials utilizing unistrut type or equal supports and hangers wherever possible shall be provided.

7.2.12.2. RIGID METAL CONDUIT

In general, hot dipped galvanized, rigid steel conduit shall be used for exposed conduit. The minimum size for power and lighting circuits shall be 22mm, and for communication and other circuits will be 16mm.

The conduit shall be securely fastened to the cable tray with clamp-type fasteners, and the end of the conduit shall be equipped with an insulated conduit grounding bushing.

Solid insulated bushings shall be provided for all conduit 42mm and smaller when not terminated in a box.

For rigid conduits enter sheet metal enclosure, threaded hubs shall be used.

When conduit passes through concrete slabs, galvanized steel pipe which extended 150mm above the surface of slab shall be provided. Conduit should be made of galvanized steel and should extend 150mm when passing through concrete. Contractor can use PVC to cross concrete slabs, and galvanized steel pipe can pass through this PVC pipe. It is mandatory to install galvanized steel pipes.

Complete galvanized support system including all hangers, brackets, inserts, anchors, steel structures and hardware shall be provided.

Unistrut type or equal conduit hangers and supports shall be used.

Conduit body shall not be used for grounding purposes.

7.2.12.3. LIQUID TIGHT FLEXIBLE METAL CONDUIT

Liquid tight flexible conduit will be used between motor terminal boxes, or other equipment subject to vibration and/or mechanical adjustment and the rigid conduit systems.

Liquid tight flexible conduit to be used will be a type fabricated of galvanized steel conduit flexible tubing with a synthetic jacket extruded over the tubing.

The length of this conduit is not less than twelve times the conduit diameter (trade size), minimum 300mm.

7.2.12.4. DUCT BANKS AND MANHOLES

In general, all underground raceway shall use heavy wall high density polyvinyl chloride conduit or hot-dipped galvanized steel conduit encased in concrete.

Where practicable, duct size shall be standardized and minimum may be 75mm .

Duct banks will be separated and supported by using non-magnetic materials to keep 75mm minimum spacing between the outer surfaces of the conduits.

All duct banks shall be provided with a reinforced concrete encasement. Minimum of 75mm concrete covering on sides, top and bottom will be provided. No waterproofing of the encasement will be considered.

The top of the duct bank encasement shall keep a minimum distance of 300mm below finished grade where applicable.

Ground wire that follows routes of duct banks may be laid directly on top or side of duct bank encasements.

Manholes shall be provided in underground duct runs so that duct lengths will not exceed those which will develop pulling tensions greater than allowed by the cable manufacturers. The maximum length of a duct bank between manholes (pulling points) will not exceed 90m. Pulling eyes will be installed on walls of manholes.

Manholes will be of the precast or cast in situ type.

Floors of manholes shall have slope to a sump in one corner or a drain pipe to the drainage system.

Where a duct bank turns upward for continuation above grade, the last 2.4m (8 feet) of the underground portion may include a transition to heavy-wall hot-dipped galvanized steel conduit and terminate approximately 150 mm above grade. If a bend exceeds 2.7m then manholes shall be provided.

7.2.13. Communications / telecommunications

7.2.13.1. OFFICE TELEPHONE AND LAN SYSTEM

The Contractor shall provide and install a complete office telephone and LAN system for the administration building, workshop and stores building and gatehouse comprising a main distribution frame, switches, routers, a full site cabling network including underground ducts, joint boxes, external and internal cabling, telephone/data sockets, telephone handsets and all other appropriate equipment.

The Office LAN shall be designed to operate at 1 Gbit/s (1000 Base-x) Ethernet. The buildings shall be connected by ring mains which end in data distribution boxes. The optical fibres shall be equipped with the pigtailed via ST-connectors which are connected to the patch bay.

The components of the Office LAN system to be provided by the Contractor shall include patch bays, cubicle for patch bays, LAN- board, data sockets, data cable cat 7, fibre optical cable, testing of the cables including test protocol, switches and routers. The virtual PBX and LAN servers are to be provided by others.

The Office telephone system shall be compatible with the requirements of local telecom company for a virtual IP PBX.

The Office LAN system shall be compatible with the requirements of the Owner's LAN servers.

7.2.13.2. INTERNAL TELEPHONE SYSTEM

The Contractor shall provide and install a complete internal telephone system for the site comprising PABX, main distribution frame, a full site cabling network including underground ducts, joint boxes, external and internal cabling, telephone sockets, telephone handsets and all other appropriate equipment.

The PABX shall be state of the art and shall have an equipped capacity for 175 extensions, extendible to 100 without major hardware or software changes.

The site telephone cabling provided shall be capable of supporting clear voice communications.

Telephone instruments shall be provided in all site strategic areas, with two telephones in the administration building and one telephone in the main gatehouse. These instruments shall be provided to suit the use and location and shall include basic wall and desk mounted units, background noise suppressed units with acoustic hoods for noisy plant areas and weatherproof units for external areas.

The Work shall include the design, supply, factory testing, delivery, installation, site testing and commissioning of the telephone system. The provision shall include full system documentation, including handbooks and drawings, spares, consumables, tools and test equipment.

7.2.13.3. SIREN SYSTEM

A station siren system shall provide a sound level to a minimum of 5dB above ambient levels throughout the site.

Consideration shall be given to the type of loudspeakers required to be suitable for the environment to which they will be subjected.

Wiring to the sirens shall generally be by means of PVC multicore armoured cables clipped to cable trays. Consideration shall be given to separation from other services.

7.2.13.4. CCTV

CCTV requirements for security shall be provided as per SEC/SA standards specified in Exhibit B, Part 1 and regulatory requirements. Process monitoring is not required unless specifically requested in SEC / SA standards specified in Exhibit B, Part 1 or in regulatory requirements.

7.2.14. Lighting requirements

Lighting requirements are defined in the following table.

Location	Service Illumination in LUX
Control & Relay rooms, desks and control surfaces (controllable lighting levels) - GLARE Index of 16 to be applied in the CCR.	400*
Offices, laboratories and First Aid Rooms	300
Workshops (General)	300
Gate House & Rest Rooms	300
Computer & Electronic Equipment Rooms	400
Switchrooms & relay rooms	300
Turbine Hall (general/operating station)	100/200
Building corridors & Stairways	200
Toilets	150
All indoor areas not specifically listed	200
Transformer bays & Boiler areas	100
Water treatment plant	200
Cableways and cable basements	50
Outdoor areas not specifically listed	50
Roadways vehicle yards & car parks	30
Site perimeter fences	20
Area between HRSGs	30
Access ladders, stairways & platforms	50

7.2.15. Earthing system design, bonding and lightning protection

The scope of supply shall include all required copper conductors, and earth rods, connecting material, and all attachment material to make up a complete, safe and reliable system. All electrical equipment such as motors, transformers, switchboards, control boards, relay and auxiliary relay boards, and all other subsidiary electrical equipment, as well as all metal parts of the civil structure or the mechanical equipment which might be subject to earth fault currents shall be connected to this network.

The Contractor will provide a potential gradient control system for the switchgear rooms and all such locations where dangerous surface voltage gradients may occur.

Each building roof or civil structure protective cover shall be provided with a lightning protection system composed of roof collectors and down conductors of adequate dimensions and spacing connected to the earthing electrodes.

Metallic structures or equipment housings located on the roofs of buildings shall be connected to the lightning protection conductors, with the equipment earthed via the earthing network.

The lightning protection system shall be bonded to the Plant's earthing system and shall be according to the applicable NFPA and/or other internationally recognised standards.

It shall be the responsibility of Contractor to ensure that proper and safe earthing is provided for the plant; Earthing calculations should be carried out (dissipate electric currents into the earth and no danger to electrical shock - touch and step voltages values).

7.3. HV sub-station (Electrical Special Facilities)

Contractor shall comply with the requirements of Volume 2 of Exhibit B, Part 1.

8. INSTRUMENTATION AND CONTROL / AUTOMATION TECHNICAL REQUIREMENTS

8.1. Degree of automation

All the processes shall be completely automated and they shall be controlled from the main control room.

The control and automation systems of the Plant shall be designed and realised using a proven, state of the art modern control system and computer technology as specified in Exhibit B, Part 1 and shall also fulfil the following requirements:

- the plant can be run efficiently and safely
- the plant shall be operated in the most cost effective manner
- the Plant operation is to be based on Central Control Room (CCR) principle
- the Plant shall be designed with a view to high reliability and availability with a minimum of operator intervention being necessary during the normal operation
- all the normal operations including start-ups and shutdown (exception can be given after long term outages) of single GT/HRSGs or whole Plant shall be performed by the operator in the permanently manned CCR without any dedicated field operations

The major Plant components such as gas turbines can have their own control systems with interfaces to the Distributed Control and Monitoring system (DCMS) so that it is fully transparent to the operator i.e. operator interface is identical irrespective of actual control system used. However, number of control systems interfaced to the main DCMS (sub-control systems) is to be kept at a minimum. All these sub-systems shall be operated through the main DCMS.

There may be separate local control rooms where appropriate sub-processes can be monitored as may be necessary time-to-time for maintenance purposes but not permanently and all necessary data and information and control shall be managed from the Central Control Room (CCR).

The Plant automation and operating concept shall be designed to enable fully automated normal operation (including start-ups and shut-downs) with two operators carrying out all control and monitoring tasks in normal operation in the main control room.

8.1.1. Software security management

In the event of damage to a contract software package or module, a back-up copy shall be readily available to reinstate the system.

Safeguards shall be included to protect Owner against the eventuality of the Contractor or any sub-contractor and/or any system software authors and/or any suppliers ceasing to exist, or ceasing to be able to support the software for any part of this Agreement. The Contractor shall include for the provision of an escrow agreement if required by Owner.

The Contractor shall carry out the following:-

- classification and listing of all contract software
- provision of the back-up software required in this Specification
- the placing of the back-up software into secure repositories
- the establishment of software upgrade services
- the provision of coded listings of software where appropriate
- maintaining back-up copies of all software on site while project is active
- providing details of Ownership, licensee, for all software supplied
- providing details of proposed escrow agent and proforma agreement

8.1.2. Lifetime system maintenance support

It is a requirement of this Contract that system support for all control systems proposed shall be available for the lifetime of the plant. Such support is to include spare parts and engineering, via an office established in the Middle East region close to the country of installation of the plant Man Machine Interface (MMI)

Functional keyboards shall be used for the Plant operation, while for all system service and management functions the ordinary alphanumeric keyboards may be used.

Free configuration graphic pages shall be available for the animated display of the Plant flow diagram, with the indication of the control parts and measurement points. These graphic displays shall be designed considering operating and monitoring of processes without having too many shifts of displays during critical phases (like start-up of the Plant).

A general summary page of all the alarms, organised in sub-pages and selectable according to priority levels, shall be available. The subdivision according to priority shall meet the requirement that the operator is warned with alarm signals of higher priority, requesting his intervention and his attention is not diverted by less priority signals not requesting any specific intervention or can be examined with longer times.

The graphic representation of the trends of freely selectable analogue variables (minimum 6 at the same time) shall be possible. The time basis could be freely selectable.

It shall be possible to obtain on a colour printer the status reports and the history of variables measured and/or calculated values, operator's actions, alarms, etc. from the operator's console.

A laser jet colour printer shall be provided for printing copies of the HMI video displays.

The automation systems of independent auxiliary plants and other minor systems dedicated to special supervision functions that can foresee local control panels, shall however supply the alarms and other important information of their plants on the man machine interface (MMI) in the control room. This information shall be in the same form as much as possible in all places of operation.

Information for the operator shall be available, such as:

- calculation of the Plant effectiveness
- mass and power balance
- storage and display of rating trends of the Plant in particular conditions
- storage of the Plant operating data, for historic file
- event chronological recording, for printing in case of malfunction

Graphic displays could contain the following animated and interactive representations:

- plant mimics, highlighting the Auto/Manual status and the position of the controlled parts, the controlled variables, the variables that can be modified, etc.
- control loop diagrams of complicated loops
- automation sequences, highlighting the progress status
- control stations to be used in particular operation conditions
- operating point displays: shall include, on an appropriate reference system in engineering units, the limit curves (with dynamic updating possibility) and the current operation point
- for each controlled part, the operation status and the presence of possible failures shall be highlighted, that could be distinguished through colour and blink attributes

OPC shall be used as common protocol for data exchange to office and plant systems (such as plant management systems, MMS, inventory and procurement systems, document management systems, etc.).

8.1.3. Engineer's programming terminal

Separate Engineering programming terminals shall be provided for the DCMS, GT, ST and other control systems to permit program modification, debugging etc., on-line. Operational utilities shall be provided to support data input-output, data storage, monitors and printers which shall be provided for the program development facilities.

It shall be possible to over-ride actual inputs with forced values for all points in the plant database. Points so forced shall be tagged to that effect. It shall be possible to develop and debug new application software packages allied to the process. This facility shall be accommodated in background mode. Debugging facilities shall permit simple and effective debugging of application programs.

8.2. Automation System concept

Automation system concept for the Plant shall consist of:

- main control system (DCMS)
- protection System (SRS)
- separate control systems (if necessary) for gas turbine, steam turbine, water treatment plant, etc. (PLCs)
- interface to other systems outside power plant
- control room devices for MMI (Man Machine Interface)
- field instrumentation and data acquisition
- CEMS

The Plant shall be controlled and monitored via a Distributed Control System (DCMS). The usage of separate programmable logic controllers (PLCs) or other separate and specialised control systems shall be minimized. Typical and acceptable reasons for a specialised control system can be high speed of response as typically required at gas turbine and steam turbine control, or special safety requirements, which may require a separate safety-related system (e.g. for boiler protection).

If specialised separate control systems or pre-packaged PLCs are supplied, they shall be monitored and operated solely by the operator workstations of the DCMS. The interface to the DCMS shall be either of conventional hardwired type (small amount of signals) or of a well-known communication data link type (e.g. Modbus or Profibus DP).

The cabinets containing the control system electronics shall be placed in a room in the control building. Electronic cabinets dedicated to signal acquisition (remote I/O- cabinets) could be installed at field. For all the equipment not installed in air-conditioned environment, provisions shall be made for an adequate protection degree (IP 55 minimum).

All indoor I&C cabinets installed in air conditioned environment shall be of min. IP 42 protection (IP 41 acceptable for cabinets located far from any source of water dripping – doors, windows, etc)

The design of the automation system shall generally comply with IEC standards.

8.3. Distributed Control and Monitoring System

8.3.1. General requirements

Control and monitoring of main processes shall be carried out with the Distributed Control and System (DCMS). The DCMS shall take care of all automation functions, such as:

- open loop drive controls, group controls, and sequence controls
- equipment protection of individual drives (pumps, valves, etc.)
- closed loop drive controls and co-ordinating controls
- man-machine interface as process displays and trend reports
- alarm annunciation
- self-diagnostics
- process interface and power supply for field equipment, signal conditioning
- interface to external systems as PLCs, computers, local area network (LAN), etc
- back-up functions
- engineering functions

The availability and reliability level of the DCMS shall not compromise the one of the Plant. For this purpose, adequate redundancy of processors, communication buses, supply devices, etc. shall be used. The following requirements shall be taken into consideration by the Contractor:

- all processes can be controlled and monitored via any operator terminal (display capacity shall be high enough)
- however, each operator terminal can be authorised to carry out the control and monitoring only of the specified processes (authorisation shall be possible)

- should an operator station, terminal, or monitor be out of operation, all functions can be carried out via other stations, terminals and monitors (each disturbed piece of equipment is replaced by a working one)
- system buses are mainly redundant
- process control stations are in general redundant
- the concentration of signals per I/O card, depending of the type of signal/cards, shall not exceed 32 channels per card and the single signals shall be isolated among them and from the field
- in the case of changing of redundant devices (input/output I/O cards, power supplies, CPU,...) or in case of partial power failure of the system for maintenance during operation, no changes to the previous conditions of the controlled plant shall be made. In general, the replacement of failed components shall be possible when the system is in operation. The systems shall enable the identification of malfunctions at least at module level, through detailed video pages and prints
- the organisation of the different units forming the system (control units, data acquisition units, computerised operation stations etc.) shall be such not to spread out a possible failure to the other units
- a single failure of a redundant component does not cause a spurious intervention of the protection system nor make it ineffective and, in any case, the failure of a single component does not cause the protection system to become ineffective
- when redundant field sensors are foreseen (duplicated or triplicate), each of them shall be acquired by an input card different from that of the other redundant sensors
- the processing of the main protections and critical trips shall be made according to 2 out of 3 logic
- there are gateway stations and appropriate protocols to connect the system to external systems and Ethernet network
- back-up function stores application programs and reloads them automatically into the stations after a power supply failure
- engineering tools to allow the user to make changes, maintenance, tests of the programs and extensions to the system are included in the delivery

The Distributed Control Systems (DCMS) and Instrumentation systems shall be designed and furnished to achieve the automatic safe and efficient operation of the Plant and equipment provided by the Contractor.

DCMS Plant interfaces shall be as per Section 2.4 of Exhibit B-1, Part I, MFS of this Agreement.

With respect to the Load Dispatch Center, the control system of the Power Units shall be equipped with automatic power generation control capability (AGC) for automatic and manual regulation of the real and reactive power output of the Plant and shall be capable of remote AGC operation by LDC

The DCMS has to provide all the required man-machine and machine-machine interfaces for monitoring and control of the Plant.

All Plant equipment and system controls shall be achieved by Distributed Control System (DCMS) stations, local control systems which interface with DCMS, stand-alone control systems and hardwired circuits. With respect to local control systems associated to GTs and ST, the HMI station enabling a local control (e.g. screens in the PCC) will not be available after the Commercial Operation Date.

The system shall enable the following fundamental functions:

- acquisition, filtering, validation and conversion of both digital and analog input values originating from the process
- processing of all input values using:
 - sequence control performing succession of logic operations depending on previous orders and presence of criteria
 - combinatory (or interlocking) control that combines authorization criteria for sending orders that switch on/off one or several equipment, that realizes security actions and that generates the necessary alarms
 - analog control (open and closed loop) that keeps the process parameters at preset values (that are functions of the load of the Plant within the limits contractually foreseen)
- creating the output information for digital (on/off) or modulating actuators;
- working out of supervision information (including but not limited to status indications, alarms) for control room operators and for maintenance, planning or statistics

The automation stations (and other control stations) shall be connected to a number of operating stations in the Central Control Room (CCR) by data buses. Data handling systems such as archiving and interfaces with other systems shall also be connected to the system.

Control/supervision units receive data from their relevant packages control systems to perform remote control and/or monitoring of those packages. This includes sets of images, alarms and hardware buttons.

A redundant OPC server with DA and HDA protocols and the necessary license to provide access to all and every data available in the DCMS system shall be part of the scope of supply. The data access of the OPC-clients to the OPC-server is limited and licensed for up to 5000 signals.

The good working of the OPC server shall be demonstrated with an OPC client which shall also be part of the scope of supply.

For the DCMS system, Contractor shall ensure the availability of a supply of spare parts for a period of 25 years from COD. For a period of 5 years the equipment shall be functionally compatible for plug in main equipment, for the remaining time the equipment is functionally compatible and as far as possible compatible for plug in main equipment.

8.3.2. Availability - Reliability - Maintenance

The DCMS is operating on a 24 hours a day basis. Verifications, checking and maintenance operation may only happen during the scheduled partial outages of the Plant (while maintenance is performed the system will still be operational in degraded mode). Beyond the outage periods no preventive maintenance shall be required.

Repairing is to be performed during operation of the Plant and shall consist only in module replacement (with minimum needs in wiring disconnection).

After detection and localization of a faulty module maintenance may be performed without disturbance on overall operation of the system.

Critical sensors will be redundant or connected in a 3 out of 3 manner to individual I/O's: only loss of both sensors or both I/O's would lead to shut down of controlled system. Critical cards (communication cards, power supply cards or devices, ..) shall be fully redundant. Contractor shall describe the principles of the segregation of redundant modules/devices to avoid any common mode fault.

The period between two successive maintenance actions (checks, controls and maintenance) cannot be shorter than 24 months.

8.3.3. Typical functions of the DCMS

For each control logic class typical diagrams/functions shall be defined by the Contractor. Moreover, inside each control logic, the signals shall be classified and processed according to the following proposed priority list :

- protection controls
- interlock signals
- manual controls
- automatic controls

8.3.4. Execution times of the main DCMS

The process requirement will determine the executing time for each control function. The process requirement dictates how fast the execution time for each control/protection function shall be.

The following performance criteria shall be achieved by the system, under all normal conditions of operation and under all emergency conditions:

- Process alarms (digital): VDU display and audible signal within 1 second from plant initiation and time stamped to no greater than 1 second resolution.
- Status display: VDU display change within 2 seconds of step change of plant transducer signal.
- SOE: VDU display within 2 seconds, all time tagged to 10ms including SOE designated points and electrical points.
- Command from CCR/LCR (local control room): status change at plant within 1 second of operator initiation.
- Analogue sampling : at plant level better than 250 ms.
- Binary sampling : at plant level better than 100 ms.

8.3.5. Checking of measurements to DCMS

Before processing the signals coming from the field they shall be submitted to a validation function. For analogue measurements, implementing of the following functions shall be taken into consideration by the Contractor:

- check of the signal validity limits
- cut-off and not supplied thermoresistance detection
- open circuit check (for thermocouples)
- filtering when needed
- derivative check between two subsequent samplings when applicable
- check of deviation between redundant measures

For important and protective binary input contacts (changeover contact) the system shall perform a consistency control.

8.3.6. DCMS control functions

The following requirements shall be taken into consideration by the Contractor :

- the control functions shall not require in their normal automatic operation any action from the operator on final regulation parts, limiting the operator's manual interventions to the minimum
- transfers from an operation mode to the other one shall be made with automatic equalisation systems and tracking functions (bumbles changeover)
- when a control loop switches to manual due to unpredictable irregular events, the operator shall have possibility to decide whether to return back to automatic mode

- the final controlling actuator, in case of failure of the control chain or failure in auxiliary power, shall be driven to a position, which does not risk the Plant safety and enable to continue the service, if possible
- at single equipment control level, interlocks and protections shall be provided to assure a safe operation, even when the remaining part of the automatic equipment at higher levels is not operating
- the DCMS shall perform the sequential functions relevant to the starting, stopping, etc. of the CCGT Plant and functional/machinery area
- for all the automatic sequences, the corresponding manual controls shall be foreseen at functional unit level, operating unit or control of the single equipment level, to be used in case of failure of the sequences.

8.3.7. DCMS supervision functions

The supervision shall enable the operator to check the information required for the CCGT Plant control in all the operation conditions, offering him a complete range of means, including:

- video functions for the process supervision
- management of alarms
- presentation and recording of status and events
- supervision of event sequence

The information available to the operator's interface in the control room shall always be clear and sufficient to enable the operator to make decisions in a short time.

8.3.8. DCMS event chronological recording

Protective, fast signals shall be put in chronological order in the events record in order to originate the actuation of one or more protection logics.

This function is proposed to be performed only on digital signals with 10 millisecond resolution.

8.3.9. Measurement supervision

The operator shall be kept alerted on the status of field signal non-reliability (invalid measurements or non-consistent contacts).

In case of double or triple measurements, information shall also be obtained when an excessive deviation is detected between measurements.

The management of the measures dedicated to control functions is proposed to follow the criteria listed below:

- in case of an invalidity of a single measurement (signal out of scale), information shall be given to the operator and in case the measurement is connected to a control loop, the loop shall be switched into manual mode
- in case of invalidity of one of the double measurement, the valid measurement shall remain used for the control (which remains in automatic) and an information to the operator shall be given
- moreover, for double measurements, the deviation between the two channels shall be monitored and when the same exceeds the tolerance limit (that could be calibrated), after an appropriate time, an alarm shall be given to the control room and the concerned control loop shall be switched to manual
- in case deviation existing among the three measurements, taken two by two, shall be monitored. Whenever the detected deviation exceeds the tolerance limit (that could be calibrated), after an appropriate timing, the invalidity of the measurement channel deviation from the remaining two shall be declared. This shall automatically select the calculation of the mean on the two valid channels left (which shall be treated as double measure) and shall originate an alarm signal and the display of comparisons exceeding the limit set
- in case of double or triple measurements any measurement(s) could be manually excluded

8.3.10. Alarm function

The following requirements shall be taken into consideration by the Contractor:

- each alarm shall always enable sound and visual devices
- the alarm displaying on the monitor shall be complete with all the indications useful for the acknowledgement of the alarm
- self-diagnostic messages relevant to the malfunction of each component of the system shall appear on the screens of the operator's console
- alarm chronological pages shall include, in the form of a table, the alarm messages that shall contain the following information :
 - event time also in cases the alarm is initiated by a subsystem
 - identification code
 - description
 - current status or value
 - description of irregular conditions or threshold value exceeded
- the presence of alarms shall also be signalled out of the alarm chronological video pages, using colour and light signals
- the acknowledgements could be made either from alarm chronological pages or from the other video pages, limited to alarm conditions clearly highlighted on the page itself
- acknowledgements of alarms in sub-automation systems shall work with the main DCMS

One part of the Plants DCMS system will be the process information system. The system shall comprise

- **Performance Monitoring System:** On-line near-real-time thermodynamic process including power and heat rate calculations, corrections to reference conditions and means for off- design simulation for online near-real-time equipment performance monitoring. All major equipment must be included, for examples, ST, GT, Generator, HRSGs, main pumps/motors.

The system will be designed to allow the monitoring of all its components in a general way and particularly of:

- the status of each plug-in module and its position in the racks;
 - the status of the actuators;
 - the status of the sensors and associated signals (among others for analog signals : out of range for transmitters, open transducer detection for thermocouples, Pt100 Ω and variation of resistance value);
 - feedback of order completion within specified time;
 - every power supply protection device.
- **Vibration Monitoring System:** for turbine/generator packages, boiler feedwater pumps fuel gas compressors and all critical pumps at 13.8kV level and as further requested elsewhere in this document.
 - **Generator Monitoring Systems**, e.g., for partial-discharge in stator and detecting of rotor winding short turn which will be independently provided with dedicated rack mounted display in the central control room as per generator manufacturer experience and design criteria.

A module for lifetime calculation according to TRD 508 for components operating in a region of high creep stresses shall be part of the system. The system shall also serve as an extensive history database having state-of-art trending and reporting features.

The system shall also have data transfer interface to Maintenance Management System.

8.3.11. External links – design requirements

Settlements meters and instrumentation

The scope of Contractor shall include: all integration work in Offtaker Authority's Load Dispatch System such as for SCADA if any, but limited to Project Site (including the meters located in GIS) under co-ordination of Owner

Links with Fuel Demand Model

Necessary input signals to FDM shall be part of the Contractor's scope of supply and shall include but not be limited to:

- ambient air conditions (in dry bulb temperature oC, relative humidity %, barometric pressure bara)
- natural Gas lower heating value (LHV in kJ/kg)
- natural Gas pressure and temperature at Plant supply point (bar g, °C)
- grid frequency (in Hz)
- generators power factor (-)
- required Plant net power output (in MW) as per dispatch instruction
- type of Fuel (low BTU or Sales)
- number of gas turbine units in operation (-)
- number of steam turbine units in operation (-)
- number of HRSG units in operation (-)
- gross power output of each gas turbine unit (in MW)
- gross power output of the steam turbine unit (in MW)
- gas turbine evaporative coolers in operation, for each gas turbine unit (on-off)
- gas turbine air preheater in operation, for each gas turbine unit (on-off)
- steam and feedwater supplied to FGP (in t/h)

The entire signal list shall be agreed during detail engineering.

8.3.12. Plant Information (PI) System

The Contractor shall supply an historian server. The system must consist of PI server license with app. 10,000 tags and 5 licences for PI-Client. The system must interface to DCMS via Ethernet network and ability to read data from it. All required tags are configured in DCMS.

8.3.13. Reserve capacity

The system shall be designed so that sufficient reserve capacity (on the average) is included, the following criteria shall be taken into consideration by the Contractor:

- 20% of free space for racks and I/O-cards in the I/O cabinets
- 10% of free channels in multi-channel I/O units
- 20% of reserve capacity in process control stations and other system stations
- 30% memory for application programs

The system resources shall enable minimum 20% increase of the control logic and loops, without increasing the processing time.

Hardware reserve capacity is calculated from the actual hardware in active usage. The system capacity is calculated from the technically accepted maximum capacity where the system still runs without problems.

8.4. Plant protection system

8.4.1. General requirements

The following requirements shall be taken into consideration by the Contractor.

The Safety Related System (SRS) shall be an emergency shutdown system that shuts down part of the Plant whenever the safety limits have been exceeded. The SRS shall include all the sensors, logic, and final elements needed for the safety function.

The SRS is required for the gas turbines, steam turbines, HRSG and the fuel systems.

The SRS system shall fulfil the requirements of IEC 61508 or equivalent:

- the SRS shall be independent from the DCMS. A fault at DCMS side may not affect the functions of safety system
- the SRS system shall be fault tolerant so that no single fault inside the SRS will cause a loss of a safety function
- the SRS shall be redundant. A single fault inside the SRS system may not cause a spurious trip of the plant/unit
- the SRS shall have a certificate that it fulfils the integrity level 3 of the IEC 61508 or equivalent. The requirements of the certification body shall be followed
- the safety functions and the safety integrity functions shall be allocated to each SRS system based on risk analyse of the process
- the major property damage shall be protected using SIL level 2 (IEC 61508). In case there is a risk of losing several human lives, the SIL level shall be 3
- response time of the logic controller: the maximum allowed response time for the safety-related functions of the logic controller is proposed to be 500 ms, that is 50 % of the proposed response time of 1 s
- in case the risk analyses indicates faster than 1 s response time, the SRS system shall fulfil that requirement

8.4.2. Safety Systems

The critical protection functions shall be realized using multichannel techniques with the 2 out of 3 redundancy principle.

Multichannel technique is provided for "Fail-safe" protection which will be limited to such systems which jeopardize the safety of persons. Such features are limited to fail safe systems of GTs, STs and HRSGs. All important, fail safe relevant measurements will be triplicated (exception: absolute bearing casing vibration, external binary trip signals). The fail-safe protection system itself shall be executed as a 1 out of 2 controller redundancy.

The system works preferably according to principle of current switch-on to actuate and de-energize to trip.

A single failure shall not prevent a protection to be activated.

A single fault shall not trigger a protection if not required.

Measures shall be taken so as to ensure that orders of such critical functions are duly sent to the actuators, in particular but not restricted:

- power distribution will be sufficiently subdivided in order to prevent a fault from having repercussions on different functions: complete selectivity of protection will be achieved
- binary and analog outputs are neutralized during initialization and start-up of the CPU's
- each signal used for critical functions shall be validated by continuous monitoring of its value and/or the variation thereof in relation to other values/signals: associated digital information is made available for any further functional processing
- interruption of a single external AC power supply won't impair operation of the DCMS (UPS power for the DCMS is redundancy provided)

Also:

- Redundant elements (motors, valves etc.) shall not be controlled from the same I/O card; the I/O corresponding to redundant devices shall be put in evidence in the lists supplied by the Contractor;
- Redundant measurements shall not be acquired on the same input card; the I/O corresponding to redundant measurements shall be clearly put in evidence in the lists supplied by the Contractor.

The Contractor shall inform the Owner's Representative how these particular points are met and shall also describe all other security features that are built in the system. with clear distinction between firmware built in security features and others that have to be realized by applicative configuration being mentioned.

ESD system

The plant protection shall be developed through an appropriate system. This one will respect Saudi Aramco ESD standard SAES – J -601 and shall develop all protective logics, from field instrument 2 out of 3 configuration.

8.4.3. User interface

The following requirements shall be taken into consideration by the Contractor:

- there shall be also manual means of shutting down the process. This shall be independent from the DCMS and SRS
- the manual emergency trip shall be hardwired from a push button to an input of the safety related control system. This function shall be totally independent of the DCMS
- if any bypass operations of the SRS are allowed, the bypass switches shall be protected by key locks
- the SRS status information is proposed to include :
 - process state: non-purged, purged, fire, trip initiated, safe state
 - criteria by-passed shall be indicated and alarmed with a repeating alarm
 - status of the voting (2/3) for logic part , the comparison fault status of the logic: normal, processor fault, I/O fault
 - status of all measurements and final elements connected to the SRS
 - status and alarm of all loss of energy
- manual acknowledge of a trip is proposed before process operation is allowed
- the process alarms shall be displayed via the normal DCMS alarm displays
- if a safety function is not causing automatic trip but initiates an alarm, then this alarm shall be done in SRS system and indicated independently from the DCMS system

8.4.4. Signal interfaces

The following requirements shall be taken into consideration by the Contractor:

- field signals shall be wired to the safety related control system (SRS) directly from the field
- signals from/to the DCMS system shall be transferred using the bus connection. Safety related functions shall not use a signal that is coming from the DCMS system and which might in special cases prevent the safety function

- in general, the philosophy shall be that instruments are not shared between control/monitoring systems and plant protection systems. In cases where this is not practical or there is sufficient technical justification (e.g. measurement redundancy) in doing so, a shared arrangement may be used subject to the approval of the Owner/Engineer. In case signals shall be HW shared between DCMS and SRS the fault in DCMS side shall not cause a fault at SRS side

8.4.5. Grounding

All the metal parts of the SRS logic cabinet shall be potential grounded according to IEEE standard..

8.4.6. Safety Integrity requirements

The following requirements shall be taken into consideration by the Contractor:

- for safety integrity definitions standards IEC61508 shall apply
- Safety Integrity Level of the central part of the SRS shall be the same as the maximum SIL level stated at the risk analyse.
- when calculating the integrity level it shall be assumed that :
 - the central logic controller has a one year test interval
 - the field equipment has a one year test interval
 - the SRS is a “low demand shutdown system”
 - if there is a need of test period shorter than 1 year, it shall be possible to do these tests during normal process operation

8.4.7. Fault tolerances of the SRS

The following requirements shall be taken into consideration by the Contractor:

- the logic controller shall have a two power supplies, one of which shall be enough for normal operation. Total loss of both supplies shall cause trip of fuel supply (GT) and related plant sections.
- field circuits shall be fed from two power supplies, one of which shall be enough for normal operation. Total loss of both supplies shall cause trip of fuel supply (GT) and related plant sections. The de-energised state of output signals shall be the safe state (tripped).
- the logic controller shall be one-fault tolerant when counting the safety functions. Fault tolerance is achieved by self-testing (diagnostics) or by multi-channel structure. Second fault (of same function) shall cause a trip.
- all binary (Boolean) data shall be wired so that a safe state is an energised state.
- all analogue (4...20 mA) signals shall have a signal range monitoring.
- the MCC contactors shall be suitable for emergency usage; the current cut-off capability shall be over dimensioned by dividing 0.6.

8.4.8. Field devices

The following requirements shall be taken into consideration by the Contractor for Plant protection systems:

- HRSG protection field device types shall be proven on use, for at least three (3) years' service. There shall be available the probability-to-fail-on-demand values for each device
- smart transmitters shall have a write protection to prevent modification from remote location
- In general, the philosophy shall be that instruments are not shared between control/monitoring systems and plant protection systems. In cases where this is not practical or there is sufficient technical justification (e.g. measurement redundancy) in doing so, a shared arrangement may be used subject to the approval of the Owner/Engineer. In case signals shall be HW shared between DCMS and SRS the fault in DCMS side shall not cause a fault at SRS side.

8.5. Time synchronization

GPS-based time synchronisation system shall be included to the scope. This system will give the common time for DCMS and each separate automation system.

The master clock shall provide real time reference to the DCMS and all other plant control, monitoring and protection systems by suitable interfaces capable of achieving a time synchronisation accuracy of 1 ms among these systems for their data acquisition and logging activities.

8.6. Power supplies for automation systems

The following requirements shall be taken into consideration by the Contractor:

- in general, supply systems shall be designed in such a way that a power failure occurring in one of the components of a control circuit causes the selective drop of the power supply only to that circuit
- DCMS shall be equipped with redundant power supply system, suitable to be connected to different supply sources.
- it shall control power lines and perform all voltage sublevels and distributions requested for the internal architecture
- the outputs of all the supply units shall be monitored and each malfunction identified
- provisions shall be made for the relevant signalling in the control room

- the internal distribution of supplies shall be designed to meet the principle of survival to the single failure
- for the power supply of the automation system equipment, the following shall be available :
 - two separate 230 V AC voltage single phase, supplied from redundant UPS (Uninterrupted Power Supply) dedicated for DCMS/control system devices (to be detailed later))
 - one single 230 V AC voltage single phase supplied from one normal Plant service network for non-critical DCMS/control system cabinet auxiliaries (lighting)
- the electrical supply that the system distributes to the instrumentation and actuators shall be taken from these sources. The supply of the instruments, when not sent on the same cable with the signal shall be distributed by the system which is acquiring the measurement signal
- the supply of process stations shall be redundant and made in such a way that they are not affected by possible failures of one of the sources. In any case, process stations shall be equipped with appropriate systems (emergency supply unit, data saving routine, non-volatile memories, etc.) enabling to keep the calibration and pre-set values unchanged, even in case of power failure, safeguarding the most significant internal variables and the codes of processing programs

8.7. Plant Instrumentation

A consistent control and instrumentation philosophy shall apply throughout the plant and shall be implemented in terms of a range of equipment exhibiting a minimum diversity of type and manufacture. The objective shall be to standardize all measurement and control equipment throughout the plant (except for the GT/ST/HRSG/Package suppliers) in order to rationalize operation, maintenance and reduce spares holding.

Generally equipment shall be supplied from one composite range of measurement and control equipment as marked by a reputable manufacturer of international standing and shall have operational use on similar projects. The equipment shall be of modern compact design incorporating the latest developments in proven technology.

Minimum protection degree IP 54 shall be provided for equipment in normal environment. IP 65 shall be provided in dusty environments, for open buildings, for outdoor installation and field installed instrument or explosion proof equipment following Eexd or Eexi when required.

All components shall be designed for the local ambient conditions. Equipment shall be designed to withstand the worst site conditions without deterioration in its functionality beyond the typical wear and tear. Field equipment shall have an appropriate degree of environmental protection.

The instruments shall not be shared between control/monitoring systems and plant protection systems. In cases where this is not practical or there is sufficient technical justification (e.g. measurement redundancy) in doing so, a shared arrangement may be used subject to the approval of the Owner/Engineer. In this case, the field signals shall be connected directly to the plant protection system where the pre-processing of the signals shall be carried out, thus avoiding the possible dependence of the protection system on the related control system for its inputs.

Analogue signals shall be standardized at 4-20 mA in the field. Direct current plant status and alarm signals shall be standardized on a DC voltage level between 22 to 50V. Instruments without separate power supply shall be chosen (excepted where impossible). Power supplies for current loops shall have a dedicated fused link for protection and isolation.

Two wire programmable smart type transmitters (e.g. HART) shall be provided. Transmitters used for control and monitoring may be hardwired to the DCMS I/O system or connected by a proven 'Fieldbus' communication system. However, transmitters used in functional safety systems shall be hardwired.

All transmitters shall have local indication of the measured value either built-in or equipped with separately mounted local indicator. Transmitters (except temperature transmitter and radar level transmitters) shall be provided with test connections and instrument isolating valve manifolds where appropriate.

Instrument ranges shall be chosen such that the nominal operating point lies between 60% and 75% of the measurement span.

Transmitters shall be used wherever possible instead of switches however, switches may be used with the approval of the Owner/Engineer in cases where their use can be technically justified. Switches shall be of the heavy-duty type with double pole changeover contacts. Mercury switches shall not be used. Transmitters and switches used for functional safety shall have traceable and appropriate SIL ratings.

Pneumatic transmission and control signals where offered, shall be of the 0.2 to 1.0 bar (3-15psi) standard range.

8.7.1. Field mounted instruments

All electrical items located in the field shall be weather-proof to IP65 or NEMA Type 4 whereas pneumatic instruments shall be weather-proof to IP54 or NEMA Type 2. Any local equipment panels used outdoors shall be weather-proof to IP54 or equivalent. For explosion proof equipment specified standards shall apply.

Components that are not line or vessel mounted shall be installed on local instrument racks and connected to the measurement source with stainless steel tubing or pipe. These racks shall be configured in a manner to group multiple devices in the same plant area to a centralised location for cable routing and to facilitate equipment servicing.

Seal chambers or diaphragm seals shall be used where fluid services are corrosive, viscous, congealing, deposit forming, or dangerous. Seals shall be located as close as possible to the primary isolating valve. Instruments in liquid services, which solidify at ambient temperature, shall also be provided with diaphragm seals.

8.7.1.1. PRIMARY ISOLATION AND INSTRUMENT VALVES

The primary isolation valve shall be located at the tapping point of the process and rated for the process conditions (e.g. temperature, pressure etc.) of the main line.

Two valves in series shall be provided for:

- applications above 25bar(g) line pressure
- hazardous services

All primary isolating and blow down valves shall be capable of being locked in the open or shut position. The direction of rotation shall be clearly marked on the hand-wheel.

Instruments measuring differential pressure shall be provided with equalising valves downstream of the instrument isolating valves to permit zero checking.

In addition to the primary isolation valve a secondary isolation valve shall be provided at the instrument itself. The secondary isolation valve may be a part of a manifold fitting with integral test connections. The instrument isolating valves shall be located on the connecting pipe at the measuring device end of the line.. Instrument isolating valves shall be rated to at least 100°C and the pressure of the main process line.

Only for the GTG Package, equipment: Root valves are not offered at the tapping point of the process. Isolation valves are offered where possible (does not compromise safety) on instruments.

8.7.1.2. INSTALLATION REQUIREMENTS

The installation of all instrumentation shall comply with the guidelines given in BS 6739 or an international equivalent standard. The piping/tubing to instruments shall be of stainless steel or a material corresponding to the piping class of the line to which it is connected. Erosion and corrosion problems shall be prevented by the use of stainless steel (316L) where appropriate.

Each tapping point shall be dedicated to only a single instrument.

All measuring devices shall be provided with instrument isolating valves, vent/test valves and instrument manifolds where appropriate to enable them to be disconnected from the connecting pipe work without having to disconnect or drain the pipe work. Instrument impulse lines shall use compression type tube fittings.

The impulse lines shall be arranged so as to avoid any gas bubbles entering piping for liquids (and liquids entering gas piping). Unless otherwise specified on the erection drawings, the slope of the impulse lines from the process shall be at least 10% of length (to the connection point when the instrument is placed higher than this connection point, and to the instrument when the instrument is placed lower than the connection point);

If necessary, pneumatic piping shall be laid in perforated cable trays supported at least every 0.6m.

All instruments and their piping shall be fixed in a way that prevents vibrations being generated or transmitted and the Contractor shall be taken into the fact that process piping and equipment may be subject to expansion or contraction.

The Contractor shall maintain a distance sufficient with the other equipment.

Primary process instrument and sampling tubing for steam and water systems shall be 316L stainless steel 12mm or more. Control air tubing shall be 6mm or more, 316 SS or sheathed copper. Air mains may be galvanised steel;

There shall not be any open blow down or drain connections inside panels or local equipment housings. All instrument vent/drain points shall be directed to a safe position away from maintenance personnel. Sample tubing systems carrying high temperature samples shall be insulated in areas which require personnel protection.

The arrangement of pipe work and connections shall be such that the temperature of the instrument isolating valve or instrument valve manifold does not exceed 55°C.

For pneumatic connections, each individual air user shall be provided with a lockable line size isolating valve immediately upstream of a dedicated filter regulator.

It shall be possible to read and access all indicating instruments and controls from floor level or permanent platforms. Enough clearance shall be provided around instruments and controls to access and service them without having to dismantle nearby equipment or pipe work.

Instrument mounting brackets or posts shall not be attached to pipe work, removable flooring or handrails. The mounting fixture shall not be directly affixed to machinery or equipment that may be subject to vibration.

Typical instrument installation hook up's sheets shall be made in order to define how instruments will be installed. Separate bill of material information shall be prepared to accompany these details.

Sufficient test and calibration points shall be provided to enable staff to carry out breakdown, routine, periodic and outage maintenance of the control and instrumentation systems.

Full details of local configuration of instruments and actuators, whether on Site or out at factory, shall be provided and all software and hardware (such as portable HART communicators/any special device/laptop) as necessary for configuration shall also be provided.

The presence of radio frequency radiation from portable radio transceivers or fixed installations such as radar and electric/magnetic interference from electrical equipment shall not be detrimental to the performance of the control and instrumentation equipment.

8.7.1.3. CALIBRATION

Before installation:

In general, each instrument covered by the Contract shall be tested regarding calibration and linearity, which tests shall be carried out by the Contractor.

The reference instruments used for calibration shall have accuracy at least two times higher than that of the instrument that is being calibrated.

The reference instruments used in calibration shall have been recently compared to a calibration standard which is also at least two times more accurate than the instrument to be calibrated. Recently means that the certificate of "gauging and stamping" must be available, and not be older than 12 months.

Each instrument covered by the Contract shall be tested for five readings in both directions: 0%, 25%, 50%, 75% and 100% of the set range. Switching thresholds shall be verified in two directions, and switching thresholds shall be adjusted in function of the thresholds (H, HH in upward direction; L, LL in downward direction).

During calibration no account shall be taken of shift in point zero as may result of installation of the instrument.

At least the following information shall be given in the calibration sheets:

- accuracy required of the instrument to be verified;
- verified readings in two directions;
- % deviation, and mA deviation for analog measurements;
- static pressure deviation (differential pressure instruments);
- verified switching thresholds and their type (H, HH, L, LL, ...) in two directions;
- hysteresis.

Instruments delivered by the suppliers with an appropriate calibration sheet may be, in case of doubt, readjusted.

Recalibration will be required only if technically justified.

After installation:

After installation of the equipments, the Contractor shall perform at least the following complementary calibration and checks:

- filling of the impulse lines, zero-shifting adjustment for manometers, p transmitters, pressure switches, with thresholds adjustment;
- post calibration on two points (0% and 100%) in two directions, for p or p transmitters;
- filling of the bottles for level measurements/switches;
- calibration of ultrasonic level switches, mass flow meters, and other specific measurements;
- calibration of analyzers.

8.7.1.4. PRESSURE INSTRUMENTS

Pressure transmitters shall withstand at least 150% of nominal input range without damage or calibration change. All differential pressure transmitters shall be capable of withstanding without damage or calibration change, full rated line pressure as a differential input.

Siphon loops shall be used for pressure gauges and transmitters employed in applications where the temperature of the process medium exceeds 70°C. Alternatively, use of impulse tubing from isolation valve to transmitter and cooling of process fluid in the impulse tubing below 70°C is also permitted. However, the length of impulse tubing shall be sufficient enough to cooldown the process medium. Moreover, where pressure measurements are subject to pulsation, snubbers (pulsation dampers) shall be fitted.

Pressure gauges with low ranges shall be fitted with over-range protection in applications where a higher containment pressure is involved. Gauges shall be fitted with vent/test valves as standard, together with blow out devices when used in applications greater than 25barg. Pressure gauges with Bourdon-tube and spring-type mechanism shall generally be used.

Pressure gauges shall be liquid filled type with 1% precision or better.

Where pressure measurements are subject to pulsation, snubbers (dampers) shall be fitted. Installation of pressure instruments at locations subject to excessive vibration shall be avoided.

Filters and strainers required for reliable, safe and continuous operation of the Plant (including main air filters) shall have differential pressure measurements providing readings and alarms in the control room. All other non-critical filters and strainers shall be provided with a suitable local monitoring device.

8.7.1.5. TEMPERATURE INSTRUMENTS

The selection of Resistance Temperature Detectors (RTDs) or thermocouples for temperature measurement shall be based on accuracy, reliability and measurement range considerations. Type K thermocouples and PT100 4-wire RTDs are preferred for normal applications.

Suitable protective sheath shall be used for sensing elements when used for air, inert gas and radiant temperature measurements. For steam, water and hazardous applications, thermowells shall be employed.

Head mounted temperature transmitters are preferred where feasible. Transmitters for thermocouples shall incorporate cold junction compensation and when mounted remotely from the sensing element shall use suitable extension cables or compensating cables.

Where thermocouple assemblies are supplied, a disconnection head shall be arranged clear of the pocket and associated lagging and shall be sufficiently long to prevent excessive temperature of the head assembly.

8.7.1.6. FLOW INSTRUMENTS

The carrier for the primary device for high pressure applications shall be of similar material to the pipeline in which it is mounted and shall be butt-welded into the pipeline. Other applications shall use flanged connections wherever possible.

Flow transmitters using differential pressure devices shall be provided with tapping point isolation valves, 3 valve manifolds and drain valves or 5-valve manifolds, as required.

Meters used for tariff/performance metering purposes shall be supplied, complete with adjacent pipe sections where necessary, with calibration certification from an agreed laboratory. Where density, temperature or pressure corrections to tariff quality performance monitoring signals are required, such measurements shall be made close to the flow meter location.

The flow measurement system shall be designed, built and installed to ISO 5167-1 or an equivalent international standard. It is acceptable only for the GT OEM to apply fuel gas performance flow metering as per ASME PTC 22. The uncertainty of the measurement systems shall be assessed in accordance with ISO 5168 and the relevant parts of ISO 5167. Turbine meters shall comply with the relevant requirements of ISO 9951.

Unless specified otherwise in Exhibit B, Part 1, the flow measurements transmitter used for various services shall be:

- gaseous fluids: annubar or Pitot tube with Δp -cell, mass flow meters or Vortex. For flow measurements not dedicated to settlement system (metering), orifices are allowed
- liquid or vapour: flow orifice, nozzle, venturi, electromagnetic
- straight length specified by Manufacturer (to be foreseen at both sides) shall be respected

8.7.1.7. LEVEL INSTRUMENTS

Level measurement shall generally be made by either displacement or differential pressure measurement, however other techniques may be considered.

Direct measuring devices shall be used for local indication of level. Unless otherwise approved by the Owner it shall be possible to remove any level measurement device without the vessel or other instruments being taken out of service.

It shall be possible with both the displacer type and differential pressure type of transmitter to isolate the transmitter from the process by suitable tight shut-off valves conforming to the pressure rating of the vessel to which they are connected and allow removal of the transmitter for maintenance without causing shutdown of the process.

Switches used for level detection shall be fitted within external chambers, which are provided with isolation valves, and suitable drains and vents.

Level gauges shall be fitted with top and bottom isolating valves with a full bore drain valve at the bottom and plugged vent at the top. Flanged connections which have the same rating as the vessel shall be used.

Float switches shall be glandless. Switches used for level detection shall provide facilities for testing the mechanical and electrical operation of the switch without its removal from the process.

8.7.1.8. ANALYTICAL INSTRUMENTS

Chemical or physical properties of process fluids and substances shall be measured by analytical instruments that are installed either directly in the process line or vessel or in the vicinity of the plant area (e.g. in an air-conditioned container) and connected by means of sampling systems. The delay time between the sample being withdrawn from the process and being available to the analyser shall be consistent with the measurement and control requirements of the particular process. The preferred method of keeping sampling lags to acceptable limits shall be by locating the analyser close to the sample take-off point. When this is not practicable the distance velocity lag of the sampling system shall be reduced by increasing the velocity of the sample.

Analytical instruments shall be selected to operate either continuously, infrequently or on a continuously interrupted cycle as most suited to the measurement and the intended application.

All analytical instruments shall be supplied with a minimum of one year's supply of consumables and operating spares at normal usage. This shall include as a minimum, reference gases and solutions for instrument calibration as well as spare sensors required for replacement or removal for recalibration or rejuvenation at regular intervals.

8.7.1.9. VIBRATION INSTRUMENTS

The required vibration monitoring system shall be provided as per relevant standards for rotating machines wherever applicable. Vibration monitoring, analysing and diagnostic for GTG, STG, Boiler feed pumps and other critical pumps at 13.8kV levels shall be as per RFP and SFR requirements. All other rotating machines (e.g Pumps, fans, etc) with a rated motor capacity not less than 200 kW shall be equipped with vibration measurement devices with indication/warning/alarm in the DCS. All rotating machines shall be capable of satisfactory operation when delivering varying quantity from minimum to the design maximum and shall be constructed to ensure satisfactory operation under all climatic conditions prevailing at site.

Vibration measuring devices shall be selected for operation in their particular intended environment and duty. It shall use non-contacting transducers (proximity type) and shall be radially mounted in the same transverse plane perpendicular to the shaft axis and 90° apart. Absolute vibration measurement shall measure the bearing pedestal vibration and shall use seismic transducers (velocity or accelerometer type). The measurement of both relative and absolute vibration shall have sufficient bandwidth so that the frequency spectrum of vibration of the rotating plant is adequately covered.

For the protection, analysing and diagnosing of vibrations in major rotating machinery, the vibration monitoring system shall be provided in accordance with ISO 10816.

The vibration monitoring system shall be reliable, accurate, easy to maintain, and suitable for use in such environmental conditions appertaining to the intended plant installation.

8.7.1.10. TRANSMITTERS

Generally two wire transmitters shall be used. If or some special purposes (e.g. analyzers) 230 V AC analysers supply is required the output circuit shall be isolated. All transmitters supplied with 230 V AC shall be individually fused. All transmitters shall be of the SMART type.

All transmitters shall be fitted with an indicator displaying appropriate physical units which may be read clearly by the plant attendant from an easily accessible position. Transmitters shall be provided with all necessary isolating, vent and blowdown valves, thermometer pockets and interconnecting pipework.

All transmitters shall be supported in a vibration free manner independently of the plant connections and shall be mounted in the correct attitude. No stress shall be imposed through the connections between plant and the transmitter.

GT OEM's instruments shall be sufficiently protected from local conditions and are rated for outdoors however; it is acceptable only for GT OEM to not always have separate enclosures for instruments. Facilities shall be provided for the connection of test instruments to the input and output of each transmitter to enable in-situ calibration to be carried out.

Bodies, primary elements and wetted parts shall be of a material that is corrosion resistant to process fluid and environmental conditions.

Differential pressure/ flow transmitters shall be furnished with a close coupled equalizing valve manifold.

Variable area type flow transmitters, if applicable, shall be of all metal construction with appropriate means of analogue signal transmission.

On large installations locations for transmitters on similar plant duties shall be grouped unless this is economically infeasible.

To maintain minimum spares holding, the range of transmitters used for general pressure and differential pressure applications shall have facilities for simple on site range changing without changing of components. In addition each transmitter shall incorporate means of zero suppression elevation allowing zero shifts equivalent to at least 100% of maximum transmitter span.

All differential pressure transmitters shall be capable of withstanding without damage or calibration change, full rated line pressure as a differential input. Pressure transmitters shall withstand at least 115% of nominal input range without damage or calibration change.

All transmitters potentially subjected to vacuum shall be capable of withstanding 100% vacuum without damage.

Transmitters for use on electrical input signals such as thermocouples or RTDs shall allow range change on site by the use of simple plug-in range cards. They shall incorporate electrical isolation between input and output.

Transmitters shall meet the following minimum requirements unless agreed or specified otherwise by the Engineer.

- DC Voltage Level: Between 22 to 50 volts
- signal current: Between 4 to 20 mA
- accuracy (of signal current with respect to measured variable): Better than 0.2%
- linearity: 0.1% of span
- repeatability: 0.1% of span
- stability: 0.1% of nominal input range per annum
- dead Band: Not exceeding 0.1% of span
- hysteresis: Less than 0.1% of output span

- discrimination: A change on input of 0.02% of actual range shall produce a measurable change of output
- maximum permissible effect of vibration on output signal: +1.0% of output span between 5 and 300 Hz at 10 m/s² (1g)
- humidity range: 5 - 95%
- ambient temp. range: 0°C -52°C
- ambient temp. Effect: Not exceeding 0.5% of maximum span per 55°C change between the limits of: -40°C and +85°C
- design pressure rating: Not less than 200% of maximum process static pressure

8.7.1.11. AMBIENT CONDITIONS MEASUREMENT FACILITIES

The Contractor shall install the Ambient Conditions Measurement Facilities at a suitable location within the Site.

The Ambient Conditions Measurement Facilities shall include all the instrumentation for monitoring the ambient air temperature (dry bulb), ambient barometric pressure, relative humidity and wind speed . Data shall be repeated to the Plant DCMS.

8.7.1.12. CONTINUOUS EMISSIONS MONITORING

The GTs, HRSGs and Reserve Boiler(s) shall be equipped to continuously monitor and record NO_x, CO and O₂ emissions. The continuous emissions monitoring system (CEMS) shall be interfaced to the DCMS for display and recording of the analytical measurements.

8.7.2. Final control elements

8.7.2.1. SOLENOID VALVES

- 5-way solenoid valves (bistable-2 separated solenoids), for actuation of bistable valves shall be provided by Contractor
- 3-way solenoid valves for actuation of fail open of fail closed valves shall be provided by Contractor
- operating voltage: shall be 24 VDC (preference) or 230 VDC, 125 VDC acceptable for the GTG package

- a built-in local manual operation allowing to force position whether solenoids are powered up or not shall be provided by the Contractor; however it is understood that local operating and monitoring of systems are not required for Plant normal operation, and for that, manual operation on solenoid valve shall not be necessary. Solenoid valves operating isolating valves without limit switches shall be provided with position detectors or with (logic) sensors on the process fluid in order to allow signaling of actual state (this point is not applicable only for the GTG package supplied by the GTG OEM wherein the GTG OEM's standard practice shall be followed)
- inputs and/or exhausts shall be fitted with built-in adjustable retarders
- electrical connection shall be made by connector on solenoids It is acceptable for the GT OEM only to use their standard terminal blocks inside junction boxes of the GTG Package
- DC Solenoids valves shall be fitted with diodes or varistors installed as short as possible to the coil to prevent overvoltage peaks due to power cut off or reverse polarity

Orifices for external connections of pilot fluid if necessary; filters and lubricators if appropriate, shall be included in Contractor's scope of supply.

8.7.2.2. CONTROL VALVES AND ACTUATORS

Irrespective of the specific function that a control valve performs in the control scheme, i.e. modulating, inching or on/off, all control valves shall have a local control facility and local indication of the actual position of the valve. The normal operating mode shall be 'remote' from the CCR via the DCMS, the selection of which shall be local to the valve and transmitted to the DCMS.

Control valves shall be supplied with suitable actuators matched to the operational and environmental requirements of the plant. In general, pneumatic actuated control valves shall be used for modulating duties. However, motorised control valves or hydraulic valves may also be used in specific control loops with the approval of the Owner. Control valves for inching duty shall be of the motorised type. On/off control valves may be motorised or pneumatically actuated with a solenoid pilot valve.

In general, all control valves shall have end-position detection by limit switches and indication in the CCR. Modulating and inching valves shall have transmitters or LVDT's to indicate the valve position in the CCR, the measurement of which shall be taken directly from the actual valve position.

8.7.2.3. MOTORIZED VALVES

Motorized valves shall consist of self-contained actuator units with motor, gearbox, electric/electronic controls, limit/torque switches and all auxiliary devices installed in the actuator housing. A lockable local/remote selector switch with open/close controls shall be located on the actuator. Only for the Gas Turbine package, for safety reasons if some valves are not lockable it is acceptable. The actuator shall be provided with a hand wheel for operation of the valve in an emergency. Only for the Gas Turbine package, for safety reasons if some actuators do not have hand wheels it is acceptable. The operation of the hand wheel shall disengage the actuating motor to prevent damage.

Each end of the actuator travel shall be fitted with suitable limit and torque switches and correctly set up for preventing over travel and consequential damage to the actuator. The stopping of the operating mechanism on limit or torque switch shall be determined based on the specific application for which the valve is used. Adjustment of maximum torque shall be performed at works. Torque shall be adjustable from 80 to 115% of rated torque value demanded by valve;

Semiconductor power-electronic switches rather than contactors shall be used in the actuator for switching the motor supply.

Motorized valves shall be equipped with Anti-condensate heater.

Control and supervision signals to DCMS or Control System:

- OPEN and CLOSE control
- open and Closed position in feedback
- open and close torque faults
- motor in “local/remote control”
- motor General “fault”

8.7.2.4. PNEUMATIC ACTUATORS/POSITIONERS

An electro-pneumatic converter shall generate a 0.2 to 1.0 bar pneumatic signal from a 4 to 20 mA electric signal for use by the actuator positioner for modulating duties. On/off applications shall make use of a solenoid pilot valve with a suitable operating voltage to minimise volt drop on the connecting cable.

Control valve actuators shall be designed to drive the valve to a safe position from the process view point on failure of signal and/or air supply. For each valve application, due consideration shall be given to whether the safe state is open, closed or for the valve to remain in the last position prior to loss of control ('stay put' design). The fail-safe position for all pneumatic control valves shall be advised during the detailed design of the system.

Volume boosters shall be provided between the actuator and the control valve where necessary, e.g. on fast-acting loops.

All positioners and electro/pneumatic converters shall be furnished with pressure gauges for air supply, signal input and control air output.

8.7.2.5. MANUAL VALVES

Manually operated valves critical to the process shall be provided with end position limit switches for indication in the CCR via the DCMS HMI.

8.7.2.6. MOTOR CONTROL CENTRES

Motor control centres (MCC) used for on/off control of pumps, fans etc. shall have a hardwired interface with the related control system. The MCC shall have the facility to select the point of control, i.e. local-MCC or remote-DCMS, on/off local controls, as well as local status indications.

The interface between the MCC and the control system shall be standardised. This shall include on/off command inputs to the MCC and on/off/unavailable/fault feedback signals to the DCMS. The command inputs to the MCC shall be electrically rated to allow a direct connection (i.e. without the use of external interposing relays) with the DCMS binary output module.

Segregation in terms of cabling and DCMS binary I/O modules shall be provided for duty/standby pumps/fans.

8.7.2.7. VARIABLE SPEED DRIVES

Electronic variable speed drives (VSD) shall be considered for flow control where large pumps/fans would be employed with modulating valves/dampers. The use of a VSD shall be based on a cost-benefit analysis in terms of energy saving and/or the need for achieving the required level of modulating control performance of the process.

The signals exchanged between the VSD and the DCMS shall be categorised into critical and non-critical. The critical signals (e.g. speed reference, speed feedback, drive trip command, drive fault, drive common alarm) shall be hardwired. The non-critical signals (e.g. internal drive status and warning signals) shall be communicated to the DCMS via a serial data communication link using a standard fieldbus communication protocol.

Segregation in terms of cabling and DCMS I/O modules shall be provided for duty/standby pumps/fans.

8.7.2.8. GAS DETECTION SYSTEM

A field proven and intrinsically safe fuel gas leakage detection system with international approvals shall be supplied. .

The gas detection system shall interface with the DCMS to warn the CCR operator that a gas concentration exceeding the alarm limit has been detected. System faults shall also be alarmed on the DCMS workstation.

A gas turbine trip shall be implemented when the gas concentration exceeds the trip level setting.

8.8. General I&C Requirements

Sun shades shall be installed for outdoor instruments that are subjected to direct sunlight.

All isolating circuits and devices used in I&C equipment shall be designed to provide an insulation resistance of not less than 20 MΩ measured at 500 VDC.

8.8.1. I&C cabling

I&C cabling shall use multi-pair cables with screens, multi-core cables and optical fibre cables as appropriate. The cables shall be of the rodent proof, flame resistant and fire retardant type. The protection method for I&C cables from physical abuse and rodent/insect attack shall be to the approval of the Owner/Engineer.

The multi-pair/multi-core cables shall be made up of copper conductors of not less than 1 mm² cross-section except for those associated with interposing relay cubicles and power supplies where higher conductor cross-section shall be chosen based on volt drop considerations. I&C cables of less than 1 mm² cross-section shall be subject to approval by the Owner/Engineer.

There shall be 20% spare conductors as a minimum in I&C cables.

Adequate separation of I&C cables from power cables and segregation of I&C cables for different systems shall be provided. Redundant cables (e.g. communication network cables) shall be installed over separate routes to minimise the possibility of a common failure.

8.8.2. Cubicles and racks

Mechanical enclosures such as cubicles, racks and panels used for I&C systems shall conform to the relevant sections of IEC 60297: Part 3 for uniformity of dimensions and arrangements. The degree of protection for indoor cubicles and panels shall be IP42 minimum while those located outdoor in plant areas shall be IP54 minimum. The cubicles and panels shall be mounted in such a manner as to minimise external vibration being transmitted to internal components, for which suitable anti-vibration mounting arrangements (e.g. anti-vibration pads) shall be used.

The earthing scheme proposed for I&C systems shall reduce the effects of electrical/electromagnetic interference and ensure the safety of personnel. All metallic parts of cubicles and panels including doors shall be earthed by earthing conductors/braids of suitable cross-section using serrated washers to ensure a positive contact. Cable screens shall be terminated in such a way that earth loops are not introduced. Cable screens shall be terminated only at one end for which an insulated 'clean' earth bar shall be provided in cubicles. The earth bar shall be connected to the site earth via the cubicle safety earth at a single point. Armoured cables when used shall have their armours connected together externally, but shall be electrically isolated from the cubicles. The cable armours shall be connected to the site earth at a single point.

The layout of the interior of cubicles, junction boxes and panels shall facilitate easy access to components and terminals during commissioning and maintenance. In general, 10% of spare capacity shall be provided as a minimum in cubicles, panels, racks and junction boxes. This shall include unused module slots, unallocated wired I/O channels, unused terminals, space for additional external cables etc. The spare capacity of 10% shall be made available by Contractor after final commissioning which shall be pre-engineered, implemented, terminated and ready for use.

If forced ventilation is to be used in cubicles and panels, 100% redundant fans shall be employed with facilities for fan failure detection and alarming on the DCMS. Thermostatically controlled anti-condensation heaters shall be installed in cubicles and panels. Maintenance facilities such as internal lighting and a power socket shall also be provided in cubicles and panels.

Labels or other suitable identification methods shall be provided for sub-assemblies and terminals in cubicles and panels. This shall include modules, racks, power supplies, pre-fabricated cables, I/O termination boards, external cables and their conductors, terminals etc. A 'danger' label shall be provided with shroud for terminals connecting 110 V and above.

External cables entering the cubicles and panels shall be adequately secured and supported by a suitable method, e.g. cable glanding. Outdoor I&C enclosures, e.g. junction boxes and plant-mounted control panels, shall only use cable glanding for this purpose. Alternate methods may be used to secure and support the external cables. In this case, the cable entry area shall be sealed by a suitable method to guarantee that the cubicle/panel has the required fire rating and is dust/vermin-proof.

9. CIVIL WORKS/STRUCTURAL AND BUILDINGS TECHNICAL REQUIREMENTS

9.1. General requirements

Buildings and structures for the Plant shall form an integrated structural theme. They shall be designed to acceptable international standards and shall be fit for the purpose in each case. The Works shall include excavation, foundations, reinforced concrete, masonry and structural steel all as is necessary. HVAC, piping, ducting etc. inside the buildings, sanitary installations, power & lighting, earthing, lightening protection etc. shall all be provided to high standards and workmanship.

For a plant that requires to be enclosed, a suitable building shall be provided which meets the requirements of the local planning authority and the Owner. A plant or system which does not require to be enclosed may stand outdoors, although the Contractor shall judge for himself whether these items may need containment to match the main building and to meet Authority and Owner requirements.

9.2. General design criteria

The design shall reflect the climatic conditions that pertain to the site and which will be considered applicable during the minimum working life of the works. Special attention is to be paid at the effects of aggressive conditions, safe and durability of the facility.

All buildings housing the plant items shall be adequately proportioned to facilitate the installations, operation, maintenance and replacement of the plant.

All structures shall be designed to facilitate erection and so that they remain stable at all stages of construction.

9.3. Foundations

Design and construction of all foundations shall be based on building/equipment loads and ground conditions on site.

Reinforced concrete foundations shall be provided and piled if necessary to transfer the loads satisfactorily to the supporting soils.

The final foundation scheme to be adopted shall be based on the soil investigations carried out by the Contractor in the exact location of the buildings and structures to ensure that the foundations shall be suitable to safely support the loads imposed on them.

Foundations for machinery shall be designed to withstand the imposed static and dynamic loads. Adjacent equipment and structures shall be isolated from the effects of vibration to acceptable level. The dynamic analysis of major machinery components will be provided for the review of the Owner.

The foundations shall be provided with appropriate bench marks for settlement monitoring purposes.

All concrete foundations and structures exposed to the ground shall be protected by bitumen based damp proofing.

9.3.1. Geotechnical / Foundations

The Contractor is responsible for performing his own subsoil investigations to establish soil characteristics for detailed design. The campaign and techniques used will conform to the “Code of practice for site investigations” - BS5930, or approved equivalent.

The Contractor shall execute compressive static loading tests on at least three piles belonging to different areas. The procedure to be followed is provided by ASTM D 1143. The maximum test load is equal to 2 Q_n (if pre-production pile) or 1.5 Q_n (working pile), where Q_n is the net vertical load of the tested pile

Foundation type will be chosen by the Contractor, based on the geotechnical investigation report and a comprehensive calculation of the structure, bearing capacities, safety factors and settlements. When piling is proposed, the piling design and works shall be in accordance with internationally accepted standards such as BS 8004, BS 8008, BS 5930, BS 6031, ASTM D1143, DIN 1054, DIN 1626, EN 1536, EN 12699, DIN 4030, DIN 17100. Any cement to be used for concrete piles shall be in accordance with the CIRIA guide and ASTM C150 latest edition. The minimum content of cement shall be 370kg/m³, the maximum diameter of the aggregates shall be 25mm. The water/cement ratio shall not exceed 0.42. The cover on reinforcement shall be 100mm.

9.4. Plant building facilities

9.4.1. General

When designing the facilities for the plant personnel the following (preliminary) number of permanent people on site shall be considered:

- office staff (admin and plant management) app. 30
- maintenance staff app. 50
- operational staff (CCR and field) app. 5 x 8 (5 shifts with 8 persons on each shift)

Also facilities (Sanitary & Showers, locker) for non-permanent staff shall be considered:

- Owner's engineers 4
- maintenance staff allowance 40 to 50

9.4.2. Administration and social building

Refer to Exhibit B, Part 1.

Administration building shall include at least:

- eight (8) offices (CEO, CFO, Plant Manager, Operations Manager, Maintenance Manager, HSE Manager, HR and Administration staff.);
- one meeting / conference room (20 people)
- one common office room for 10 people
- one data room
- one prayer room (minimum 20 people)
- server room and separate archive room which shall comply with all IT safety standards applicable in accordance with Good Utility Practice
- one equipped Kitchen and one Canteen for 15 people and for light meals (a combined canteen and kitchen may be proposed to the Owner's Representative for Owner's Representative's approval)
- sanitary (western type)
- first aid/medical room

The Administration and social building shall be fully equipped (HVAC, electricity, IT and telecom connections, water, waste water) without (tables or chairs) or any other equipment (computer systems, phones), except for the Central Control Room (CCR) in respect of which furniture, computer systems, phones, and other appliances and furniture shall be provided by the Contractor.

9.4.3. Workshop and warehouse buildings/areas

New Workshop & stores building and Main stores building shall be provided.

Workshop and warehouse building shall be ready at Early Steam Operation Date.

In the mechanical workshop, power supply and fume extraction shall be provided for installation of a welding bay.

The workshops and warehouse building shall provide facilities for the CCGT Plant's day-to-day maintenance and operation. As a minimum the accommodation shall be provided for the following functions:

- mechanical workshop and stores
- instrumentation workshop
- electrical workshop
- the ware house shall accommodate at least two complete sets of GT spare parts including combustors
- warehouse and indoor storage facilities for spare parts and special tools.
- fenced outdoor area of 2000m² suitable for heavy load storage. Surface shall be adequate for truck access and heavy load storage
- applicable storage facilities for flammable materials and hazardous (chemicals) materials separated from other storage facilities

Warehouse shop should be large enough to accommodate all the spare parts storage. Chemical storage should be large enough for all the chemicals, and should have waste storage areas for both non-hazardous and hazardous material storage. Chemical storage tanks have to be big enough to make possible unloading with tankers.

Dedicated and suitable constructed areas for the storage of compressed and or flammable gases, chemicals and oils shall be integrated into the warehouse. Where required bunding, ventilation and specialist fire protection shall be provided.

Contractor shall provide the storage area (warehouse) for the storage of spare parts at their arrival on site.

In the case the workshop and warehouse is not ready on time, the Contractor shall provide temporary adequate storage for theses spare parts up to COD.

The workshop and storage buildings, including their associated offices, shall be:

- fully equipped (HVAC, electricity, IT and telecom connections, water, waste water)
- fitted with overhead travelling cranes

- without the specific workshop equipment (tools, benches) and without the storage racks or shelves
- without furniture (tables, chairs)
- with hard flooring for movement of mobile crane/forklift (which mobile crane/forklift are not in the scope of Contractor)

The minimum requirements for the area of workshop and storage are 2,000m² accordingly, plus 400m² of common facilities (offices, toilets etc...). In storage and warehouse, elevation of the building shall be at minimum 10 meters high (suitable for storage shelves installation) at 80% of the surface.

The workshop and storage buildings shall comprise:

- one workshop area
- three offices
- one Hazardous Equipment and Gases Store Area
- one General Stores Area
- one Hazardous Materials Store Area
- one Hazardous Chemical Storage Area
- sanitary (western type, including shower)

Note: All mechanical/electrical/instrument workshop equipment and hand tools for use after COD are excluded from Contractor scope of supply.

9.4.4. Gate house

The gate house shall be:

- Fully equipped (HVAC, sanitary (western type), electricity, IT and telecom connections, water, waste water)
- Without furniture (tables and chairs)

9.4.5. Central Control and Switchgear building

A Central Control and Switchgear building shall be constructed by the Contractor as per the location identified by item 213 on Annex 05a.

The Central Control Room (CCR) shall be provided with all necessary furniture, computer systems, phones, and other appliances as required.

The CCR flooring shall be at a single level, including accommodation of the anti-static raised access floor. The doors to the CCR shall be sufficiently sized to facilitate easy installation and maintenance of all panels and control room equipment. The doors and ceiling height shall be suitable for future removal of panels and equipment without the need for adjustment or removal.

Equipment such as work stations for non-operations related work, meeting rooms, printers, engineering work stations, electrical cubicles, etc. shall be located in adjacent or separate rooms on the same floor with easy access from CCR.

Amenities shall be provided suitable in size to accommodate the largest number of staff to change shift on site at any one time. The facilities shall include toilets, showers and messing facilities (with lockers).

All building layouts shall be provided for the approval of the Owner and the Contractor shall describe the proposed layout philosophy of the control room and equipment. Drawings of the outline layout of the CCR and the Central Control and Administration Building shall be included with the proposal.

The buildings shall meet the necessary Saudi Aramco standards as required.

9.4.6. Laboratory

Laboratories (mechanical, I&C, electrical etc) shall be provided by the Contractor.

The Laboratories shall be fully equipped (HVAC, electricity, IT and telecom connections, water, waste water).

The chemical laboratory shall also include a chemical storeroom and shall enable the Operator to monitor among other things:

- the water/steam cycle water quality
- produced demineralised water quality
- effluents quality as per requirements in Exhibit B, Part 1
- oil analysis (transformer oil, lube oil, hydraulic oil)
- fuel oil quality testing
- cathodic protection systems implemented (reference electrodes)

The Contractor shall prepare a list considering the analysis requirements of the suppliers of the main equipment according to their operation and maintenance instructions.

The laboratory shall be at least 6m long and at least 3m wide.

An emergency shower, a flexible eye shower, a first aid box, safety lock and doors with inside panic opener/bars shall be installed in the laboratories. Concerning the fire detection and fire fighting system to be installed at the laboratories, the NFPA requirements shall be implemented (and as a minimum smoke detection and one fire extinguishers shall be installed).

Suitable clothing for laboratory works shall be provided.

The list and data sheets of the measuring instruments, of the chemicals and reagents and a list of glassware and other equipment shall be provided for Owner's Representative's approval. The list and characteristics of the furniture for the laboratory shall also be submitted to the Owner's Representative for approval.

Consumables for the laboratory after COD (specifically excluded furniture) shall be provided by the Owner.

9.4.7. Other buildings

Other buildings shall be adequately sized and fully equipped (HVAC, electricity, IT and telecom connections, water, waste water).

Other buildings include:

- storage building
- security building
- fire brigade building
- gate house

plus others as deemed necessary, refer to Exhibit B, Part 1.

9.4.8. Design criteria / structural design

As per Codes and Standards (refer to section 4) and Exhibit B, Part 1 and Good Industry Practice, the design for the Plant shall incorporate the worst load combination (as per applicable code).

In particular:

- Erection loads (including loads during hydraulic testing of equipment);
- Wind loads shall be in accordance with ASCE 7-10 "Minimum Design Loads for Buildings and Other Structures". The basic wind speed for the site, as defined in ASCE 7-10 standard, shall be determined by the Contractor, but shall not be less than 45 m/s. The wind exposure C is to be considered for calculating the wind pressures in any direction;
- Seismic Loading – Refer to Exhibit B, Part 1.
- Temperature Loading – The temperature difference considered in the design, shall not be less than: $\frac{2}{3} \times (\text{maximum ambient temperature in the shade} - \text{minimum ambient temperature})$ and 35°C.

- Special loads such as:
 - impact, braking and dynamic loads from cranes, hoists, elevators;
 - impact loads in silos, bins and hoppers due to arching from stored material;
 - dynamic loads from vortex shedding in wind fields;
 - dynamic loads from rotating and reciprocating machines;
 - short-circuit loads.

Roads shall be designed to DIN 1072 - SLW 60 and/or to accommodate trucks capable of delivering the heavy equipment. Minimum radius at kerb shall be 5 m.

In the case of no specific requirement in Exhibit B, Part1, the Contractor shall apply NFPA codes for the protection of steel structures for the Steam Turbine building as detailed below:

- Required area: The protection of the areas is subject to the systems contained within (i.e lube oil, electrical, fuel etc). Contractor shall refer to sections NFPA 850 Sections 5 -7 (particularly sections 5.3 and 7.7) for guidance
- Fire rating : 2 hours (for column, beam/girder only)
- Fire proofing type: There are no specific coating requirements. NPFA 850 requirements shall apply for the building materials e.g. cementitious fireproofing

9.5. Architectural / structural treatments

9.5.1. Structural framework for buildings

All the structures below the ground floor level will be made of reinforced concrete.

Steel framed buildings shall be provided with metal cladded roofs and walls. Reinforced concrete framed structures shall be provided with metal cladded roofs and walls or with concrete roofs and concrete/concrete blockwork/brickwork walls. Steel framed buildings bounded by trafficable areas, shall be provided with concrete blocks perimeter walls at least 1.6m high.

All supporting structural steelwork, un-galvanized handrails and chequered plates shall be protected against corrosion by a coating system and decorative paint system.

The whole plant shall be designed and provided with adequate detection and protection of structures as far as required by the NFPA regulations and it shall be consistent with local fire authority and Insurer's requirements.

The paint protection shall be based on a minimum of 15 years to first maintenance. Colour coding system shall be standardized.

9.5.2. Flooring

Floors shall be structural ground slab on grade, suspended reinforced concrete slab or steel open grid flooring, in accordance with the loading and requirements of usage. Adequate falls and drainage outlets to be provided to prevent the standing water to collect.

The CCR flooring shall be at a single level, including accommodation of the anti-static raised access floor. The doors to the CCR shall be sufficiently sized to facilitate easy installation and maintenance of all panels and control room equipment. The doors and ceiling height shall be suitable for future removal of panels and equipment without the need for adjustment or removal.

9.5.3. External Wall Cladding

The main cladding material shall be coloured profiled metal sheeting in standard factory colours to be selected by the Owner and/or Authority/architectural officer. Appropriate system may be utilised incorporating acoustic insulation as necessary. The choice of finish shall be made considering the specified service life under the specified site conditions.

9.5.4. Roof Cladding

The Contractor shall make his proposals for the roof cladding construction. Rainwater roof outlets shall be fitted and detailed to prevent the ingress of moisture.

With regard to fire, manually operated hatches for smoke ventilation shall be located around the roof plan as necessary.

9.5.5. Internal Walls and Partitions

Where required for structural reasons, sound transmission or fire prevention, internal walls shall be of concrete block work/brick masonry. Otherwise internal walls will be of appropriate lightweight panel construction. All penetrations through fire partitioning walls and floors shall be sealed to achieve fire endurance equal to the respective structure.

9.5.6. Doors

Personnel and escape doors and frames shall be mainly of painted steel. Vehicular access and erection purpose doors shall be plastic coated steel roller shutter doors or folding gates with personnel access doors. Internal office doors and frames in the admin building and in control and electrical building shall be of aluminium with necessary furniture and closers.

All locks shall be in harmonised/suited according to process areas and/or functional requirements or possible confined spaces. At least 3 level master key system is required. The Contractor is to provide a lock proposal for Owner's approval.

9.5.7. Windows

Areas with permanent staff occupancy shall be provided with metal framed windows with double glazing. Consideration shall be given to the provision of daylight in all buildings with permanent working places consistent with function, acoustic and aesthetic considerations.

9.5.8. Stairways

The main stairways shall be enclosed in fire protected shaft. External emergency stairs shall be of steel according to local regulations.

9.5.9. Internal Finishes

Internal surfaces of the buildings shall be selected to suit the particular requirements of each building, its function, requirements of occupancy and operation requirements.

Only the use of architectural finishes and materials with a proven track record of satisfactory installation in similar operational and climate conditions will be permitted.

The Control Room shall have better quality finishes to provide good wearing properties and visual appearance. Where raised flooring is provided the base structure shall be sealed against dust.

Hard finishes shall be provided for the majority of floor areas throughout the Plant. These shall provide durable surfaces that enhance the building environment and are easy to clean. Protective floor finishing shall be provided to Plant areas, ceramic tiling to toilets, kitchen and lockers. In the areas with constant movement of vehicles or risk of oil or chemicals spillage the floor shall be provided with surface hardening or chemical/oil resistant finishes.

Wall finishes shall conform to normal practice for each type of building and facility. Concrete block surfaces to be usually painted.

Offices, kitchen, locker room and control room shall have suspended mineral fibre ceilings on metal strip profiles.

General architectural finishes shall conform to the standards outlined in the table below. Furniture's will be supplied by Owner except for specialist areas such as control rooms and other technical buildings.

Location	Finishes		
	Floor	Wall	Ceiling
Workshop & Store	Heavy duty floor surface hardener Anti-dust finish	Profiled metal insulated cladding with dado wall under. Cladding to have protected colour coating. Wall rendered externally, emulsion painted fairface block internally	Not required
Battery Rooms	Acid resistant ceramic tiles. For sealed batteries - Terrazzo tile	Plaster and emulsion paint	Suspended fibre cement board
Demineralisation Building (as applicable) Laboratory (as applicable)	Heavy duty floor surface hardener Anti-dust finish Acid resistant finish in chemical areas	Fairface block and emulsion paint. Acid resistant tiles in chemical areas	Not required
Admin / Office Building	Vinyl tiles	Plaster and emulsion paint	Suspended acoustic mineral fibre board
Control Rooms/Computer Room Electronic Equipment	Anti-static vinyl tiles on raised computer type floor	Plaster and emulsion paint	Suspended acoustic mineral fibre board
Switchgear Rooms	Terrazzo tile, or Anti-static vinyl tiles on heavy duty raised floor system	Plaster and emulsion paint	Emulsion paint on concrete
Air Compressor Building Gas Reduction Station (only if applicable)	Heavy duty floor. Surface hardener, Anti-dust finish	Fairface block and emulsion paint	Not required

9.5.10. Furnishing and Fittings

The Steam Turbine building shall be furnished with an overhead travelling crane with sufficient lifting capacity in order to ensure effective maintenance. The removal of generator rotor shall be possible without need to demolish any fixed structure. Regarding the maintainability of various components of the Plant certain practical means will be considered:

GT area shall be provided with adequate lay-down and service areas during the major overhaul.

ST zone shall be provided with adequate lay-down and service areas during the major overhaul.

The mechanical workshop shall be equipped with a crane of sufficient capacity taking into account the possible heaviest component (certain pumps and motors) to be maintained.

For the electrical and instrumentation workshops a manual chain hoist shall be provided.

The stores shall be provided without racking system. The warehouse shall be equipped with at least a 5 ton crane or hoist.

9.5.11. Fire protection / separate fire compartments

The Plant shall be divided into fire compartments. Special attention shall be paid to fire barriers and penetration seals in separating walls and floors all over the Plant.

From every exit areas there will be two exit routes to the exit stairways or directly outdoor or to another fire compartment.

Exit routes will be provided with emergency and exit lighting and panic bars. The exits and the access to them shall be marked.

The fire detection and fire fighting systems are specified in section 6.8.

9.5.12. Sanitary and cold water plumbing

Sanitary plumbing

The sanitary, drainage and vent system shall be designed in accordance with the Local Requirements. All floor drains within toilets and tea and mess room areas shall discharge into the external sanitary sewer as per requirements of waste disposal. .

Cold water plumbing

The cold water plumbing shall be designed, installed and tested in accordance with BS6700 (1997): "Specification for the design, installation, testing and maintenance of services supply water for domestic use within buildings and their curtilages", and installed in accordance with local water supply regulations and approved by the Governmental Instrumentality. In general for drinking, cold water will be provided at a temperature below 20°C according to BS 6700:1997 (2.2 Cold water service).

As a general guideline, Contractor shall provide:

- For washing (to WWC, urinals, shower and sink) water at ambient temperature will be provided.
- For safety eye washer, tepid flushing water will be provided according to ANSI code (ANSI Z.358.1). All piping for the safety eye wash stations that are exposed to direct sunlight shall be insulated.

Hot water plumbing

The hot water plumbing shall be designed, installed and tested in accordance with CP342, Part 2 (1974): "Code of Practice for centralised hot water supply to buildings other than individual dwellings", and installed in accordance with local water supply regulations and approved by the Governmental Instrumentality. Hot water shall be supplied in accordance with local requirements for each building and its purpose of use. Water heater used shall be of electric water heater type.

Floor and equipment drainage

Plant rooms shall be provided with floor drains for general floor washing. In addition, individual floor drains shall be provided for condensate drainage. Floor drains for washing water and condensate shall be provided with water seals and connected to building perimeter drains.

9.6. External Works

Refer to Exhibit B, Part 1.

9.6.1. Transformer compound design

The transformer shall be supported on large reinforced concrete bases. Where recommended by NFPA the transformer shall be enclosed by fire walls on three sides.

The fourth side shall be provided with a removable security palisade type fence fitted with a lockable personnel access gate or as required under the SSDs.

In addition to normal loading design requirements, the transformer walls shall be designed to contain and confine damage caused by an explosion of the transformer, as far as practical.

The compounds shall be sized to permit safe working and provide room for installation, maintenance, removal and cooling of the transformer. Provision shall be made to collect the oil from any transformer in the event of a major explosion or other cause leading to the loss of the full oil contents.

9.6.2. Turbine foundations

The foundation design shall comply with the German Standard DIN 4024 Parts 1 & 2 or an equivalent international standard. The design concept of the foundation block shall ensure that there are no structural resonant frequencies closer than + or - 20% of the machine operating frequencies. For the pedestal foundation the model used for the dynamic analysis must be three dimensional and include the lower foundation and soil stiffness. As far as possible the structure shall be symmetrical about the centre lines both in plan and elevation.

The Foundation design shall be subject to the turbine supplier's requirements and approval of the Owner/Owner's Engineer, and will ensure that:

- the machine can operate efficiently and reliably without unacceptable vibration or misalignment
- the foundation itself suffers no damage or settlement sufficient to cause the machine to malfunction or to affect alignment
- the waves propagated through the soil by vibration of the foundation should not be harmful to persons or to adjacent structures, pipework and sensitive machinery or processes

The manner in which the machine parts are supported on their seatings in the structure top shall provide facilities to enable adjustments to be made, if found necessary, at a later date, for maintaining correct casing alignment and obtaining the required curvature of the shafting.

The supporting structures shall be set into position under the supervision of an appointed representative of the turbine generator manufacturer, who shall check and approve the settings and the relevant tolerances prior to and upon completion of casting the surrounding concrete.

The turbine-generator manufacturer is to supply a set of datum level monitoring plates which shall be grouted into the foundations. The number and spacing of these mounting points on the foundation block and on any other suitable surfaces shall be recommended by the turbine generator manufacturer and be subject to approval. When each block is complete, prior to the erection of the turbine, the Contractor shall record an agreed set of datum levels. Subsequent level measurements shall be recorded at key stages of turbine erection in order to monitor any foundation settlement.

9.6.3. HRSG foundations

A reinforced concrete slab shall be provided under the HRSG. All drains from all equipment in the area shall be contained and routed properly so that they are discharged to the waste water system. Bunding shall be adequate to prevent any overspill of blow-down water under any conditions.

The supporting structures shall be set into position under the supervision of an appointed representative of the HRSG manufacturer, who shall check and approve the settings and the relevant tolerances prior to and upon completion of casting the surrounding concrete.

Any floor mounted equipment within the Plant buildings shall be mounted on at least 150mm high concrete plinth

9.6.4. Roads

Contractor shall provide all roads as per Exhibit B, Part 1.

Contractor shall take into consideration the loading capacities and clearances of existing bridges and connecting roads when carrying out transportation from the ports or from other places to the sites and vice-versa.

The Contractor shall provide all permanent roads within the Plant Sites. The pavement construction shall be suited to the type of traffic likely to use the roads during construction and if applicable, during the lifetime of the Plants. All the internal roads shall be provided with concrete lining on either sides.

The sub grade earthworks for these roads shall be compacted to 95% maximum dry density for the top 600mm. All other associated earthworks shall be compacted to at least 90% maximum dry density. Local load tests will be performed during the construction of the roads

The road surface shall be asphalt. Roads for common and frequent use by the Plants shall be designed of sufficient width and load.

Hardstanding areas shall be provided around all plant, where mobile cranes or equipment will be used for maintenance. The compaction will be function of the load require for the erection.

As GT is located outside, Contractor shall also ensure unhindered crane approach for any major maintenance.

The general Plant layout, including laydown areas, shall be submitted for Owner's approval in the detailed design stage. Changes, if any, requested by Owner shall be carried out by the Contractor without any cost or time impact to the Owner.

9.6.5. Paved areas

Concrete paving

Concrete paving shall be used for areas subject to high risks of fuel spillage/leakage. They shall be designed to support the heaviest wheel loads of the vehicles that will operate at or frequent the power and water station.

Gravel paving

Gravel paving shall be constructed with 40mm max. crushed rocks or gravels of approved grading and quality, densely compacted to 100mm thickness. If the gravel paving is applied as driveway, the surface shall be blinded with rock chippings to form a reasonably smooth serviceable driveway.

Other areas

The areas not to be paved shall be levelled, and finished with a 75mm minimum thickness of 20mm single sized pebbles or stones.

9.6.6. Permanent fences

All security fences for the FCHPP and ESF as required in Exhibit B, Part 1 shall be provided by the Contractor.

9.6.7. Cable trenches

Where direct burial is not suitable, underground cables and pipes shall be in trenches. The trenches shall be designed and constructed to provide adequate access for maintenance purposes. Trenches shall be provided with removable reinforced concrete cover slabs. Electrical cables may be laid in PVC-cable ducts. Routes shall be provided with suitable, appropriately located manholes. Electrical cable pull pits shall be designed and tested for 100% water removal.

The above ground pipe racks shall provide headroom of 4.5m as minimum in the trafficable areas.

Cable trench bottoms shall be laid to falls and drained at regular intervals along their lengths. Trench covers shall be suitable for the general area loading through which the trenches run. All necessary equipment for lifting shall be provided for single trench cover heavier than 50kg.

9.6.8. Site installation

Contractor shall provide temporary off site lay-down areas and housing compounds and to bear the respective costs if the Plant Sites and the allocated temporary areas to be provided by Saudi Aramco/SEC are not sufficient for Contractor's camp, equipment lay-down, and office and/or administration requirements. In any case, Contractor welfare camp shall only be located outside Saudi Aramco area, location and cost of which shall be borne by the Contractor.

The Contractor shall optimize and include in its Proposal its requirement for the temporary and future use of the Plant Sites and shall propose corresponding time schedules for the use of these areas. Backfilling and grading of required areas shall be provided by the Contractor, as far as required.

Upon completion of construction, Contractor shall remove from the Contractor's camp site and other temporarily used areas all the buildings, rubbish, unused materials, storage tanks, temporary power distribution lines and all other temporary facilities. Contractor shall fill and dress all holes and cavities made for its convenience and leave all areas in good order and condition.

For avoidance of doubt, Saudi Aramco/SEC shall not be responsible for the provision of land for such off-site areas. The temporary installations include labour camps, offices and car parking for the Owner's Representative and for the Contractor workshop facilities, laydown areas, etc.

9.6.9. General levelling and grading of Site

As per Exhibit B, Part 1, Saudi Aramco shall grade and level the Plant area to MSL + 30.60 m. Any additional levelling and grading as required shall be done by the Contractor. The Contractor shall include for importing any further suitable material that may be required and for levelling and compacting the sites to acceptable standards and grades.

For the final grading of the Sites, the requirements of each complex, the requirements of the general layout of the Entire Plants, the storm water drainage system and other external installations shall be considered.

The finished ground floor slab levels of buildings shall basically be 0.20m above ground level.

Unsuitable material and debris shall be removed from the sites. Pockets of soft or unstable subsoil shall be excavated and replaced by compacted fill material. All filling work shall be carried out in 250 mm layers and compacted before proceeding with the next layer.

9.6.10. Fuel Oil Storage

Fuel oil storage tanks and bunding shall be founded in accordance with the soil investigations and structural analysis and shall comply with the current regulations for the handling, storage and use of flammable liquids. The impervious bunded area for the tanks will be at least 110 % of the total capacity of one tank. This area run-off shall drain to oil water separator.

9.7. Drainage

Refer to Exhibit B, Part 1.

9.8. HVAC systems

9.8.1. General

The functions of the ventilation/HVAC systems are following:

- to provide fresh air to occupied areas
- to control room temperature and RH
- to remove excess heat released from process equipment
- to remove noxious fumes and chemical vapours
- to keep proper pressure differences between certain rooms to prevent the ingress of dust and noxious fumes

Where forced ventilation is provided, the design shall ensure that the total room volume is swept. The supply air shall be taken from outside. Supply air to offices, control and electrical rooms and workshops shall be filtered. Forced mechanical exhaust ventilation can be used in process areas. Sound attenuators shall be installed to prevent excess noise to the neighbourhood (if required).

The main materials of the ventilation equipment and duct shall be galvanised steel.

Special care shall be taken in sound attenuation of the control room air-conditioning system.

The battery rooms have their own exhaust fans.

Air conditioned areas and performances as per Exhibit B, Part 1

Power Control Centers (PCC) associated to the GTs and STs, the Contractor's proven standard shall be applied with respect to indoor design conditions; No return air shall be taken back into the system from battery room.

As a part of the commissioning, a performance test of the air conditioning and ventilation systems shall be carried out, based on a procedure agreed between the Owner's Representative and the Contractor.

The fresh air shall be provided in occupied areas (e.g. control room, offices, meeting rooms, prayer room, computer rooms, sanitary rooms, laboratory rooms, etc) with 30 m³/h person, For non-occupied areas like electrical & switchgear, etc, forced fresh air supply is not required to be provided.

The air handling unit (other than those specifically mentioned in the RFP, which shall follow the RfP requirements) shall be air cooled and suitable for mounting at the roof or wall or the window balustrade with horizontal air discharge. The unit shall comprise:

- casing of galvanized sheet steel, inside sound and heat insulation, outside stove enameled finish.
- centrifugal fan with double inlet. Single inlet may be provided with Owner's approval
- panel filter, filter material washable and removable without removing cabinet, dust weight separation efficiency according to ASHRAE Standard
- air cooler for direct expansion made of seamless copper tubes mechanically expanded to fix aluminium fins, including expansion valve and drain-pan complete with siphon and drain-pipe
- built-in compressor of hermetic or semi-hermetic type.
- complete internal switch and control panel with thermostat, on/off switch, fan state-selector switch
- external louvre for air intake and discharge
- protection of fire areas by means of fire dampers, according to the NFPA guidelines

9.8.2. HVAC automation

The control system of the HVAC equipment shall be a centralised computer based system with an alarm printer. The operator station shall be located in the main control room.

The system shall have appropriate software for operation of the HVAC systems and building services. Full-colour graphics shall include following:

- HVAC application schematics
- equipment images (fans, dampers, etc.)

- sensor, indicator and actuator symbols (temperature, humidity and pressure)
- animation symbols (damper position)

The system shall provide a reporting facilities, such as alarm histories, operator activity log etc.

9.9. Electrical Special Facilities / HV sub-station : civil works

Refer to Volume 2 of Exhibit B, Part 1.

10. GENERAL OPERATIONAL & MAINTAINABILITY REQUIREMENTS

10.1. Uniformity of Supply

The Contractor will endeavour to obtain (together with its sub-Contractors) a uniformity of supply for items such as electrical equipment, motors, relays, main pumps, valves, instrumentation and control equipment, as far as reasonable, to facilitate the training of the maintenance staff and reduce the number of spare parts.

With respect to the package control system when standard manufacturer packages are implemented, the package control system may be designed partly by different manufacturers in order to get the best subsystems and functionality.

All equipment placed in the control room (such as recorders, indicators, push buttons, etc) and electrical rooms (such as electronic modules, controllers) shall be as far as practicable of one manufacturer.

10.2. Interchangeability

As far as possible, equipment of the same type shall be identical and interchangeable, with, eventually, the small necessary adjustment work with an item of the same type.

10.3. Selection of Materials

Particular care shall be taken to corrosion prevention by, among others, choosing adequate materials, limiting heterogeneous welding to the minimum and protecting the equipment against galvanic and electrostatic couples.

All materials shall be new (subject to prior usage and storage as required under this Agreement) and of best quality, free from defects or damage and shall be of a class suitable for working under the local conditions.

The material shall withstand all expected physical working conditions without undue distortion, stress or deterioration.

Materials to be selected by the Contractor shall be proven to be eligible and sufficient for the complete design lifetime of the Plant.

In material choices, main attention shall be paid to compatibility of materials and operating conditions. Also mechanical, manufacturing and maintenance characteristics (e.g. strength and weldability) of materials shall be considered as important criteria of choice. The materials to be chosen shall be standardised and represent long-term operational experience.

Components of the CCGT Plant which are subject to abrasion or potential high rates of wear shall be selected from proven grades of high corrosion/erosion resistant materials.

To prevent galvanic corrosion, use of improper material combinations (e.g. carbon steel/stainless steel) shall be avoided.

In circulating water-steam cycle copper-based alloys are not accepted.

Asbestos is not allowed to be used.

10.4. Operating and troubleshooting criteria

10.4.1. Overall Maintainability

The ambient conditions (particularly high temperature and sand storms) and possible outdoor installation principles are to be taken into account when planning the maintenance and service arrangements for the Plant equipment and systems:

- valves, instruments and other equipment shall be easily and safely accessible for maintenance and operation.
- the main equipment and systems should have dust and rain protection.
- places requiring regular maintenance shall have proper weather and sun protection

The steam turbines shall have one permanent suitable overhead travelling crane for the heaviest lift with a weather protected “parking” cover. The GTs will require scheduled maintenance every year and thus a permanent service/dismantle/lay-down area is to be provided. The Steam Turbine crane shall be equipped with main and auxiliary hooks.

As no building is foreseen for the Gas Turbines, the Steam Turbine hall and overhead crane shall be:

- suitable to perform ST overhaul and maintenance
- suitable to perform full overhaul GT maintenance (including GT de-stacking if required)

The weather protection of the service space is to be taken into account whether it is temporary or permanent one.

The maintenance of each GT/HRSG unit shall be possible when the other GT/HRSG unit(s) are running on combined cycle mode.

The Contractor shall, as far as practicable, harmonise and standardise on the manufacturer, size, type, rating, lubricants, control systems, materials of construction of equipment and components supplied under the Contract. This applies especially to shut-off valves, actuators, field instrument, switchboards and electrical motors. Such standardisation shall in no way compromise the suitability of each item to perform its required duty. The objective of standardisation shall be to rationalise the spares inventory, to minimise the maintenance requirements and to ensure that safety of operation is not compromised.

The requirement for standardisation shall extend to all platforms, steps and ladders to ensure their uniform function and appearance throughout the Plant.

The scope of Work shall include all necessary special tools for scheduled maintenance or scheduled inspections. This includes also required routine inspection software equipment.

Attention shall be paid to the proposed strategic spare parts by the Contractor: e.g. 50m of each type of HV cables applied in Plant shall be included in the Spare Parts proposal by Contractor.

10.4.2. Maintenance Access Criteria

It shall be assumed that all Plant mechanical, electrical and control equipment will require access for maintenance purposes periodically during the life of the Plant. The following general criteria shall be used as a guideline for locating devices either as singular components or packaged systems from vendors:

- one person without hindered access can lift about 30 kg
- two persons each without hindered access can lift about 45 kg

If access is hindered or weights are greater than 45 kg, means of lifting like permanent monorail, lifting beams and hoists with a chain block or via chain fall, or with fork-lift with unicorn attachment, or in the event of special cases, mobile (any mobile crane or fork-lift post COD is to be provided by the Owner) or permanent cranes shall be anticipated. In the case of some electrical equipment, trolleys or special handling gear are available from the Contractor and these should be included in the Work if other means are not available.

Disassembly of components into sub-assemblies shall not generally be accepted as a means around the weights above. For instance, while it is acceptable to take the diaphragm operator and yoke off a control valve as a sub assembly, it is not acceptable to break the diaphragm operator into several pieces for meeting the weight criteria.

A clear identified path from the device to a maintenance corridor for hauling/transport purposes shall be specifically identified if not obvious from the arrangement. If the device is at an elevated location in the Plant, a clear method of getting the component to a service platform or road shall be identified if not obvious.

Where mobile cranes are required for maintenance purposes, there shall be specific plans to identify placement of the crane and specific make and model rigs shall be identified for the duty. Use of greatly over capacity rigs for light components because of long reach shall generally be avoided.

Back clearance from the work face area shall be a minimum of 600 mm at a point, and preferably be greater than 750mm. In the case of pumps (or other rotating equipment), these clearances should be maintained on each side of the pump centreline. An unobstructed path from the work area to a plant maintenance corridor shall be established in the plant design.

The lifting equipment shall conform with the latest edition of the international standards.

Material test certificates shall be provided and dynamic and static load test shall be performed on site in the presence of the owner's representative.

10.4.3. Operating and Troubleshooting Criteria

Beyond the routine access for observation of equipment operating parameters and maintenance (including lubrication and chemical fills), the following specific criteria shall be followed:

- all control or remotely operated valves shall be accessible from floor level, stairway or permanent platform meeting the maintenance criteria above. elevated platforms shall be accessible from stairs, inclined ladders or vertical ladders (in order of preference)
- all electrical/electronic panels containing relays or solid state circuitry potentially requiring replacement or troubleshooting shall be protected with a local sun/rain cover to protect technicians

- primary control sensors and their sensor point valving do not need to be accessible to the above criteria as long as they can be reached via an extension ladder of less than 5 meter length or a man lifting platform. if access to these devices needs to be from a ladder from an elevated platform, the ladder length shall be limited to 5 meters and the platform from which it will be supported shall be accessible via stairway

In congested local areas (subject to Owner's review), it is permissible to assume that local rolling ladders/stairs can be used for access to elevated devices less than 5 meters above a permanent platform or floor for either operational or maintenance purposes. Maintenance corridors shall not restrict movement of such stairs.

10.5. Plant Operation and Manning level

The Contractor shall take into account the O&M manning level throughout the Work including especially, but not limited to, automation concept, control room, training, office rooms, meeting rooms, staff room, sanitary rooms, car parking slots, workshop and temporary office space during construction.

11. STUDIES / ENGINEERING

As per Codes and Standards (refer to section 4), Exhibit B, Part 1 and Good Industry Practice.

12. FABRICATION/MANUFACTURING / CONSTRUCTION

In addition to Exhibit B, Part 1 and Good Industry Practice, the Contractor shall also undertake any necessary studies (e.g. power system studies, load flow calculations etc.) to make the system design complete.

13. PACKAGING, TRANSPORT, STORAGE AND HANDLING

All parts shall be carefully packed to ensure their protection against shocks and weather conditions during their transport and outdoor storage if necessary before being put into place.

All metal parts subject to corrosion shall be protected by layer anti-rust paint/temporary transport protection or by air tight plastic sheets. Bags of SILICAGEL or equivalent drying agent shall be placed, in sufficient number, in cases containing equipment sensitive to moisture.

All tube, header and pipe ends shall be protected by plastic caps.

If temporary coatings should be necessary, they shall be provided and completely removed before or during commissioning, except for insulated parts.

14. ERECTION / SITE ACTIVITIES / SITE MANAGEMENT

The Contractor shall provide a Site Organization scheme with nominated personnel, their names, attached with qualifications and key personnel CVs with identification of responsibility and authority illustrating how the management structure/lines of authority will function. The organization shall be such as responsibilities are clearly and effectively divided to each organization level and project personnel. The Owner shall have clear understanding of Contractor's organisational responsibilities and decision process in Contractor's organisation.

14.1. Site Facilities

14.1.1. Contractors Central Administration Area

The Contractor shall submit outline drawings showing proposed locations of its and its sub-contractors central office area together with any other temporary buildings, storage compounds, car parks etc.

14.1.2. Contractor's Accommodation

Before the erection of any temporary site accommodation commences, the Contractor must obtain approval from the Owner of the size, type, condition, location, access and services. The Owner's approval shall be obtained before any accommodation is dismantled or removed from the Site. The Contractor's accommodation shall be reasonably maintained and suitably identified with the Contractor's name and that of his Site Engineer.

14.1.3. Owner's Accommodation and Facilities

The Contractor shall provide near to Contractor's site offices:

- project site temporary offices for the owner's project team *and* O&M
 - from NTP: five (5) –employees - and owner's engineer team – ten (10) employees, - *and O&M representative -1 employee*, including at least 2 conference rooms, 7 office rooms, , drawing office facilities with ample storage for records as well as accommodation for 2 clerical staff with filing and reproduction facilities. Washroom with 1 shower shall be included, as well as all facility connections (water, sewage, electricity, telephone, internet...)
 - *12 month after NTP, 4 additional O&M employee in 2 additional office rooms*
- access to contractor's food facilities will be available for the owner's and Owner's Engineer's site staff

- computers, telephones, fax, printers are not included however, the Contractor shall provide necessary LAN network and internet connectivity (at least 8 Mbps) for the Owner and Owner's Engineer's site staff
- costs of consumables (water, electricity, phone...) and cleaning will be borne by the Contractor (international calls will be paid by the Owner)
- site preparation and access roads for temporary offices

These temporary offices will be made available to the Owner at the same time as Contractor's first site offices.

The quality of the Owner's temporary offices shall be equivalent to Contractor's site offices.

14.1.4. Site Records Office

The Contractor will be required to maintain a suitably staffed records office to maintain updated records of all relevant documentation and test certificates and to see that these are available to meet obligations to all statutory bodies and to any commissioning committees, working parties or test teams which are established to meet the needs of the Contract. These records shall be maintained and retained until handed over on completion of all works on Site.

The relevant documents envisaged will include:

- site register for lifting appliances, cranes, slings, hooks, lifting beams and tackle for construction of the works either mobile or fixed.
- site register for lifting appliances, cranes, slings, hooks, lifting beams and tackle supplied as part of the permanent works
- scaffold register
- site works permit system records
- site labour records
- site working drawings
- commissioning documents
- test certificates from works and site.
 - civil
 - mechanical
 - electrical
 - control and instrumentation
- operation and maintenance instructions
- quality assurance documents

14.1.5. Telephones & IT

The Contractor shall provide his own on-site/off-site telephone, fax and LAN facilities.

14.1.6. First Aid

The Contractor shall, in collaboration with and to the requirements of the local health authorities, ensure that medical staff, first aid facilities, sick bay and the ability to obtain ambulance services are available on the site at all times.

These arrangements are not intended to relieve the Contractor of his law and statutory obligations other than his obligations to provide such first aid post under “Site Security”.

The Contractor shall arrange a security facility for the Plant and temporary storage areas, consisting of road access gates manned by uniformed security staff. The Contractor shall submit details of his proposed site security policy and procedures.

At a time of the Taking Over the security for all or part of the Site will be taken over by the Owner.

The Contractor shall ensure that only personnel necessary for ongoing work are permitted onto the site and that all admitted personnel are aware of all precautions to be observed.

14.1.7. Site security

The Contractor shall arrange a security facility for the Plant and temporary storage areas, consisting of road access gates manned by uniformed security staff to safeguard the materials and stores in the area allocated to him, together with the Work on the Site. The Contractor shall submit details of his proposed site security policy and procedures.

At a time of the Taking Over the security for all or part of the Site will be taken over by the Owner.

The Contractor shall ensure that only personnel necessary for ongoing work are permitted onto the site and that all admitted personnel are aware of all precautions to be observed.

The Owner or Saudi Aramco or SEC will not accept responsibility for any losses or damage which may occur during the execution of the Work.

14.1.8. Contractor's Storage Areas

Contractors shall submit details of their requirements for laydown areas. These areas shall be surfaced with self-draining materials for weather protection.

The hardstanding for external storage together with foundations and drainage for the covered storage shall be provided by the Contractor.

14.1.9. Sign Boards

The erection of sign boards or posters will not be allowed without the Owner's approval of the size, type, location, wording, etc.

14.1.10. Site Cleanliness and Waste Disposal

Throughout the period of construction of the Work the Contractor shall maintain the whole area of his operations in a clean, tidy and safe condition and shall arrange his materials in an orderly manner. The Contractor will be responsible for waste collection and transfer to a suitable disposal site.

All rubbish, etc. shall be systematically cleared from the working areas and, if not removed directly from the Site, shall be deposited at general collecting points provided by the Contractor pending the removal from the Site. Where practicable, screening shall be provided to prevent the ingress of dust and dirt to any part of the Work.

The Contractor shall ensure that during the Contract work he does not cause pollution of the Site land drainage system by either dumping rubbish, fill materials or waste products from any source whatsoever.

In the event of an incident whereby the pollution is attributable to negligence on the part of the Contractor then the Contractor shall clear this pollution as soon as possible and make early arrangements to remedy the cause.

The Contractor shall prohibit the committing of nuisance by his employees within the Site or upon either the Owner's land or others, and any employee found violating this provision shall be immediately discharged and not be employed again on this Contract.

In addition to his statutory obligations as stated in the Conditions of Contract the Contractor shall make himself familiar with all requirements relating to the discharge of effluent from the station and shall take all steps to ensure that these are not contravened by the disposal of effluent from the site.

No dangerous or noxious waste products, chemicals or materials shall be disposed of, on or off the Site without the Owner's approval.

The Contractor shall take all necessary measures to ensure the safe collection and disposal of waste oils pertaining to those Work or plant for which he has responsibility and to ensure the prohibition of:

- any discharge of waste oils into surface waters, ground waters or drainage systems of any kind

- any deposit and/or discharge of waste oils on to the ground or floors of buildings and any uncontrolled discharge of residues resulting from the processing of waste oils or the cleaning of oil contaminated surfaces with solvents
- in order to ensure compliance with this requirement the contractor shall provide suitable vessels for the collection of all waste oils. where large quantities of waste oils need to be stored prior to being disposed of in bulk, suitable storage tanks shall be provided by the contractor for this purpose
- any processing of waste oils causing air pollution which exceeds the level prescribed by existing statutory provisions. the burning of all waste oils should only be carried out following written permission being granted by the owner's site engineer

14.1.11. Sanitation and Drainage

The Contractor shall be responsible for the foul drainage and the disposal of all sewage from the Site during the construction period in accordance with all consents and licences or instructions from the local authority. Foul water and sewage shall not be discharged untreated from the site.

14.1.12. Sanitary Facilities

The Contractor shall provide temporary changing facilities, sanitary facilities and amenities. The Contractor shall maintain and operate these facilities in a clean and tidy condition for the duration of its use at the Site.

14.1.13. Scaffolding

Scaffolding shall be erected by qualified Scaffolders working under the supervision of experienced supervisors. Should the scaffold be required to remain in position for some time, it must be inspected and certified at weekly intervals by a competent engineer of the Contractor and the accredited representative of the scaffold Company. Should any fault be found on either the initial or subsequent inspections, these faults must be corrected before any personnel may work on the scaffold.

14.1.14. Use of Motor Vehicles and Parking

Parking facilities shall be provided by the Contractor in a compound within the Site boundary but outside the security fence of the construction area and as near to the Site access as possible. The Contractor's employee's private vehicles will not be permitted to enter the construction area. Only Authorised vehicles shall enter the construction area and these must display the vehicle pass issued by the Contractor.

Designated parking spaces shall be allocated local to the main site office for the Owners Project Manager, Owner's Engineer team and other senior project staff.

14.2. Health & Safety Requirements

The Contractor shall prepare health, safety and environmental (HSE) instructions for the project for Owner's review. The Contractor's activities at Site shall meet all requirements set out in the Owner's HSE guidelines as provided in Exhibit T to this Agreement, the EIA, the local legislation and World Bank Group Environmental, Health, and Safety Guidelines and shall be carried out according to a well-structured plan and under a comprehensive set of safety rules and procedures.

The Contractor shall familiarise all people working at Site with any HSE related aspects. In general terms the following instructions shall apply:

- each member of any site teams shall wear helmet and protective shoes
- goggles shall be used when entering to or working at areas where eye protection is required
- safety harness with lifeline shall be used when entering to or working at high level (safety belt is not accepted)
- each member of site team shall wear proper work clothing
- other personal protective devices shall be case by case, e.g. hearing protection over 85db noise
- no intoxicants are allowed or shall not be used at site or at any of project areas
- proper courtesy rules and good conduct shall be maintained in all relations between the owner's personnel and the contractor staff and his sub-contractors and with others involved in the project

Each Site team member shall respect the HSE requirements and follow the project rules.

14.2.1. HAZOP study

An overall Hazard and Operability Study ("HAZOP") and SIL assessment and calculation shall be carried out by the Contractor for all plant systems by appointing a group (or groups) of multi-disciplinary specialists familiar with the HAZOP technique and the type of plant or system being assessed to evaluate problems that may represent risks to personnel, equipment, environment or prevent an efficient Plant operation. In addition, all interfaces with other parties (e.g. Saudi Aramco / SEC) shall also be subjected to detailed HAZOP studies.

P&IDs and other required documents shall be systematically and in order studied at the system component level for parameter excursions outside the normal design mode in terms of (but not limited to) radiation, pressure, temperature, flow, composition or electrical power. Each parameter shall be reviewed via guide words such as 'more', 'less', 'high', 'low' 'no' or 'reverse' to establish the causes and consequences of such variations. Then, current safeguards shall be analysed and, if required, additional measures should be proposed to reduce the risks of the analysed equipment or system. Results shall be recorded in an orderly manner.

HAZOP's shall be scheduled 8 weeks in advance with Owners and their Engineers and the Owners shall be represented at all HAZOP sessions. As a minimum, Contractor shall perform a preliminary HAZOP study after the basic engineering to incorporate its findings into the basic design and a final one during the latest stages of the detailed design.

HAZOP findings shall be followed up and design changes incorporated where deemed necessary. HAZOP computer based aids can be used to assist the HAZOP process.

14.2.1.1. HAZARDOUS AREAS CLASSIFICATION

The Contractor shall carry out a hazardous area classification of the Work, and comply with any mandatory norms and standards. The Contractor shall provide drawing and the list of rooms/areas defining the classification of the Plant and areas near equipment. The classification of the areas shall include all necessary information on:

- temperatures (ambient, surface, internal)
- dust and water conditions
- fire hazard
- explosion hazard (incl. analysis and descriptions in accordance with ATEX requirements)
- requirements for electrical equipment (IEC 60079)

It is acceptable only for the GT OEM's equipment to use alternative protection techniques as defined in NFPA 70 Article 500 for equipment in hazardous areas.

Both normal and exceptional conditions shall be considered including exceptional weather conditions, any equipment or pipe breakage as well as operational errors.

14.2.2. Safety Management

Industrial Health & Safety of the highest standard is required of all contractors. Safety must form an integral part of the line management control of the site construction works.

The Contractor has full responsibility for the management of health and safety on site (including Plant site and Share Facilities sites), and shall co-ordinate the activities of all sub-contractors to ensure that the requirements of the contract are satisfied, compliance with all aspects of Health & Safety legislation and authoritative guidance, and the establishment and maintenance at all times of a safe and healthy environment.

The Owner reserves the right to audit the management and control of health & safety on site at any time.

All contractors shall co-operate with the Auditor may be appointed by the Owner and shall make available to the Auditor such information records and access to places of work as required by the Auditor in pursuance of the Audit.

The Contractor shall bring the requirements of the contract to the attention of all sub-contractors.

14.2.3. Safety Plans and Method Statements

The Contractor shall prepare a detailed Site Safety Plan and submit it to the Owner for review. The Contractor shall prepare method statements for work which they consider requires detailed instruction and will be responsible for agreeing these, where necessary, with their sub-contractors. Before any site construction work begins, the Contractor shall submit several typical method statements, to provide an example to the Owner that a suitable standard will be achieved. Subsequently where deemed necessary by the Contractor and in order to meet his safety obligations, method statements will be produced and submitted to the Owner at least 30 days before work commences on the activity covered by the statement.

The Contractor shall require all sub-contractors to submit Site Safety Plans and method statements to him for approval before sub-contract work commences. This process is open to Audit by the Owner's Auditor.

Site Safety Plans should include all the relevant subjects, and specify how safety is to be achieved in each case.

14.2.4. Safety Rules

The Contractor shall comply with the requirements of the following documents and all relevant Country legislation where appropriate, and with the rules relating to the issue of 'Safety Documents' contained therein.

- Saudi Aramco and SEC Safety Rules

- Safety Rules related to construction in Saudi Arabia.
- Owner's General HSE guidelines as per Exhibit T to this Agreement

14.2.5. Safety Personnel

The Contractor shall notify the Owner of his Directors and Managers primarily responsible for implementation of the safety system.

The Contractor shall appoint a full time Site H&S Officer to co-ordinate the safety activities of his own and sub-contractors activities.

Contractor shall appoint one safety supervisor for every 50 workers.

Details of the proposed appointment together with curriculum vitae shall be submitted to the Owner before site work commences.

14.2.6. Safety Training

The Contractor shall ensure that no individual commences work on site unless he has successfully completed a Safety Induction training course specific to that site. The training is to include all sub-contractors or personnel working on site (including CCGT plant SF).

The Contractor is responsible for ensuring that all required specific activity training, e.g. scaffolders, crane drivers, etc has been carried out.

Where work on site interfaces with the grid or gas/liquid fuel operating system, the Contractor is responsible for ensuring that relevant Safety Rules are applied, and that all such training and nomination as required by the said rules are completed.

The training process is open to audit by the Owner's Auditor.

14.2.7. Safety Meetings

The Contractor shall establish and chair at senior management level a monthly site safety meeting, to which management representatives and safety officers of Contractor and his sub-contractor shall attend.

The purpose of the meeting is to discuss, co-ordinate and resolve all safety matters of concern on site, and examine current and future activities to ensure a consistently high standard of health & safety on site.

The Contractor shall be responsible for collating all site accident statistics, and these shall be considered and reviewed at the meeting.

The Contractor shall bring to the meeting's attention for comment and action, any reports arising from Audits undertaken by the Owners Auditors.

The Owner reserves the right to observe the proceedings at site safety meetings.

14.2.8. Typical Items for Inclusion in a Site Safety Plan

The list below is not intended to be exhaustive, and is provided for guidance purposes only. The Site Safety Plan should specify the measures proposed to meet the safety requirements in each areas below.

Management of the Contract

Detail the management and supervisory structure responsible for safety, together with reporting procedures to monitor safety performance. Detail the arrangement for site safety training, especially induction training.

Safe Access to Work Areas

Specify safety precautions for work in hazardous areas, i.e. above ground, steel erection, confined spaces, excavations, underground works and basements. State the preferred access system to be used, e.g. access platform machines.

Management of Scaffold Safety

State contract strategy and detail aspects of design, loading and scaffold management to ensure compliance with Statutes, Standards & Codes. Include scaffolders qualification requirements, identification and inspection procedures.

Plant and Equipment, cranes, Hoists & Lifting Gear

Detail the arrangements to ensure the management and safe use of all such equipment, e.g. method statements are required for complex lifts. Include provisions for statutory test, certification and inspection, and maintenance procedures. Specify operator training and authorisation to ensure competency.

Electrical Equipment

Specify site voltages, maintenance and inspection of all fixed and portable electrical equipment. Describe systems of work and management control of equipment.

Welding & Burning

Describe arrangements to ensure safe use of gas and electric welding equipment. Include control and storage of gas cylinders, fire and explosion precautions. Note: flashback arrestors are mandatory on the Site.

Substances Harmful to Health

Detail how compliance with Country Regulations is to be achieved. Provide a list of all hazardous substances, as defined in the regulations, with a copy of the site specific assessments.

Personal Protective Equipment

Detail situations requiring the use of Personal Protective Equipment (PPE) and specify policy regarding its wear, use and control.

Emergency, Welfare & Health Provisions

Describe facilities provided for health and welfare, rescue, fire fighting, first aid and control of site emergencies.

Company Safety Policy

This should be included as an addendum to the Site Safety Plan. A copy shall be provided with the Proposal, it should be signed and dated by the Director/Manager responsible for company safety.

Common Hazards

The following list indicated some of the hazards commonly encountered on construction sites. Describe measures and precautions to achieve safety, this may be in the form of a company safety manual/booklet e.g.:

- Legionnaires disease
- abrasive wheels
- lead poisoning
- compressed air
- insulation materials
- buried services
- flammable liquids & gases
- glass containers
- airborne dust & particles
- waste products
- working near water fire
- noise poor illumination
- ionising radiation sources
- site transport
- high voltage apparatus explosives
- Leptospirosis

14.2.9. Permit to Work system

The Contractor shall establish its own permit to work system applicable for the Plant until handing over to the Owner. This system is subject to Owner approval and shall as a minimum comply with the Owner's safety rules.

Where work on Site interfaces with the Owner's, Transmission System Operator's or Associated Works Contractors' operating systems, the Contractor is responsible for ensuring that the relevant safety systems and procedures (or other such safety procedures operated by the Owners or their Operations and Maintenance Contractor, Transmission System Operator or Associated Works Contractors) are applied, and that all necessary training, authorisations and nominations are completed. These do not absolve the Contractor for ultimate responsibility for all safety matters on the work area under its control.

14.2.10. Obstruction to access

All work at the Site shall be carried out in such a manner as not to obstruct the operation of any other Contractor on the Site, or in the latter stages of the Project the operation of the Plant equipment.

The Contractor shall also allow the Saudi Aramco, Owner, and Owner's representatives unhindered access to all parts of the Work in progress and where normal access is obstructed he shall provide temporary alternative access at his cost.

The Contractor shall provide fencing and barriers to segregate the construction areas from operational areas within the Plant.

14.2.11. Clear road access

The Contractor shall take precautions to keep all temporary and permanent roadways clear of any spillage from his traffic to the satisfaction of the Owner's Engineer. All such spillage that occurs shall be cleared immediately at the Contractor's expense, including excessive earth brought in on wheels or tracks of the traffic, or other materials.

14.2.12. Site buildings

The Contractor shall provide and maintain an office on or near the Site and other buildings as necessary for the due performance of the Work. The office shall be open at all reasonable hours to receive instructions and other communications in the form of manual delivery, post, telephone, telefax and email.

14.2.13. Site labour

The Contractor shall include for the provision of all necessary supervisory, skilled and unskilled labour for erecting, testing, commissioning, and maintaining the Work together with all necessary transport, accommodation where required, and other services necessary to support his supervisory and labour force.

14.2.14. Sanitary and health control

The Contractor shall implement a programme of health education to ensure that all appropriate personnel working on the Site are aware of health risks associated with living and working together with necessary protective measures.

All persons connected with the Work shall be obliged to use the sanitary facilities provided and maintained on the Site by the Contractor.

The Contractor shall be responsible for the removal from the Site and ultimate disposal of his refuse in accordance with any local regulations.

The facilities provided and the standards maintained shall be in accordance with the WHO Guidelines as a minimum.

Special precautions shall be taken by the Contractor to keep the incidence of mosquito borne and other diseases to a minimum. These shall include spraying with approved insecticides the insides of all buildings under his control, and avoiding the formation of pools of water and other likely mosquito breeding places within and adjacent to the Works.

14.2.15. Dumping

The Contractor shall ensure that the methods employed in handling, storing and disposal of all dangerous or noxious substances are such that pollution or contamination of the Site or surrounding areas shall not occur. Waste materials shall not be dumped in rivers or on the Site but shall be deposited in an approved off-site pit or disposal area or otherwise disposed of in accordance with the approved environmental procedures. The Contractor shall be responsible for provision of the approved disposal facilities, including obtaining of all necessary permissions.

14.3. Radio Telephone Systems

The Contractor shall obtain permission from the appropriate authorities before any radio telephone system is used on the Site. This includes cordless and cellular telephones as well as other radio systems.

The Contractor should ensure that the proposed system will not cause interference to control and instrumentation equipment installed on the station during commissioning or subsequent takeover.

The Contractor shall provide details of the equipment during the Project construction phase.

14.4. Noise

Construction processes shall be planned at the outset to minimise noise emissions.

The noise levels shall be measured according to ISO standards.

To meet the noise level requirements the Plant is required to be equipped with standard noise features.

All machinery which exceeds the permissible noise limits shall be equipped with silencers or with lagging or specially designed acoustic enclosures.

The ventilation systems of acoustic enclosures shall be provided with silencers before air rejection to the outside.

Exhaust to atmosphere for safety valves and steam vents shall be equipped with silencer.

Contractor will provide an Ambient Noise distribution report.

14.5. Damage to Property

Damage to any property belonging to Saudi Aramco or SEC or any other contractors due to accident or exposure during and after construction activities and operation, shall be restored by, and for the account of, the Contractor. The Contractor shall be held responsible for any damage done to adjoining property.

The Contractor shall draw particular attention to the Laws of the Kingdom of Saudi Arabia including applicable SSDs, concerning damage to property or disruption of services.

14.6. Consents and approvals

14.6.1. Monitoring of Environmental Impacts

The Contractor shall take care of all monitoring which is required to satisfy authorities' requirements and to fulfil the conditions of the any construction/environmental consents.

14.6.2. Approvals Procedure

Subject to the provisions in the Contract the following shall be followed by the Contractor:

- The Contractor will prepare and maintain a detail programme for getting necessary approvals and supplementary consents for his project implementation. This programme will form the basis for liaison with all relevant authorities and will also describe the information to be provided.
- The Contractor will prepare all necessary material (drawings, descriptions, study reports etc.) which is needed for getting necessary approvals from the relevant authorities.
- The Contractor and his subcontractors are responsible of communicating directly with the authorities. In order to ensure well-coordinated communication with the different authorities, the Contractor has to name a person(s) in charge of coordinating the different tasks related to licensing between the Contractor and the authorities and the subcontractors.
- Prior starting of the design for licensing and permitting the Contractor shall liaise with proper local authorities to clarify the permitting procedures and detailed requirements for design documentation in order to secure fluent permitting procedure and compliance with statutory requirements.
- Prior sending any formal applications a mutual understanding should already be negotiated. Any dispute or further requirement by the authorities has to be reported to the Owner within a reasonable time. The Contractor will keep the Owner informed of meetings with the authorities and give the Owner the opportunity to participate.
- The Contractor will ensure that copies of the complete applications and agreements related to consents are handed over to the Owner for review.
- The Owner and the Contractor will formally request in due time a written confirmation from a relevant authority that the necessary approvals to start the works and progress the works are in place by target dates, if this has not become clear from the correspondence between the Contractor and the authority. The Owner will check that works subject to these approvals to the Contractor shall not commence before the approvals have been issued.
- The Contractor shall comply with the approved design information, and no deviation is permitted, unless a change is approved by the relevant authority and agreed with the Owner.

- The Owner will monitor but the Contractor shall be responsible for whether an approved design change or equivalent reason will lead to a need to change an existing consent condition. Discussions will be held with the Contractor and the relevant authority and thereafter the Variation Procedure shall be followed.

15. GUARANTEED PERFORMANCES

Refer to Exhibit J-1 of the Agreement.

16. COMMISSIONING, START-UP AND ACCEPTANCE TESTS

Refer to Exhibit I of the Agreement

17. QUALITY ASSURANCE – QUALITY CONTROL

17.1. General requirements

At all stages of the manufacture the Contractor shall make use of an effective control system for maintaining the agreed standard of quality. The quality system of Contractor and his subcontractors shall follow the requirements of ISO 9001 Standards (or equivalent).

Quality management plans shall include also the deliveries of the subcontractors.

The Contractor shall carry out quality audits to each of his subcontractors to ensure that the quality of the subcontractors will be compliant with the Contract requirements and Contractor's own quality requirements.

The Contractor or his subcontractors may be subject to quality audit by the Owner. The Owner or his nominated representative shall have the right of visiting the Contractor's or his subcontractors' manufacturing facilities in order to audit their quality control procedures, to observe the tests, inspections and works in progress and to review the test results. The Contractor shall provide any facilities, including access, which may be required by the Owner for monitoring activities.

The Contractor shall establish and maintain a documentation inspection system, which shall be capable of producing objective evidence that all manufactured parts, materials and assemblies comply with the quality requirements of the Contract.

The Contractor shall prepare his Quality Assurance Program arrangements defining a programme of quality control and inspection activities which he or his sub-contractors will perform in order to ensure that the manufacture and completion of the Plant complies with the specified requirements.

The Contractor shall provide necessary facilities and all necessary measuring apparatus for carrying out required factory/on-site inspections and tests.

If applicable, non-destructive tests of the various components, also mechanical tests on the materials thereof shall be carried out at the maker's works in accordance with the appropriate standard. Steel makers certificate of chemical and heat treatment shall be provided in all cases.

17.2. Quality Assurance programme

Contractor shall prepare a Quality Assurance Programme (QAP) dedicated for the project to ensure that the managerial, technical and quality requirements of the Contract between Owner and Contractor are acknowledged and met by the Contractor.

The scope of the QAP shall cover all participating organisations. The basis for QAP is the Quality System of the Contractor (EN ISO 9000 or equivalent). Detailed rules and measures are defined in the Contractor's internal procedures as well as in the Project Plan adjusted by the specific requirements as defined in the Contract.

Generally as a guideline the Quality Assurance Programme shall contain following three levels of procedures:

1. Project Quality Assurance Programme (PQAP)
2. Description of Quality Control Programmes for different components. These include the required inspections, witness of inspections and types of required records.
3. Detailed Technical Requirements and/or QC-Plans for different components and for specific inspections and test items. These QC-Requirements are applied between the Contractor and the subcontractors.

17.3. Quality Control Plans and programmes

Quality Control Plans shall be prepared which identify the requirements for manufacturing, the necessary inspections and tests including complete information on the acceptance criteria and the records specific to the items covered by those QC-Plans.

The Contractor shall be responsible for identifying in a suitable manner, any inspections and test required for statutory purposes.

The Contractor shall keep the Owner informed in due time of starting and progress of the work so that the Owner will have sufficient time to make arrangements for the inspection.

The Contractor shall identify and isolate all items which do not comply with the required Quality Plan or Contract requirements. All such items shall be reported to the Owner immediately.

Inspections by the Owner do not relieve the Contractor from any responsibility regarding defects or any other failures which may be found and are subject to Contractor's defects liability.

The Contractor shall adhere to the approved Quality Control Plan. Any difficulties, deficiencies or other conditions adversely affecting quality shall be reported to the Owner.

The presence of the Owner or his representative does not release the Contractor from any responsibilities towards the Owner under the Contract.

The Owner will mark each Quality Control Programme to indicate:

17.3.1. Notification Points for Inspections and Tests (NPIT) (or “Witness Point”)

Notification of an inspection or test shall be given in writing by the Contractor to the individuals or organisations designated by the Owner.

If, following receipt of timely notification, the Owner decides not to attend, the inspection or test may proceed.

17.3.2. Hold Points (HP)

A Hold Point is a stage in the material procurement or workmanship process which requires documented proof of approval and beyond which work shall not proceed without written approval by the designated individuals or organisations designated by the Owner.

Subject to the provisions in the Contract the Contractor shall notify the Owner two (2) weeks before the start of all HP and NPIT points. Verification shall be given five (5) working days prior to the start of the event.

17.4. Quality Records

Quality records for the project consists of documents witnessing the effectiveness of the quality system (e.g. review protocols) and documents prepared for reaching the defined quality of the products, i.e. inspection results and test reports.

Test reports showing detected defects shall be provided with charts or drawings showing the exact locations and magnitudes of the defects. Complete information about the corrective measures must be provided to the Owner.

On completion of the manufacture or work and completion of inspections and tests the Contractor shall provide to the Owner and include with the Plant, equipment, or completed work a Certificate of Conformity/Compliance or a Quality Release Note.

17.5. Third party certification

The Contractor shall arrange for manufacturers' certification of the design, materials and construction of the equipment in his scope to support Owner to comply with legal requirements and for insurance purposes. The certification shall include all pressure vessels and lifting equipment, including slings, and shall cover both fabrication at the manufacturer's works and tests at the Site.

17.6. Workshop Tests / Inspections

The equipment to be supplied under the Contract shall be subject to works inspection and workshop tests.

The inspection and tests shall prove whether the equipment fulfils the requirements of the Contract in respect to:

- safety conditions according to respective specifications
- considerations of the applied standards and regulations
- execution of workmanship
- conformity with the present state of technology
- conformity with the requirements of the detailed design.

Whenever inspection or tests are carried out in the manufacturer's workshop, all intermediate and final certificates, in accordance with the agreed upon test schedule shall be made available to the Owner's Representative for the inspection. The latest versions of the related drawing and/or documents indicating the state of approval by the Owner's Representative shall be made available to the Owner's Representative.

All test certificates in respect of the final works inspection or workshop tests shall be submitted to the Owner's Representative.

At least fifteen (15) days' notice shall be given in writing to the Owner's Representative of the dates upon which the inspection or testing shall be carried out.

The approval by the Owner's Representative of any such inspection or test will not prejudice the right of the Owner's Representative to reject the Plant in accordance with the provisions of the main body of this Agreement.

18. PROJECT DOCUMENTATION REQUIREMENTS / DESIGN REVIEW AND AUDIT

18.1. General requirements

Contractor shall submit to the Owner, for the Owner's review, a detailed document submittal schedule listing all documents the Contractor will produce in line with the requirements specified in Exhibit B, Part 1 and this Exhibit B, Part 2 of the Agreement.

The Contractor shall be responsible for submitting all Plant-related information, documentation, calculations, drawings, and schedules to the Owner on a schedule consistent with the Project Implementation Schedule. The documents shall employ an equipment numbering scheme based on the KKS, or an equivalent system. The Contractor shall prepare a comprehensive plant or equipment identification schedule using the above mentioned numbering system. The schedule shall include this numbering system and the respective flow sheet or diagram identification numbers.

The Contractor shall be responsible for submitting all documents in accordance with a program to be prepared by it for review by the Owner. This schedule shall provide enough time for the Owner to check, assess and comment upon the Documents within a period of twenty-one (21) days from initial submittal, unless otherwise specified in this Contract.

The Owner shall use its reasonable efforts to complete such activities in a period of less than twenty-one (21) days, with fourteen (14) days as the objective under normal circumstances.

The quality of all documents submitted by the Contractor shall conform to acceptable international practice and allow expeditious review by the Owner.

18.1.1. Documentation Review Procedure

When submitting Documents for review, including any prepared by a Sub-Contractor, the Contractor shall certify in each case that it has examined such Documents and that they comply with the requirements of the Contract.

Except as otherwise provided in the Contract, within the relevant period stated above, the Owner shall either:

a) return to the Contractor one copy of any Documents submitted to the Owner with no comments or queries endorsed thereon;

b) shall return one such copy with any comments or queries which it may have endorsed thereon; or

c) a Document Review Comments Sheet (into an agreed format) shall be issued to facilitate rapid transfer of such comments or queries.

If the Owner fails to take any such action within the said period, then the said documents shall be deemed to have been returned to the Contractor with no comments or queries endorsed thereon.

The Contractor shall, within fourteen (14) days of receipt of any comments or queries on any document amend such document in order to implement and accommodate the Owner's comments or queries and resubmit such Documents for review as above, except that the period for review of resubmitted documents by the Owner shall be fourteen (14) days.

Review of a document by the Owner will be carried out according to the criteria set out in the Agreement, and will imply that:

- general arrangement and layout drawings and key diagrams have been examined and appear to be in accordance with the basic design concept of the Plant and meet the requirements of this specification
- other drawings of plant and equipment have been examined in relation to compatibility of the items and equipment with such specifications and in respect of interconnections with other items, equipment or systems

Any review by the Owner shall not relieve the Contractor of its obligations under the Contract.

Before submittal of any documents for the Plant, the Contractor shall submit to the Owner for the Owner's review a detailed document submittal schedule comprising all documents the Contractor will produce. Based on this list the Owner will decide which documents will have to be submitted for review, information or any other purpose. The document submittal schedule shall be maintained and updated during the course of the Works.

The Contractor shall ensure that documents for review are forwarded to the Owner in sufficient time to allow adequate time for examination of such documents. The Contractor shall also ensure that documents are submitted early enough prior to the date required for review to permit amendments to be made and the documents re-submitted for review without delaying the Project Implementation Schedule.

Drawings must be reviewed by the Owner before any equipment is workshop tested or before erection work is started at site.

18.1.2. Drawings

All drawings, including diagrams, schedules, performance curves, data sheets, test reports, Quality Assurance documents etc. submitted to the Owner shall comply with the requirements of the Agreement.

The drawings shall indicate that:

- general arrangement and layout drawings and key diagrams have been examined and appear to be in accordance with the basic design concept of the Plant and meet the requirements of the specifications set forth in this Minimum Functional Specification
- other drawings of plant and equipment have been examined in relation to compatibility of the items and equipment with such specifications and in respect of interconnections with other items, equipment or systems
- drawings shall be to scale and fully detailed with the maximum size of A0 (840 mm x 1190 mm). Major dimensions shall be given and the material of which each part is to be constructed shall be indicated
- drawings and as built documentation shall be black lines on a white background with revisions clearly marked and shall be prepared in AutoCAD compatible CAD system or pdf-format
- drawings shall bear the approved project references and title block drawings shall be prepared in the English language and with S.I. units

The Contractor shall comply with the following general requirements for electronic submissions:

- for design review purposes the contractor shall set up a dedicated electronic project file system which can be used as the delivery link for the documents to be reviewed for design review purposes and place of the latest and most up-dated project design documents and files where for both parties project members and the owner's engineer has a limited access to see and get those project documents as necessary
- contractor shall propose, apply and provide a system for a suitable "intelligent" electronic system (searchable) for the O&M manuals and files
- all documents referred in this section (created using MS Project/ Word/ Excel, AutoCAD, Micro Station) shall be submitted also in CD-rom in native file format.
- the delivered documents, especially pictures and drawings, shall be in such high quality condition that a necessary amount of copies of good quality can be made of them
- all project documents will be easily traceable (e.g. by adding a cover sheet) to the subject involved and shall contain the following minimum information (final version to be agreed between the parties) :
 - the name of the contractor
 - the name of the project
 - identification code of the document
 - identification number or letter of the modification

- type or the name of the document (unit drawing, operation instruction etc.)
- when performing changes new versions of the documents will be issued and each change in the document will be clearly marked and identified
- in the drawing sizes the relevant standards shall be observed
- plant identification system shall be used for component identification.
- in addition to this coding the drawings shall include a short characteristic name/description of the content

18.2. Design documentation

The Contractor shall submit documentation for the Owner's review taking into consideration logical phasing of design based on Good Industry Practice.

The Contractor shall submit to the Owner all such Documents as may be necessary for the Owner to verify that the equipment and materials to be included in the Plant complies in every respect with the requirements of the Contract.

Contractor shall submit the following information to the Owner, as a minimum, during the design phase of the Project Implementation Schedule:

Document	Owner/ Owner's Engineer
1. Documents to be submitted to Government Authorities	A
2. Data and calculations:	
• design data, design criteria, performance calculations	A
• correction factor data sheet (correction curves) plus those relevant to plant operation	A-
• stress and elasticity calculations of the piping system (for main/essential systems only)	A
• pressure loss calculations	A
• electrical studies, short-circuit and voltage drop calculations	A
• equipment sizing calculations - basis / details	A
• electrical load summary (switchboard wise)	A
• control valves and measuring orifice calculations	A
• data sheets	A/I
3. Diagrams and graphs:	

Document	Owner/ Owner's Engineer
● piping and instrument diagrams for all media	A
● flow diagrams for all media	A
● performance diagrams for all equipment	A/I
● pump characteristic curves	A/I
● functional diagrams for all control loops	A
● characteristics of all control equipment	A
● mimic diagrams for control panels and desks (or equivalent HMI graphic displays)	A
● electrical single line diagrams	A
● electrical interlocking - scheme / diagrams	A/I
● wiring diagrams	A/I
4. Drawings and plans:	
● Layout drawings including equipment locations	A
● arrangement drawings including indication of dismantling and sectional views	A
● pipework arrangement drawings	A/I
● isometric drawings for all piping systems (to be submitted on system basis only if requested by Contractor)	I
● drawings of cable routing	A
● earthing layouts	A
● lighting layouts and panel schedules	A
● arrangement drawings of control rooms including arrangement of instruments on desks and cubicles	A
● interior arrangement drawings of cubicles (modules)	A
● (outline/arrangement drawings of the cubicles)	
● drawings of the valves, control valves with actuators and safety valves (cross sections with material information)	A
5. Drawings and Information for Civil Works & Buildings:	
● architectural drawings including external finishings	A
● drawings showing layouts and details of the supply and disposal systems (potable water,	A

Document	Owner/ Owner's Engineer
stormwater drainage, sewage drainage, etc.)	
● connection details of the equipment with the necessary supply and discharge installations	A
● foundation engineering report	A
● topographic survey	A
● plot plans (above and under ground)	A
● drawings showing layouts and details for roads and pavements	A
● loading plans (or other drawings providing foundation loads)	A
● foundation drawings (layouts and details)	A
● structural calculations and drawings of all buildings and structures	A
● “Room Books” for all rooms and areas containing information on interior finishings	A
● drawings for HVAC	A
● design of HVAC system	A
6. Lists and Schedules:	
● Overall time schedule	A
● drawing lists regularly updated	A
● consumer (user) lists for power, water, air and other auxiliary media	A
● equipment lists using a special code system (e.g. KKS) including ratings and settings	I
● list of pipework including operating and design data, pipe class (including materials), types and reference drawings and/or catalogues	A
● list of control valves including operating and design data, materials, makes, types and reference drawings and/or catalogues	I
● list of instruments (measuring points, measuring devices, transmitters, recorders etc.) including location plans	A
● I/O list with schedule of control/indication loops (measuring points, measuring devices, transmitters, control equipment, actuators) including protection, interlocking and monitoring	A
● list of actuators	I

Document	Owner/ Owner's Engineer
● list of alarms and trips including settings	A
● electrical protection relay coordination drawings and setting schedules	A
● fault printer sequence list (event recorder input list)	A
● cable lists including cable numbering	I
● terminal point lists	I
● list of binary and analogue computer inputs	A/I
● equipment numbering/cable numbering system	I
● lubricant list	I
● chemical list	I
● vendor list	A
● consumable list	I
● list of special tools and inspection equipment	A
● spare parts list (recommendation of contractor)	A
● motor list	I
7. Reports	
● workshop inspection and site test reports	A
● performance test reports (including all necessary type testing and factory test reports)	A
● monthly status reports	A
8. Description, documents, manuals	
● specifications of electrical equipment	A
● painting specification	A
● installation standards and manual	A
● vendor drawings (subject to review or for information in accordance with principles of this listing)	I/A
● description of design concept for each system	A
● description of control and monitoring equipment	A
● description of control and safety circuits	A
● description of protection against fault conditions	A
● description of computer operation	A
● description of computer language	I
● computer programs descriptions and logic	I

Document	Owner/ Owner's Engineer
diagrams	
quality assurance manuals (including quality control procedure for factory and site construction)	A
list of Codes and Standards	A
lubricant list	I
testing procedures for factory and site tests	A
9. Commissioning schedule & procedures	A
10. Training manuals	A

A = submitted for review of compliance with MFS

I = submitted for information

18.3. Training documentation

The Contractor shall submit to the Owner for review documentation concerning training including items such as:

- Detailed training schedule with the final dates and places of training courses (for Owner's approval)
- Training material for each of the training courses

18.4. Quality Assurance documents

Quality assurance documents of Contractor and sub-suppliers shall be according to the relevant ISO 9001 standard and Contractor's QA instructions. Documentation to be handed over to the Owner for review covers items such as:

- project Quality Assurance Programme
- quality Control Programmes and Plans
- health & Safety and Environmental (HSE) plan (refer to Attachment 5: "HSE documents to be prepared by Contractor")
- manufacturing Record Book
- factory and on-site test results and certifications of the Plant and Goods
- declarations of Conformity

18.5. Documents for commissioning and testing and Contractor's performance Tests and SEC acceptance tests

The Contractor shall submit to the Owner for review the program for Commissioning and Testing including also the Contractor Performance Tests and Acceptance Tests as per the Exhibit I-1 of the Agreement.

18.6. Operation and Maintenance (O & M) Manuals

18.6.1. General Requirements

The Contractor shall provide the O&M manuals for review including all information necessary for the safe, secure and economical operation and maintenance of the Power Plant.

The content of O&M documents shall be in compliance with the relevant practice and industrial requirements and subject to Owner's approval.

The operational and maintenance documentation shall cover the whole Plant and all systems and equipment one by one within the scope of Work.

The manuals shall include items such as :

- operation instructions
- maintenance instructions
- equipment lists, settings, etc
- drawings, diagrams , etc
- spare parts, special tools

18.6.2. Operation Manuals

The operation instructions shall include operational situation such as:

- preparations for start-up
- start-up
- operation in power and heat generation
- shut-down
- disturbance situations
- standstill and preparation for maintenance
- preservation

As far as applicable, the following items may also be included in the operational instructions:

- technical structure and functioning of the system in question
- theoretical background of functioning of the system concerned
- electrical part of the system/equipment
- automation/instrumentation part of the system/equipment
- water and steam part system/equipment
- steam/water chemistry
- material questions
- operational limits for measured values
- modes of operation to be avoided, if any, and the reason for restrictions
- precautions to be taken for normal and abnormal operating conditions

The operational instructions may include photographs, videos, drawings, pictures, functional diagrams and other material necessary for understanding and visualization of the instructions.

The operational instructions shall be divided according to the above operational situations of each individual system so that the above items are included in each operational situation.

18.6.3. Maintenance Manuals

The target of the maintenance documentation is to ensure that the Owner will be able to execute the maintenance of all equipment and systems, including tasks such as:

- design of equipment register
- design of preventive maintenance programs
- design of annual maintenance programs
- design of inspection system
- design of test programs
- determining the numbers of spare parts and materials, e.g. lubricants
- acquisition of special tools; dimensioning of the quantity of labour

The maintenance documentation to be supplied by the Contractor shall include items such as:

- list of drawings (summary list on drawings separately on each object)
- assembly drawings including a list of parts with data on dimensions and materials
- manufacturing drawings on fragile and wearing parts including data on dimensions and materials

- drawings, measurement and inspection charts, Site erection protocols, (including data on erection dimensions with tolerance limits), and installation instructions and documents needed for complete dismounting, control and mounting work.
- spare part lists and drawings
- data on maintenance is required separately for mechanical, electrical and instrumentation maintenance.
- all maintenance documents shall be delivered also in electrical form to the extent available

The Contractor shall provide in electronic form to be agreed equipment records for the Plant's maintenance management system and spare parts documentation as follows:

- the equipment record (data sheet) with exact identification code shall contain complete technical data on each process equipment (valve, pump, motor etc.) with reference of proper document and drawing numbers
- spare part lists, spare parts recommendation with technical data filled out is required for each equipment.
- maintenance instructions and schedule for preventive and periodic maintenance
- safety and isolation instructions
- templates and instructions for data input will be provided later by O&M Contractor

18.7. As-Built Documentation

The Contractor shall prepare, and keep up-to-date, and submit to the Owner for review a complete set of "As-Built" Contractor's Documents of the execution of the Work, showing the exact As-Built locations, sizes and details of the work as executed including documents such as :

- design drawings with final process and equipment information
- equipment, devices, signals, etc. with final settings
- certificates, manufacturer's manuals, erection dimension protocols, test protocols, QC documents

18.8. Required documentation for proposal

In addition to the requirements specified in Exhibit B, Part 1, the Contractor shall submit the following initial technical information for the Proposal.

18.8.1. Cogeneration plant concept

The Contractor shall submit, as part of its Proposal, the following technical information:

Complete data sheets to be provided to the Owner as a separate Appendix within the Agreement.

Description of main features of the proposed plant such as:

- general description and main features of the plant and use of the plot including laydown areas
- detailed time schedule
- construction approach, phases of construction, commissioning and operation of plants constructed at the various phases
- operational philosophy of power units, steam generation units and desalination units (start-up, shutdown and failure of individual units)
- maintenance philosophy, maintenance schedule activities including major plant overhauls
- details of operation, maintenance and training
- standards, codes and regulations: listing of major standards, codes and regulations (international, supplier's own country) which will be applied for the design and construction of plant and equipment and facilities, for civil works and for testing of expansion plant equipment

Description of major equipment and systems including auxiliary plant:

- main mechanical equipment and systems including auxiliary systems, such as: reserve boilers; steam turbine generator plant; water treatment plant, etc.
- electrical equipment and systems such as: generators; generator protection; electrical power system philosophy; main electrical equipment including auxiliary systems, etc.
- ESF and HV grid connections (protection, metering and control interface);
- instrumentation and control systems, such as: general control system philosophy; description of central control room; local control rooms; facility control system and control of common plant and auxiliary systems, etc
- emission control measures
- civil works, including site development buildings, seawater piping and discharge water systems
- plant drawings and diagrams
- plant location map with areas temporarily used during construction
- site plan, general layout and arrangement drawings for on- and off-site facilities indication of fenced areas
- equipment configuration in the area of the power plant, ccr, water treatment plant, etc.

- PID for main systems;
- single line diagrams for
 - facility main electrical systems
 - facility electrical auxiliary systems
- protection diagrams
- heat balance diagrams for the plant at the reference site conditions (guarantee)
- heat balance diagrams for the plant for the following operating points shall be provided but not limited to
- preliminary correction curves for part loads and deviating ambient conditions
- start-up and loading diagrams (from cold, warm and hot conditions) of units and plant
- operation curves of the units
- degradation curves of the units
- generator performance charts

Environmental Aspects and Data

The Contractor shall submit, as part of its Proposal, the description and details of the following:

- guaranteed air pollution emission rates and type of emission control and monitoring measures
- nature and estimated quantities of waste water, waste water treatment and disposal methods
- ambient noise levels

Sub-Contractors, Suppliers

The Contractor shall submit, as part of its Proposal, details and references of the nominated and/or proposed sub-contractors and suppliers/manufacturers (where applicable).

The Contractor shall submit a schedule of all major proposed subcontractors and suppliers detailing the type of work or services to be subcontracted and the country of origin.

Deviation/Exceptions from Exhibit B, Part 2

The Contractor shall declare deviations/exceptions from this Exhibit B, Part 2, any undeclared deviations shall be deemed to be accepted by the Contractor.

19. PROJECT ADMINISTRATION AND MANAGEMENT

19.1. Introduction

The Contractor shall be responsible for the management and control of all phases of his work and that of his sub-contractors and suppliers.

19.2. Project Schedule

The Contract Programme shall be the overall schedule of key dates agreed in Exhibit A of the Agreement.

19.2.1. Project schedule

The Contract Programme shall be the overall schedule of key dates agreed prior to award of the Contract, and included in EPC Agreement. The schedule which shall not be altered except as otherwise agreed by Owner shall be broken down into major elements and shall clearly show the key dates relating to:

- payment milestones
- engineering submissions and reviews
- access to the site area for preliminary activities
- access to the site for construction activities
- availability of services (water, fuel gas and fuel oil, electricity)
- exchange of information with other parties concerning terminal points
- completion of all work relating to terminal points
- start of testing and commissioning
- completion tests
- take over of each unit

19.3. Project Control Programme and Reporting

19.3.1. Levels required are: Planning Network Programmes

The Contractor shall prepare a detailed critical path network and bar chart that will fully describe the work to be carried out. The Contractor will use his standard systems for planning subject to their meeting the requirements stated herein.

The critical path network shall include all principal activities and milestones. The bar chart shall be based on the same information as the network and shall be capable of presentation in both monthly and weekly increments as may be required by the Owner.

The Agreement shall be monitored and controlled using the planning system, which shall be tiered to enable presentation at various levels of detail, as may be required for specific purposes during the Agreement, and whereby all activities can be uniquely identified and related to activities at other levels.

Level 1

Overall programme showing the project milestones, payment milestones and containing major key dates and activities

Level 2

A more detailed network programme including construction interfaces clearly identified as individual activities with both documentation and physical tie-ins

Level 3:

Project Master Schedule. The Contractor's detail level of programming with a record for each activity. Each major sub-contractor shall maintain and update his own schedule and pass all updated information to the Contractor who shall keep an overall programme on which the 90 day look ahead programmes are based

Level 4

Contractor's working schedules produced and maintained by the Contractor, for internal purpose, if available, and;

Level 5

Contractor's and sub-contractor's work definition and progress measurement level for internal purpose, if available.

Issue of Programmes

The complete Level 3 programme together with updated versions of the Level 1 and Level 2 programmes shall be issued to the Owner within 60 days of the Notice to Proceed.

19.3.2. Progress Measurement

The Contractor shall develop and maintain a formalized progress measurement system based on physical progress (not cost).

It shall be the Contractor's responsibility to measure his own progress in line with and at the levels of detail defined within this Exhibit B, Part 2 and elsewhere in the Agreement.

19.3.3. Progress Reporting

The Contractor shall submit a detailed progress report for each month according to the Agreement. The format of the monthly report shall contain but not be limited to:

- covering letter and executive summary
- an update of all safety statistics including a statement of injury or death of personnel engaged in work on site and the number of accidents or other incidents resulting in damage to equipment or plant, or incidents that represent a near miss to these events, together with details of such accidents and incidents, and statements of corrective actions
- document index marked up to show document status
- a statement of the number of site personnel engaged in the work during the reporting period including an update of manning histograms and details of erection equipment in use or held in readiness
- a listing of late activities compared to the respective originally planned (early finish) dates
- explanations for late activities and other problem areas which are having, or are likely to have, impact on the project schedule
- details of measures proposed to resolve problem areas and bring the overall project progress back on schedule or further improve on current progress
- outstanding interface data and measures proposed to expedite the issue of critical interface data
- demonstration of the achievement of all milestones
- purchasing schedule associated with major or significant items of equipment marked up to show status of procurement activities
- copies of those inspection reports which identify any deviations from the quality standards in the epc contract and a statement of corrective actions
- copies of quality assurance audit reports which call for corrective actions and evidence of the implementation of corrective actions
- colour photographs showing the progress of construction;
- update of progress 's' curves
- correspondence schedule listing all correspondence between contractor and owner and showing outstanding actions required
- a statement of progress made in the preparation of all documentation to be handed to the owner at milestone dates or substantial completion
- shipping documentation
- the level 3 "project master schedule" shall be marked up on a monthly basis. the contractor shall indicate the progress achieved against each activity and submit this as part of the monthly progress report which shall be issued to the owner one week prior to the monthly progress meeting

- the progress report shall address each of the following project phases, as appropriate:
 - engineering
 - procurement
 - expediting and fabrication
 - construction and erection
 - commissioning and tests on completion

19.3.4. Progress Meetings

During the engineering and procurement phases, all progress meetings, which shall be held monthly and be chaired by Owner, and any other meetings deemed necessary to resolve technical and commercial problems, shall be held in Owner's offices or Owner's Representative offices, or at another location agreed by Owner. Once the main construction activities commence, weekly site management progress meetings will be held at Site. Additional meetings to resolve technical and commercial issues will continue to be held, as required, during the construction phase.

19.3.5. Project Management and Co-ordination Procedures

The Contractor shall, within thirty (30) days after the Notice to Proceed Date, submit to the Owner a comprehensive set of project management and co-ordination procedures that shall be mutually agreed upon, which describe both the Contractor's project management organization and procedures, and co-ordination and communication with the Owner. Based on the Contractor's project instruction manual, the procedures shall cover the items detailed below.

Contractor's project management organization including sub-contractors (the status and responsibilities of Contractor staff (only) shall be included):

- reporting
- planning and work control
- communications
- administration
- other topics needed to describe the project management and co-ordination

19.3.6. Communications

All correspondence, communications and submissions relating to the Agreement shall be in English.

The Contractor shall have a Site Manager fluent in the English Language for routine administration with the Owner/Owner's Representative at the site.

For all documents, goods, materials, maintenance equipment, spare parts, instruments, etc. to be handed over to the Owner, the Contractor shall obtain a signed receipt by an authorized representative of the Owner that the goods, etc., have been received in a satisfactory condition as required under the Agreement.

19.3.7. Diaries for Daily Work Measurements for construction site

The Contractor shall keep suitable diaries in an approved form to which at all times the Owner and /Owner's Representative shall have access. The Contractor shall record all work progress as completed on the day on which the records are made, together with full particulars and details of all obstructions or modified or extra works or incidents, and the number of men employed in each of the several positions of the work in progress.

The diaries shall be available at all times for Owner/Owner's Representative to verify and record, and shall sign all entries therein which it verifies, and have full liberty to make entries therein but no such signature or entry shall imply acceptance of/or agreement with any area of work material used. The diaries shall, when signed by both Parties, be deemed to be a true record of the work executed and shall become the property of Owner.

20. SPARE PARTS

20.1. General

The Contractor shall provide all spares necessary for discharging his responsibilities in carrying out the Works, including commissioning and up to Commercial Operation Date. Any spares used during the period up to the end of the Commercial Operation Date shall be promptly replaced. The Contractor shall notify the Owner in advance on using the spares supplied before the commercial operation date and obtain permission from Owner for using them. However, any Spare Parts used by the Contractor to carry out warranty repairs including those for gas turbine generators shall be supplied free of charge and should be promptly replaced if used from Owner's stores.

The Contractor shall provide a priced list of spares which he recommends the Owner should hold. These recommended spares should be classified as follows:

Initial Operating Spare parts:

Maintenance spares and consumables

These are items for which the Contractor anticipates that demands will arise in normal operation of the plant.

Overhaul spares and consumables

These are items for which the Contractor anticipates that demands will arise during the programmed minor and major overhauls.

Strategic spare parts:

These are items for which the Contractor anticipates that demands may arise through breakdowns which could jeopardise the availability or safety of the plant. The Contractor shall state the proposed stock holdings for such items together with the prices of each.

All spares shall be directly interchangeable with the corresponding parts installed in the plant and shall meet the requirements of the specification of the original plant.

Spares shall be warranted for an equivalent period to the original plant.

Spares list shall be in the format of a schedule including the following data:

- description
- identification
- price
- delivery (weeks)

The Contractor shall replace at his own expense all spares taken from the station stores and used during construction, commissioning, testing and warranty (Defects Notification Period).

Unless specified elsewhere in this Exhibit B, Part 2, the Contractor shall provide a minimum 10% installed spares for electrical and instrumentation systems such as:

1. Terminal blocks for Junction boxes, Cubicles, panels, Marshalling cabinets, racks etc.;
2. Breakers for AC & DC distribution boards;
3. Breakers for MCC's, MV and LV Switchgears etc.,

21. TRAINING

21.1. General Requirements

The details of the training program shall be agreed between the Contractor and Owner's.

Emphasize shall be on “on-the-job” training and instruction for O&M personnel and building up the training programs (2 separate training tracks for operation staff and 1 for maintenance staff) accommodating O&M staff's mobilisation program.

All programs will be scheduled for a normal 5 day works week, excluding holidays observed by the Owner. Due to the technical content of the programs, instruction will be limited to 8 hours per day.

The Contractor shall advise the Owner of all sub-contractor representatives on site to allow the Owner's personnel to co-ordinate and receive on-the-job-training.

Here below is highlighted the overall framework and philosophy the O&M training should be based on when Contractor starts to plan the detailed training program for the Owner's O&M personnel.

O&M Operator's personnel shall be trained in due time by the Contractor to prepare them both in theory and practice for the management, operation and maintenance of all parts of the Plant.

Each site personnel shall be included in Overviews and Operation training.

English language shall be used for all training purposes.

The aim of the training is to teach the operation and maintenance personnel of the Owner to such an extent that after completion of the training they are able to:

- safely and economically operate the Power Plant with all its auxiliaries without the support of the Contractor
- carry out regular daily and weekly maintenance routines independently and in a correct way
- carry out maintenance and repair works independently and in a correct way

Also “system walk-downs” with groups of 6 or less students shall be conducted to familiarize the student with the physical location of the equipment and acquaint the student with system and equipment features, controls, and displays. If possible actual operations, routine maintenance and routine checks of all equipment shall be demonstrated.

In addition to the above classroom training, an essential part of the training shall be the participation in the operation and maintenance during the commissioning phase of the Plant. The trainees will obtain real experience by attending the Plant operations under the supervision of the Contractor’s commissioning engineer. The personnel will become familiar e.g. with remote/local control and monitoring, equipment operations as well as with the technical characteristics of the Plant. Detailed plan including shift schedule for the same shall be pre-agreed between the Contractor, Owner and Operator 180 days prior to start-up of any commissioning activities in the plant.

Training shall be provided by the Contractor’s commissioning team and/or a nominated professional training team. The Contractor shall nominate a training coordinator. The training documents shall be based on the Plant instruction manuals. Drawings / schematics of the Plant shall be extensively used in the training.

The training shall consist partly of classroom lectures supported by audio and/or visual training aids. However, the practical ‘hands on’ exercises utilising the supplied equipment will form a major part of the training.

The above training duration shall be considered to be an adequate prerequisite for the Plant personnel to acquaint themselves with the correct operation and maintenance requirements of the installed Plant.

The training courses and programme shall be split between those which are general or dedicated for operating staff or for maintenance staff.

The proposed training programme as described in the following shall be taken into consideration by the Contractor when preparing the training programme. The training program and its content are subject to approval by the Owner.

Training courses may be delivered in the administration building or at any other place suitable for the purpose, provided by the contractor.

Trainer’s qualifications

Trainers shall be suitably qualified, through experience and education, to teach their respective subject matter.

Evaluation of training

Owner will review and audit the Training Programme and individual performance at any time for Contractor's compliance before accepting the successful completion of any part of the Training Programme.

Competence assessment

Trainees will undergo competency assessment at two points within the Training Period. Owner will first undertake a competency assessment prior to commencement of commissioning and secondly at the completion of training.

The Contractor shall keep comprehensive records of all the personnel who have been trained in a format agreed by Owner and these records shall become the property of Owner, to be used in the final assessment of staff.

Facilities and training materials/aids

Suitable facilities and training accommodation to support Owner's and Contractor's training shall be provided by the Contractor on site to meet the requirements of the jointly agreed training programme.

The Contractor shall provide for training materials, textbooks, manuals and other materials required for the Contractor's Training Programme. Classroom equipment such as overhead projectors, white/blackboard, monitor etc. shall be provided by Contractor.

Training aids, such as films, slides, interactive videos, software packages materials and supplies considered necessary by the Contractor to support, maintain and successfully supplement the Contractor's Training Programme shall be furnished by the Contractor for the Training Period.

Training manuals provided by equipment suppliers, for operations and maintenance shall form part of the basic reference material in addition to any other references as required.

The Contractor shall arrange for representatives from selected major equipment suppliers to provide specific instruction concerning the construction, engineering, operation and maintenance of the equipment as specified in the Contractor's technical proposal.

Review process

Owner reserves the right to review and discuss with the Contractor all training materials/ aids, course outlines, training schedules, steps, procedures, guidelines and other training matters.

22. ATTACHMENTS

Attachment 1: Preliminary plant layout;

Attachment 2: Heat Balance Diagrams;

Attachment 3: Water Balance Diagrams;

Attachment 4: Correction curves;

Attachment 5: GT Supplier's Guidelines for storage and inspection;

Attachment 6: GT Supplier's Marginal conditions

Attachment 7: Specification for P91 piping

Attachment 8: Building risk assessment